B4 — Chemistry in the Atmosphere

Part II · Michaelmas & Lent Term Lecture Notes

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Preface

These are the lecture notes for B4 — Chemistry in the Atmosphere (Cambridge Chemistry Part II), covering the Michaelmas and Lent term modules:

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Each module has a paired, pre-executed Jupyter notebook in `notebooks/`, cross-linked from the relevant lecture.

Michaelmas Term

Basic Physical and Chemical Structure of the Troposphere and Stratosphere

Module 1 · Basic Physical and Chemical Structure of the Troposphere and Stratosphere

Learning aims By the end of this module you should be able to: 1. Describe the temperature-based layering of the atmosphere and explain why the troposphere and stratosphere have such different mixing behaviour. 2. Derive and apply the hydrostatic equation and scale height to calculate pressure, density and number density at a given altitude. 3. Define and calculate the column amount (e.g. in Dobson Units) of an atmospheric constituent. 4. Outline the basic composition and role of the troposphere and stratosphere, including the importance of ozone in each.

1.0 Atmospheric layers — temperature, pressure and density

Most planetary atmospheres are heated by outgoing long wave radiation from the planetary surface, and so in general planetary atmospheres get cooler as one ascends through them. However, the atmosphere of our planet is different. Atmospheric temperature decreases with height initially, before undergoing a transition leading to increasing temperature with height. These changes in temperature gradient mean that the atmosphere can conveniently be divided into a number of layers (or spheres), depending on the temperature profile (Figure 1.1). The turning points in Figure 1.1 are known as "pauses" and are located at ~10–18 km (tropopause) and ~45–50 km (stratopause).

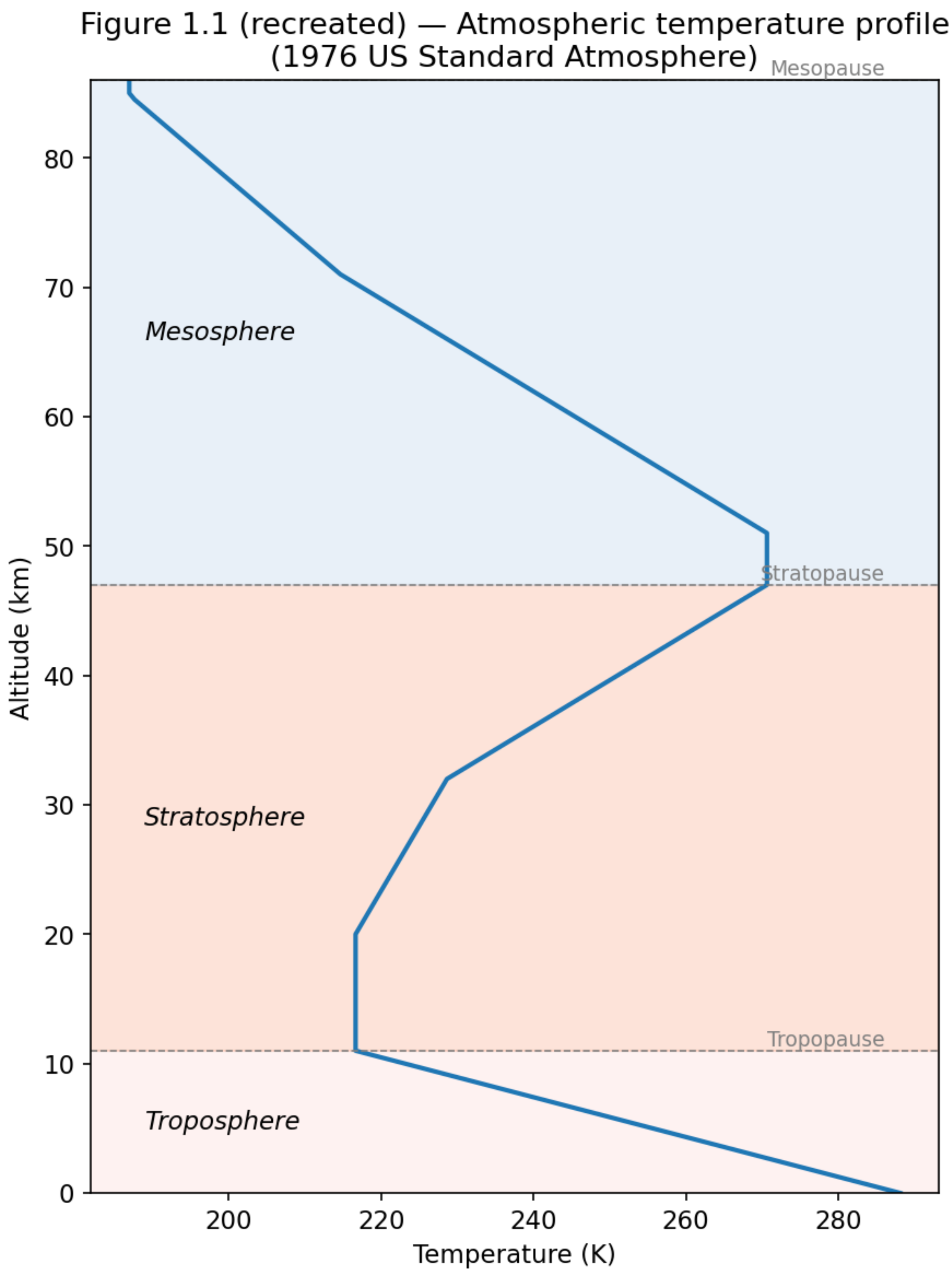


Figure 1.1 — Atmospheric temperature profile, generated from the 1976 US Standard Atmosphere reference data (see *data/us_standard_atmosphere_1976.csv* and *notebooks/01-atmospheric-structure.ipynb*).

In this course we will be primarily concerned with the lowermost layers, the stratosphere and troposphere, which make up approximately 99.9% of the atmospheric mass.

Figure 1.2 below shows that the zonal mean (latitude vs. altitude profile) atmospheric temperature profile is not only strongly altitude dependent (as depicted in Fig 1.1), but in the lower atmosphere also depends strongly on latitude. Two key features stick out in Fig 1.2: (i) the coldest part of the atmosphere is in the tropical upper troposphere; (ii) the height of the tropopause is latitude dependent (dashed line).

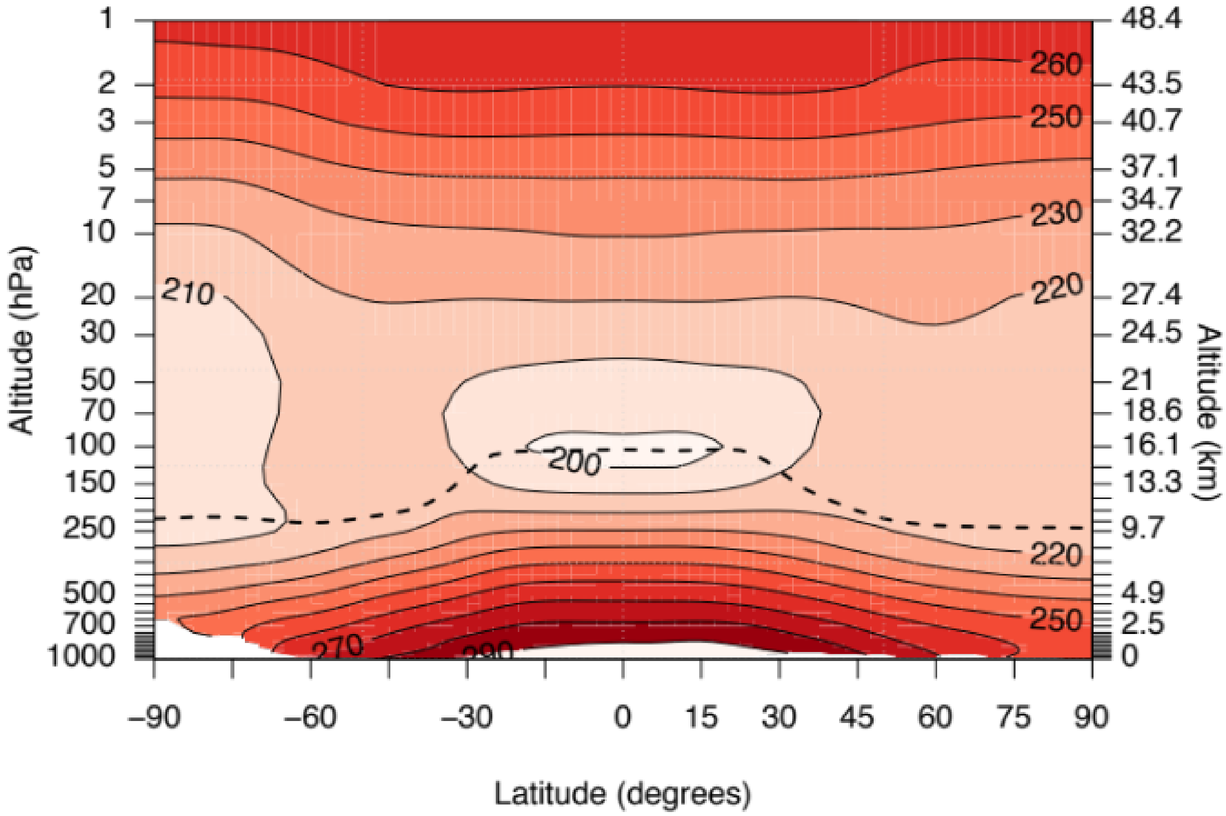


Figure 1.2 — Zonally averaged annual mean temperature.

Energy balance models can be used to determine the resulting temperature from the balance of incoming solar radiation and outgoing longwave radiation, but in order to calculate how temperature varies with altitude a radiative transfer model is needed. These sorts of calculations are beyond the scope of this course. What is important to appreciate is the general structure of temperature in the atmosphere, noting the different temperature environments found in the different layers. Indeed, these different temperature environments allow air to be well vertically mixed in the troposphere but poorly so in the stratosphere. However, mixing in the stratosphere is very rapid zonally. The temperature inversions associated with the pauses give rise to the unique chemical environments found in the different layers of the atmosphere.

1.1 The hydrostatic equation

Figure 1.2 shows that as we go up in the atmosphere pressure decreases. The hydrostatic relation is used to relate ambient pressure to altitude. If ρ is the ambient air density at altitude z with pressure p, we can write, for an incremental change in z:

$$dp = -\rho g dz$$

(assumes g is constant — not a bad assumption)

For an ideal gas we can write:

$$\rho = \frac{Mp}{RT} = \frac{mp}{kT}$$

where M and m are the relative molar mass and molecular mass respectively. Thus,

$$dp = -\frac{Mpg}{RT}dz$$

Defining $H = \frac{RT}{Mg}$ as the scale height (approximately constant with altitude) we can re-write the above expression to give:

$$\frac{dp}{p} = -\frac{dz}{H}$$

and integrate, setting the surface pressure as $p = p_0$, to yield:

$$p = p_0 e^{-z/H}$$

This equation describes the fall-off of pressure with altitude.

The pressure drops by $1/e$ over the scale height, H. In the Earth's atmosphere the average $M \approx 28.9$ g mol^{-1}, and so the scale height varies between 6 and 7.5 km, depending on atmospheric temperature.

Making use of the ideal gas equation, it is also possible to define an approximate relationship for number density ($n = n_0 e^{-z/H}$, where n often has units molecules cm^{-3}). In the rest of this course we will refer to number density as [M].

So, with increasing altitude above the Earth's surface, the density (and pressure) of the air falls by about a factor of 10 for approximately each 16 km (10 miles) of altitude (Figure 1.3).

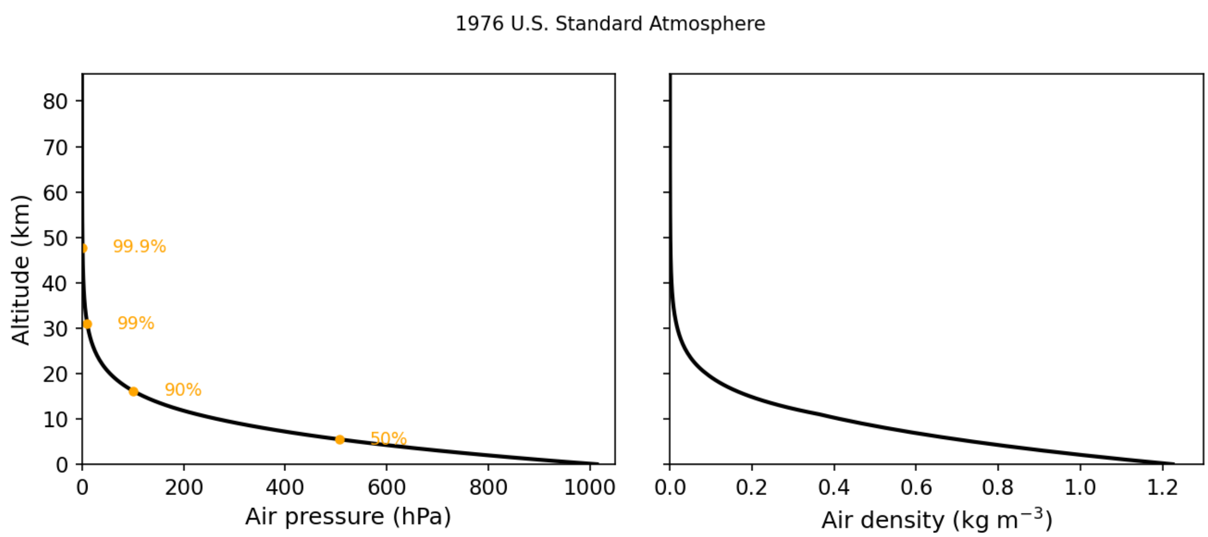


Figure 1.3 — Variation of pressure and density with altitude.

This has important consequences for the rate of some atmospheric reactions — especially termolecular reactions, but note that some bimolecular reactions are also pressure dependent. Up to an altitude of 80–90 km (the mesopause) the bulk composition is essentially $N_2 : O_2 = 4 : 1$.

Above this height atomic oxygen becomes the dominant oxygen species, the concentrations of ions and free electrons become significant and ion–molecule reactions are important.

At still higher altitudes, above 100 km, diffusive separation can occur, with light molecules such as hydrogen reaching higher altitudes (and even leaving the atmosphere altogether).

We should finally consider the timescales (τ) of mixing air across the atmosphere. The general circulation of the atmosphere is complex, but the key points to note are the different τ for mixing north–south, east–west and vertically.

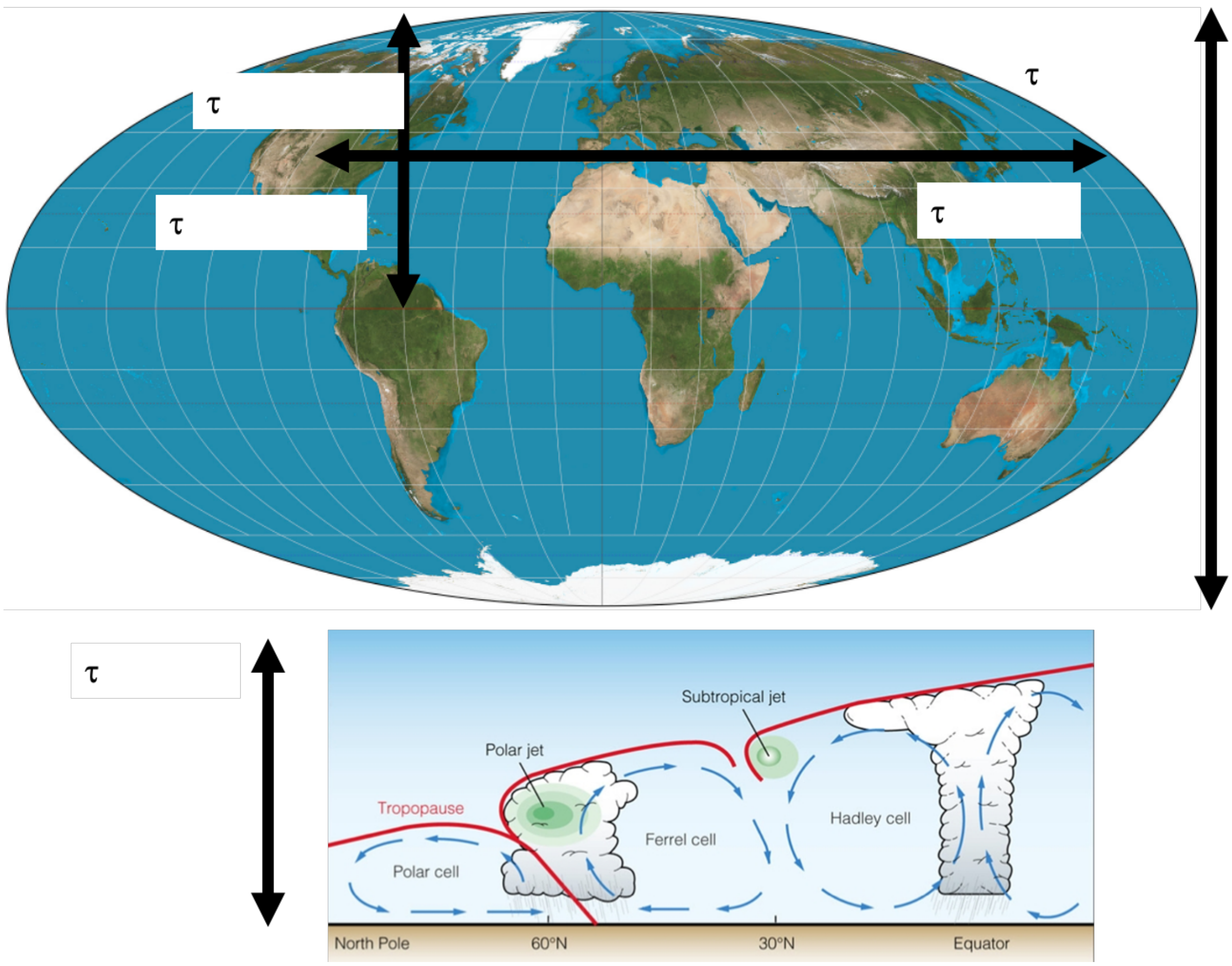


Figure 1.4 — Some typical mixing timescales (τ) in the horizontal and vertical across the atmosphere.

Exercise 1 — Calculation of the number density at different altitudes

The number density is another word for concentration, in molecules cm^{-3}.

We can make use of the ideal gas equation to calculate the concentration of a gas:

$$pV = nRT \quad \Rightarrow \quad \frac{n}{V} = \frac{p}{RT} \text{ (moles per volume)}$$

To get into molecules, use k_B rather than R (writing N for number of molecules):

$$\frac{N}{V} = \frac{p}{k_B T}$$

At 298 K and 1 bar pressure:

$$\frac{N}{V} = \frac{100{,}000}{1.38 \times 10^{-23} \times 298} = 2.43 \times 10^{25} \text{ molecules m}^{-3} = [M] = 2.43 \times 10^{19} \text{ molecules cm}^{-3}$$

1.2 The Troposphere — basic role and composition

As we've seen, the troposphere is that part of the atmosphere which lies between the surface of the Earth and about 10 to 18 km, depending on latitude (see Fig 1.2). While air in the troposphere is reasonably well mixed, it is inhibited from rapid exchange with the stratosphere because the tropopause is a relatively impermeable barrier to transport. The troposphere is also the region where most of what we regard as weather (clouds, precipitation, etc.) occurs. Radiative processes in the troposphere are the dominant processes affecting surface climate and climate change.

Chemically, the troposphere serves some very important functions. Chemical destruction (by OH (day) and NO_3 (night)) in the troposphere is the main mechanism preventing many gases emitted at the surface of the Earth from accumulating to amounts that are toxic to life or ultimately damaging to the stratospheric ozone layer or the climate system. This destruction (or usually the concentration of the oxidants) is an expression of the 'oxidizing capacity'. The troposphere provides a system of transport and chemical transformation for the natural biogeochemical cycles.

Table 1.1 shows the principal gaseous constituents of the troposphere, their mixing ratios and their atmospheric lifetimes. Disregarding the highly variable amounts of water, 99.9% of the atmosphere is composed of N_2, O_2 and the noble gases, principally Ar. These gases have been present at constant levels over geological timescales (although there is some debate over the fluctuations in O_2).

Table 1.1 — Composition of the Troposphere

Constituent	Mole fraction	Lifetime (yr)
N_2	0.781	1.6×10^7
O_2	0.209	9000
Ar	0.0093	4.5×10^9
CO_2	0.000416	50–150
H_2O	0–0.04	5 days
CH_4	2100 ppb	10
H_2	550 ppb	4
N_2O	320 ppb	150
CO	50–200 ppb	0.2
O_3	20–80 ppb	20 days
C_2H_6	1 ppb	0.2
SO_2	0.1 ppb	5 days
NO_2	0.1 ppb	2 days
OH	0.1 ppt	0.1 s

The remaining gases, representing less than 0.1% of the atmosphere, are diverse but have an important influence on a number of atmospheric processes. The gases with long lifetimes (CH_4, N_2O, etc.) are fairly uniformly distributed in the troposphere. The minor trace species, e.g. SO_2, NO_2, organic gases, all have abundances of ≤1 ppb (parts per billion by volume, or nano moles/mole, e.g. 10^{-9}), at the global average scale, but are very variable in space and time (NB on Lensfield Road the NO_2 mixing ratio is probably 20 ppb). O_3 has a concentration usually in the range 20–80 ppb with some variability.

Figure 1.5 shows the mixing ratios of a range of important gases in the atmosphere. This is a key figure we will come back to.

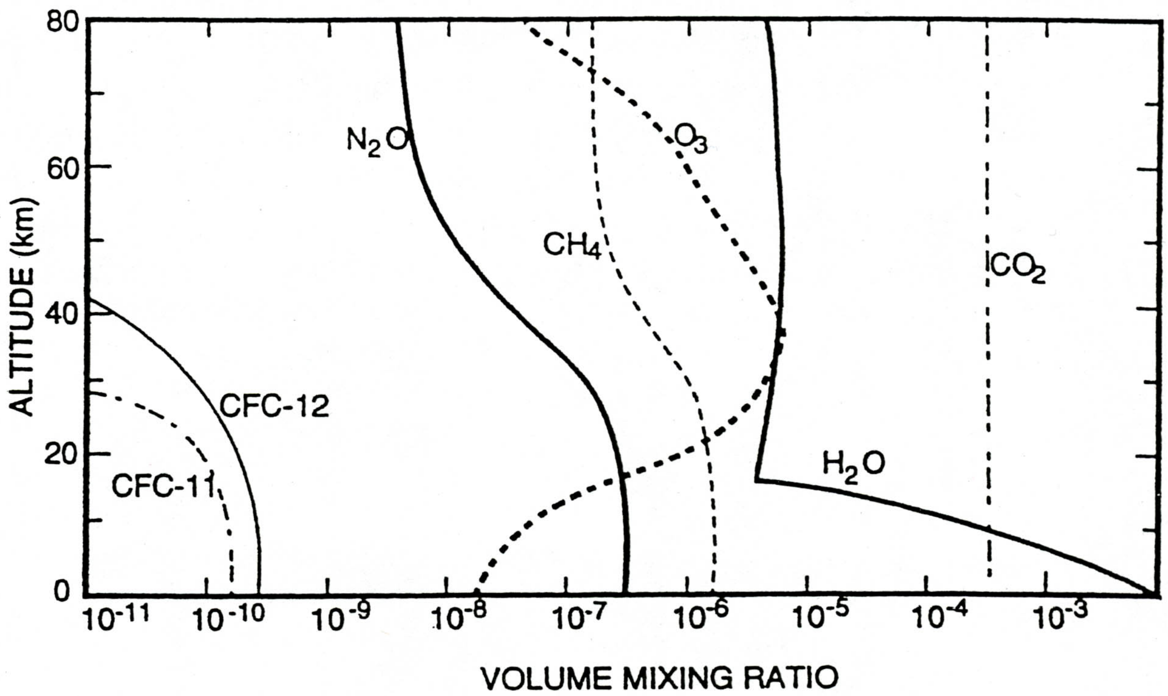


Figure 1.5 — Some typical mixing ratio profiles of a range of species.

> ℹ Key points in Figure 1.5
>
> - Mixing ratios tell us how [X] varies with [M] (NB [M] decreases exponentially with increasing height).
> - Long-lived gases have nearly constant mixing ratios (e.g. CO_2, N_2, O_2, He).
> - Increases in mixing ratio of a compound reveal to us that it is undergoing chemical production.
> - Decreases in mixing ratio denote chemical destruction (or loss).

1.4 The Stratosphere — basic structure and composition

The stratosphere lies above the troposphere, bounded by the tropopause and the stratopause at about 50 km. Temperature increases with altitude in the stratosphere — a stable situation which leads to a long removal time for pollution reaching the stratosphere.

Ozone (O_3) is perhaps the most important stratospheric constituent. Ozone absorbs ultraviolet radiation, which leads to the observed stratospheric temperature structure and also protects life at the surface from harmful radiation at $\lambda < 300$ nm. Ozone is also an important infrared absorber — it's a greenhouse gas involved in climate change.

Stratospheric chemistry is mainly concerned with the sources and sinks of ozone. Ozone has a peak stratospheric mixing ratio of about 10 ppm. Its local abundance is determined by a balance between photochemical production following photolysis of molecular oxygen and catalytic destruction by a variety of radical species (e.g. OH, HO_2, NO, NO_2, Cl, ClO, etc.), which are present at much lower concentration (usually ppb or less). (In those regions where the chemical lifetime of ozone becomes long, transport will also be important in determining the local concentration.) The radicals are produced following the breakdown of source gases (source gases shown in Fig 1.5) including H_2O, CH_4, N_2O and the CFCs ($CFCl_3$, etc.).

Throughout the bulk of the stratosphere gas phase reactions dominate. However, ozone loss in polar latitudes is initiated by reactions on the surface of particles or droplets — polar stratospheric clouds — in the lower stratosphere. Reactions on aerosol droplets (mainly sulphuric acid) are also now known to be important in the middle latitude lower stratosphere. (We will cover this chemistry in detail in IDP1.)

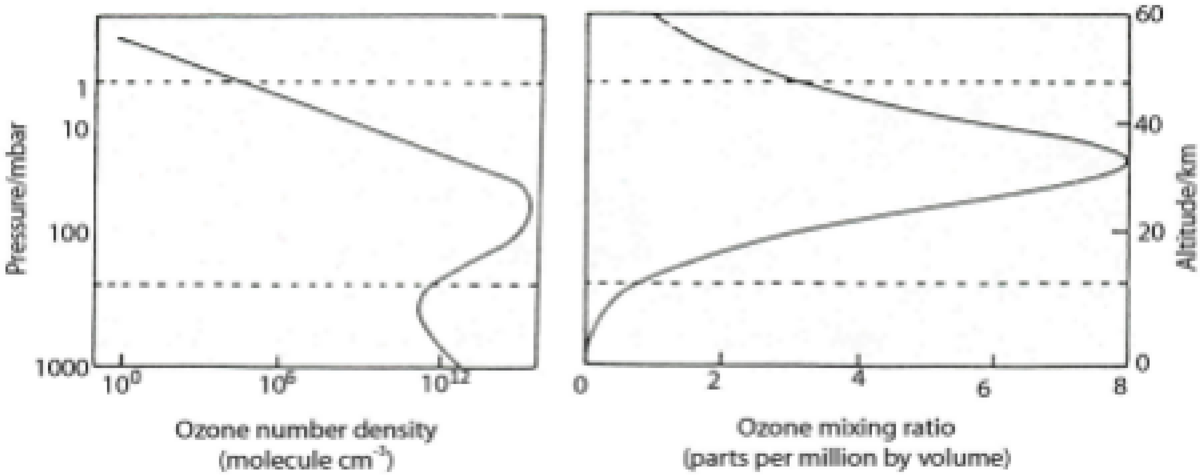


Figure 1.6 — Variation of ozone concentration with altitude, expressed as an absolute number density (n_{O_3}, with peak just above 10^{12} cm^{-3}) and as a mixing ratio (mole fraction, n_{O_3}/n_{air}).

We will see later that it is also possible to measure the amount of a species integrated between the earth's surface and the top of the atmosphere. This integral, e.g. $\int [x] dz$, is loosely called the column amount, the column density or the total column. It commonly has units of molecules cm^{-2}. For the special case of ozone, the unit is named after Gordon Dobson, an early measurement pioneer.

1 Dobson Unit (DU) = 2.69×10^{16} molec cm^{-2}. A typical column density for ozone is about 300 DU — i.e. an ozone column that would be 3 mm thick when shrunk down to the surface.

Exercise 2 — Calculation of the column of a well-mixed gas

Definition of the column:

$$\text{Column} = \int [x] dz$$

(Eq 1)

For a well-mixed gas the concentration decreases exponentially with altitude following the barometric equation:

$$p = p_0 e^{-z/H} \quad \text{(just replace } p \text{ for } [x]\text{)}$$

Thus, we can re-write Eq 1 as:

$$\text{Column} = [x]_0 \int e^{-z/H} dz$$

(Eq 2)

where $[x]_0$ is the concentration at a fixed point in the atmosphere. We then integrate Eq 2 between z and infinity (the top of the atmosphere) to get:

$$\text{Column} = [x]_0 \cdot H$$

(Eq 3)

From Exercise 1 we can work out $[x]_0$ as being the number density at the surface multiplied by the mixing ratio (which will be constant over all altitudes).

Try it yourself

Open `notebooks/01-atmospheric-structure.ipynb` to:

- Compute number density at any altitude/temperature/pressure (Exercise 1), and plot the barometric fall-off of pressure, density and [M] with height.
- Fit a scale height to a modelled or real pressure profile, and compare it to $H = RT/Mg$.
- Compute the column amount of a well-mixed gas (Exercise 2), reproduce the ~300 DU ozone column, and see how mixing ratio vs. number-density profiles (Fig 1.5/1.6) relate to the column integral.

Next: Module 2 — Chemical Kinetics in the Atmosphere

Chemical Kinetics in the Atmosphere

Module 2 · Chemical Kinetics in the Atmosphere

Learning aims By the end of this module you should be able to: 1. Apply first-order and pseudo-first-order kinetics to atmospheric decay problems, and define the associated time constant. 2. Apply the steady-state approximation to a simple reaction chain, and judge when it is (and isn't) valid. 3. Use Arrhenius parameters to describe the temperature dependence of a bi-molecular rate coefficient. 4. Explain why most apparently bi-molecular association reactions are actually ter-molecular, and describe the low-/high-pressure limiting behaviour.

2.1 Time scales and lifetimes

As atmospheric chemists we are fundamentally interested in understanding how the constituents of the atmosphere (the gases and particles that make it up) change with time. For example, the questions of how the concentration of O_3 will change in the future and how it has changed in the past are crucial to the science of the Antarctic ozone hole and air pollution. In order to understand these types of questions we rely on a solid knowledge of (i) what the constituents (chemical and aerosol species) are in the atmosphere and (ii) how the constituents in the atmosphere react over time. Hence, it's fair to say that chemical kinetics is at the heart of atmospheric chemistry!

For simple chemical systems, we can write out differential equations that describe how the components of the system will change over time. Let's start with a trivial example. Suppose that we study, in a closed constant-volume system, a reaction whose rate depends on the concentration of one reacting substance, A, only. We can write:

$$\frac{d[A]}{dt} = -k[A]^n$$

where in this case $n = 1$ and this reaction is said to be first order with respect to A (molecules cm^{-3}). Plots of the change in concentration of A with time are shown in Fig 2.1 for different values of rate coefficient, k (s^{-1}). Fig 2.1 highlights that the larger the value of the rate coefficient (sometimes called rate constant, and erroneously "the rate" — naughty!) the faster the loss of A with time.

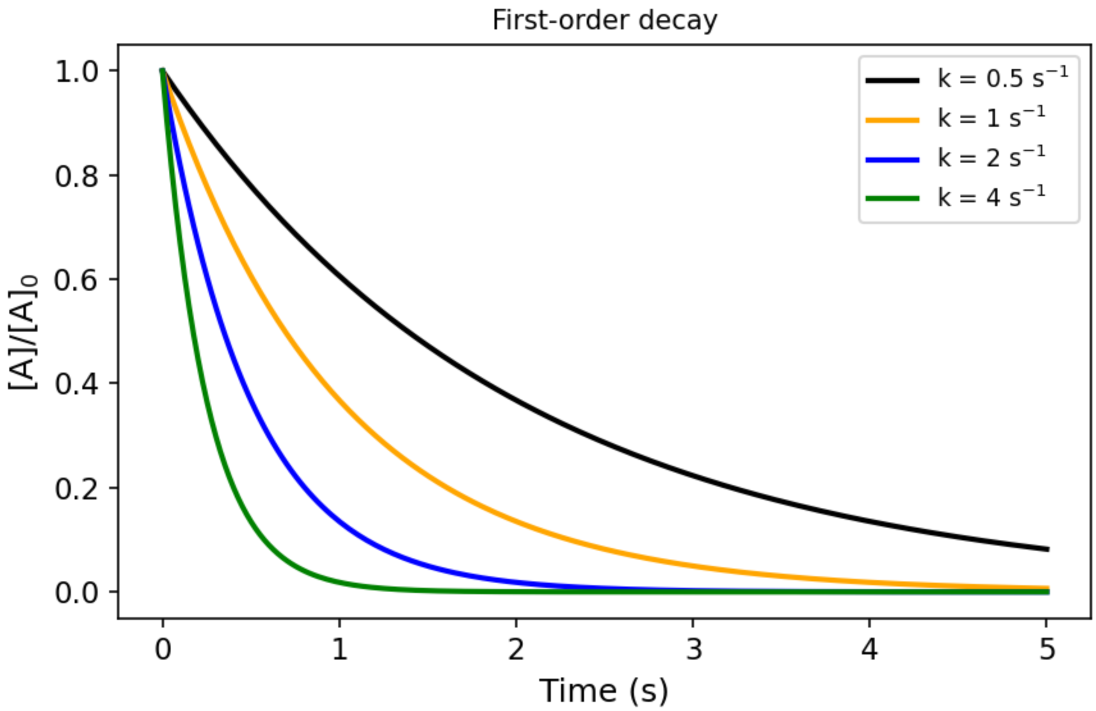


Figure 2.1 — Plots of the first-order decay of A with respect to time.

We can define a new expression by integrating our rate equation for A and noting that our initial conditions are $A = A_0$ at $t = 0$:

$$\int_0^t kdt = -\int_0^A [A]^{-n} dA$$

Integrating the rate equation

Given that k and n are constants we can then write:

$$[A]_t = [A]_0 \exp(-kt)$$

As Fig 2.1 highlights, the decay of A with time can be described as a simple exponential process. We can use this graph and the equations above to define the time constant - or lifetime - (τ) as the time taken for the concentration of A to be reduced by $1/e^{\text{th}}$ of its value:

$$\tau = 1/k$$

The smaller this time constant, the greater the loss of A with time and hence the shorter its lifetime.

Pseudo first-order kinetics

In the atmosphere there are several very important reactions that are first order and, fortunately, many that can be described as pseudo first order. For example, consider the slightly more complex reaction:

$$A + B \xrightarrow{k_2} C$$

On first glance we expect this to be a second-order process (first order in A and first order in B, so second order overall). However, if the concentration of $[B]$ doesn't change much over time (for example because B is present in large excess) then we can make the approximation that $d[B]/dt \simeq 0$, which allows us to consider A as undergoing pseudo-first-order kinetics ($[B] \sim$ constant, so it can be folded into our second-order rate constant k_2 to yield a pseudo-first-order rate constant, $k_2' = k_2[B]$). The time constant for A is now given by:

$$\tau = \frac{1}{k_2[B]} = \frac{1}{k_2'}$$

A chain reaction: $A \to B \to C$

We will now consider a slightly more complex example but one which you may be familiar with. Let's consider the chain reaction:

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

where A is our initial reactant, B an intermediate product and C the final product of the reaction. Following the same procedure as above we can write:

$$\frac{d[A]}{dt} = -k_1[A]$$

$$\frac{d[B]}{dt} = k_1[A] - k_2[B]$$

$$\frac{d[C]}{dt} = k_2[B]$$

Solutions to these equations for $k_1 = 1$ s⁻¹, $k_2 = 2$ s⁻¹, and (separately) $k_1 = 1$ s⁻¹, $k_2 = 10$ s⁻¹, are worth plotting side by side (see the notebook for details).

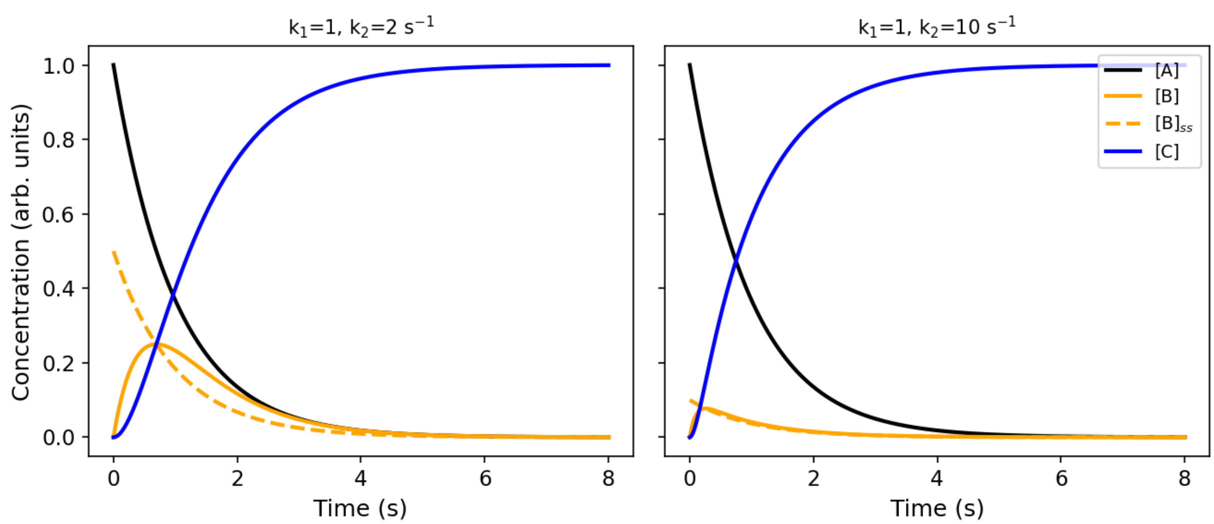


Figure 2.1a — The A → B → C chain reaction for $(k_1, k_2) = (1, 2)$ s⁻¹ (left) and $(1, 10)$ s⁻¹ (right), comparing the exact $[B]$ (solid) against the steady-state approximation $[B]_{ss} = k_1[A]/k_2$ (dashed).

What you will notice is that the intermediate B is short-lived relative to C, and that as the value of k_2 increases, the peak concentration of B decreases. The time it takes for B to reach an almost-steady concentration is $\sim 1/k_2$. In the limit where $k_2 \gg k_1$ we can make the approximation:

$$\frac{d[B]_{ss}}{dt} \simeq 0$$

where the subscript ss denotes that we are treating the intermediate B as being in steady state.

Following this through, we can re-write the remaining equations describing our reaction (substituting $[B]_{ss} = k_1[A]/k_2$):

$$\frac{d[A]}{dt} = -k_1[A]$$

$$\frac{d[C]}{dt} \simeq k_2 \cdot \frac{k_1[A]}{k_2} \simeq k_1[A]$$

In doing so, we have greatly reduced the complexity of this reaction system.

In the atmosphere there are many examples of compounds that can be considered to be in steady state. For the steady-state assumption to apply, it is necessary that the rate constants for destruction of the intermediate greatly exceed those for its formation, so that its concentration remains low and (quasi-)constant.

More on the steady state

In the chain reaction example it is clear that the steady-state approximation (SSA) will work when $k_2 \gg k_1$.

But when can we use the SSA, and when does it not work?

We need to consider several factors. First, the SSA will only work for an intermediate — in this course we can read that as meaning a radical species. So your first task is to identify the radicals and consider putting them into the SSA. Next, we need to examine the relative rates of formation and destruction. This is where things get tricky: the rate of destruction depends on the concentration of the species we are putting into steady state, so we probably don't know it in advance. However, we can get a sense of the numbers involved in the rate of production, and start to gauge whether the source of our intermediate is tiny, small, medium, large or massive. Our loss rates will need to balance these for the SSA to be valid. For example, the SSA will fail when we have a really large source of our intermediate but only a small amount of loss.

Finally, we can put some quantitative bounds on this — for example, by calculating the timescale over which the SSA starts to break down.

Consider the following reaction scheme:

$$O_2 + h\nu \xrightarrow{J_1} O(^3P) + O(^1D)$$

$$O(^1D) + M \xrightarrow{k_q} O(^3P) + M$$

$$\frac{d[O(^1D)]}{dt} = J_1[O_2] - k_q[M][O(^1D)]$$

Let's assume $[O(^1D)]$ is in steady state:

$$[O(^1D)]_{ss} = \frac{J_1[O_2]}{k_q[M]}$$

Compare that to the full solution of the ODE:

$$[O(^1D)]_t = \frac{J_1[O_2]}{k_q[M]} \left(1 - \text{exp}(-k_q[M]t)\right)$$

We note that the timescale for the system ($[O(^1D)]$) is given by $1/(k_q[M])$, and that the full solution and the steady-state solution only differ by an exponential term. This is useful because we can appeal to the graph of $y = \text{exp}(-x)$ to gain some insight into the problem: exp$(-4.6) \simeq 0.01$ — in other words, once $k_q[M]t = 4.6$, our SSA solution is within 1% of the full ODE solution. Rearranging, the time for the SSA to be within 1% of the full solution is:

$$t = \frac{4.6}{k_q[M]}$$

For $k_q = 3 \times 10^{-11}$ molecule^{-1} cm^3 s^{-1} and $[M] = 3 \times 10^{14}$ molecule cm^{-3}, this gives $t = 5 \times 10^{-4}$ s — i.e. very short, so it is very likely that the SSA is valid.

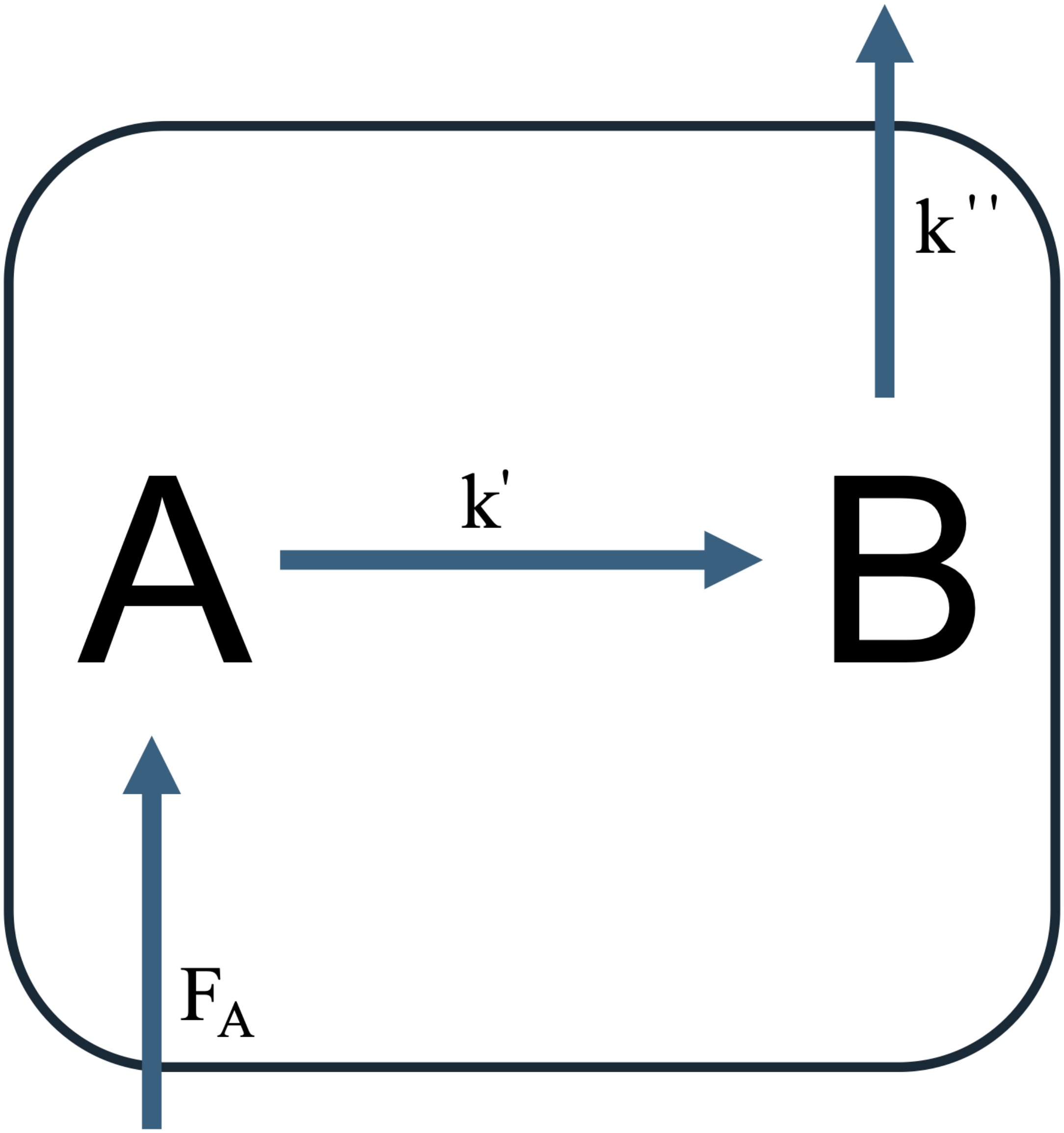


Figure 2.2 — Schematic of an air parcel containing two species, A and B.

Often we will consider the concept of an air parcel throughout this course (schematically shown in Figure 2.2). An air parcel can be thought of as a "box of air." Suppose there are two species in the box, A and B. There are emissions of A, F_A, and A can undergo reactions (conversion) to form B, which subsequently is lost via other processes. We can consider what the inputs into the box are, and what the outputs are — and appeal to steady state to write down steady-state solutions for $[A]$ and $[B]$, and hence an expression for the ratio $[B]_{ss}/[A]_{ss}$.

Worked derivation: steady-state ratio [B]ss/[A]ss for a source of A feeding into B

Consider the box model described above: A is emitted at a constant rate F_A and is lost with pseudo-first-order rate constant k' (converting it to B); B is in turn lost with pseudo-first-order rate constant k''.

$$\frac{d[A]}{dt} = F_A - k'[A]$$

$$\frac{d[B]}{dt} = k'[A] - k''[B]$$

Step 1 — steady state for A. Setting $d[A]/dt = 0$:

$$0 = F_A - k'[A]_{ss} \quad \Longrightarrow \quad [A]_{ss} = \frac{F_A}{k'}$$

Step 2 — steady state for B. Setting $d[B]/dt = 0$ and substituting $[A]_{ss}$:

$$0 = k'[A]_{ss} - k''[B]_{ss} \quad \Longrightarrow \quad [B]_{ss} = \frac{k'[A]_{ss}}{k''}$$

Step 3 — take the ratio. Dividing $[B]_{ss}$ by $[A]_{ss}$, the $[A]_{ss}$ cancels:

$$\frac{[B]_{ss}}{[A]_{ss}} = \frac{k'[A]_{ss}/k''}{[A]_{ss}} = \frac{k'}{k''}$$

Step 4 — rewrite in terms of lifetimes. Since k' and k'' are just the pseudo-first-order loss rate constants for A and B respectively, they define the corresponding lifetimes $\tau_A = 1/k'$ and $\tau_B = 1/k''$ (Section 2.1). Substituting:

$$\frac{[B]_{ss}}{[A]_{ss}} = \frac{k'}{k''} = \frac{1/\tau_A}{1/\tau_B} = \frac{\tau_B}{\tau_A}$$

So the steady-state concentration ratio of a product to its precursor is simply the ratio of their lifetimes: a short-lived product (small τ_B) sits at a low steady-state concentration relative to its precursor, regardless of the details of F_A — this is the same intuition as the A → B → C chain above, just recast for a species with continuous emission rather than a fixed initial concentration.

Integrating $d[A]/dt$ with the boundary condition $[A]_0$ at $t_0 = 0$ shows that, for a species with constant production F_A and pseudo-first-order loss k':

$$[A]_t = \frac{F_A}{k'}\left(1 - \text{exp}(-k't)\right)$$

Note that for a system of two first-order (or pseudo-first-order) reactions,

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$$

the system approaches equilibrium with a time constant of $(k_1 + k_{-1})^{-1}$ (s). So we only need k_1 or k_{-1} to be large for τ to be small — which is what lets us apply the steady-state approximation.

Exercise 3 — Derivation of the timescale for equilibrium

Consider a very simple reversible system $A \rightleftharpoons B$, with forward rate constant k_f and backward rate constant k_b:

$$\frac{d[B]}{dt} = k_f[A] - k_b[B]$$

(Eq 1)

Let x denote the amount of A molecules present as B molecules, and $a = [A]_0$. Then:

$$\frac{dx}{dt} = k_f(a - x) - k_b x$$

(Eq 2)

Setting $y \equiv dx/dt$:

$$y = k_f a - (k_f + k_b)x$$

(Eq 4)

By the chain rule, $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$. From Eq 2, $dx/dt = y$; from Eq 4, $dy/dx = -(k_f + k_b)$. So:

$$\frac{dy}{dt} = -(k_f + k_b)y \quad \Rightarrow \quad \frac{dy}{y} = -(k_f + k_b)dt$$

(Eq 7)

Integrating:

$$y = k_f a \cdot \text{exp}\left(- (k_f + k_b)t\right)$$

(Eq 8)

and, since y relates to x through Eq 4, combining Eq 4 and Eq 8 gives the exact solution:

$$x = \frac{k_f a}{k_f + k_b}\Big[1 - \text{exp}\left(- (k_f + k_b)t\right)\Big]$$

(Eq 9)

From this we see the timescale for the system to reach equilibrium is exactly $\frac{1}{k_f + k_b}$ — compare this exact result against the steady-state approximation numerically in the notebook.

We will consider many examples of using steady state in the supervision questions, but in general a good opening argument for whether or not to put something into steady state is to consider whether or not it is a radical.

Chemistry vs. transport timescales

So far we have considered some very trivial examples. Whilst these have been trivial, they are still very useful for our studies of atmospheric chemistry. But an important complication for us as atmospheric chemists is that the reaction vessel we use in our studies is the atmosphere itself! You only need to venture outdoors to note that the reactions taking place in the atmosphere are doing so under far-from-controlled conditions. The gases and particles we are interested in are affected not only by chemical change but also by transport — for example through the wind. In this course we won't dwell much on atmospheric transport (that's covered in greater detail in the Part III course). What is important, however, is to have a feel for the relative time scales of chemistry and transport.

The time constants (lifetimes) of a range of constituents of interest to the atmospheric chemist are presented in Fig 2.3. As we've seen, compounds with very short time constants change in concentration very rapidly with time — so one would expect that their concentrations will be very different over short spatial scales.

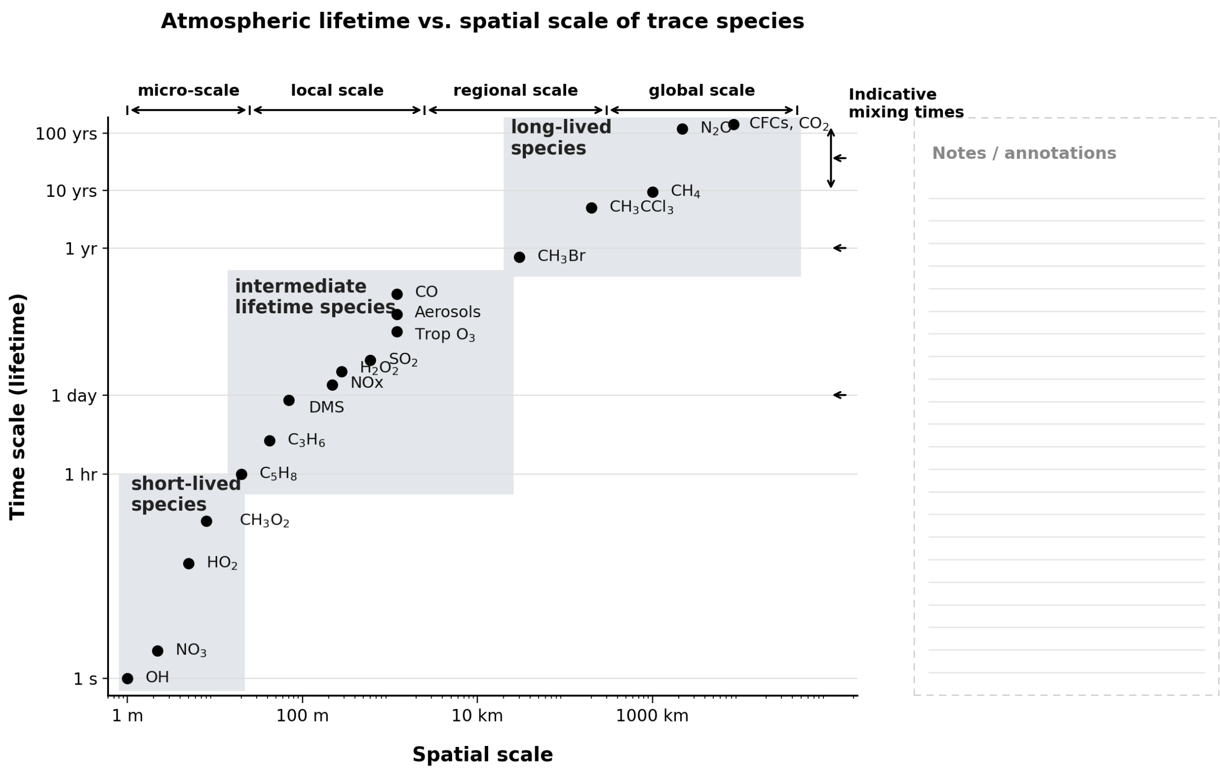


Figure 2.3 — Comparison of spatial and chemical scales of selected gases in the atmosphere.

2.2 Bi-molecular rate coefficients

Elementary reactions that involve two constituents can be written as:

$$A + B \xrightarrow{k_{obs(T)}} \text{products}$$

Often it's found that the rate coefficients for these types of reactions show temperature dependence. Whilst we can use tools like quantum mechanics and statistical thermodynamics to predict rate constants and their temperature dependence, in this course we will just make use of experimentally observed data.

Second-order, or bi-molecular, rate coefficients are usually expressed using the following form in atmospheric chemistry:

$$k_{obs(T)} = A \exp\left(-\frac{E_a}{RT}\right) \quad (\text{molecules}^{-1}\ \text{cm}^3\ \text{s}^{-1})$$

where A is often referred to as the Arrhenius pre-exponential factor, E_a the reaction activation energy (J mol^{-1}), R the gas constant (J K^{-1} mol^{-1}) and T temperature (K). Many bi-molecular reactions in the atmosphere show temperature dependence in their kinetics. A good example is the reaction of OH with methane (one of the most temperature-dependent reactions known).

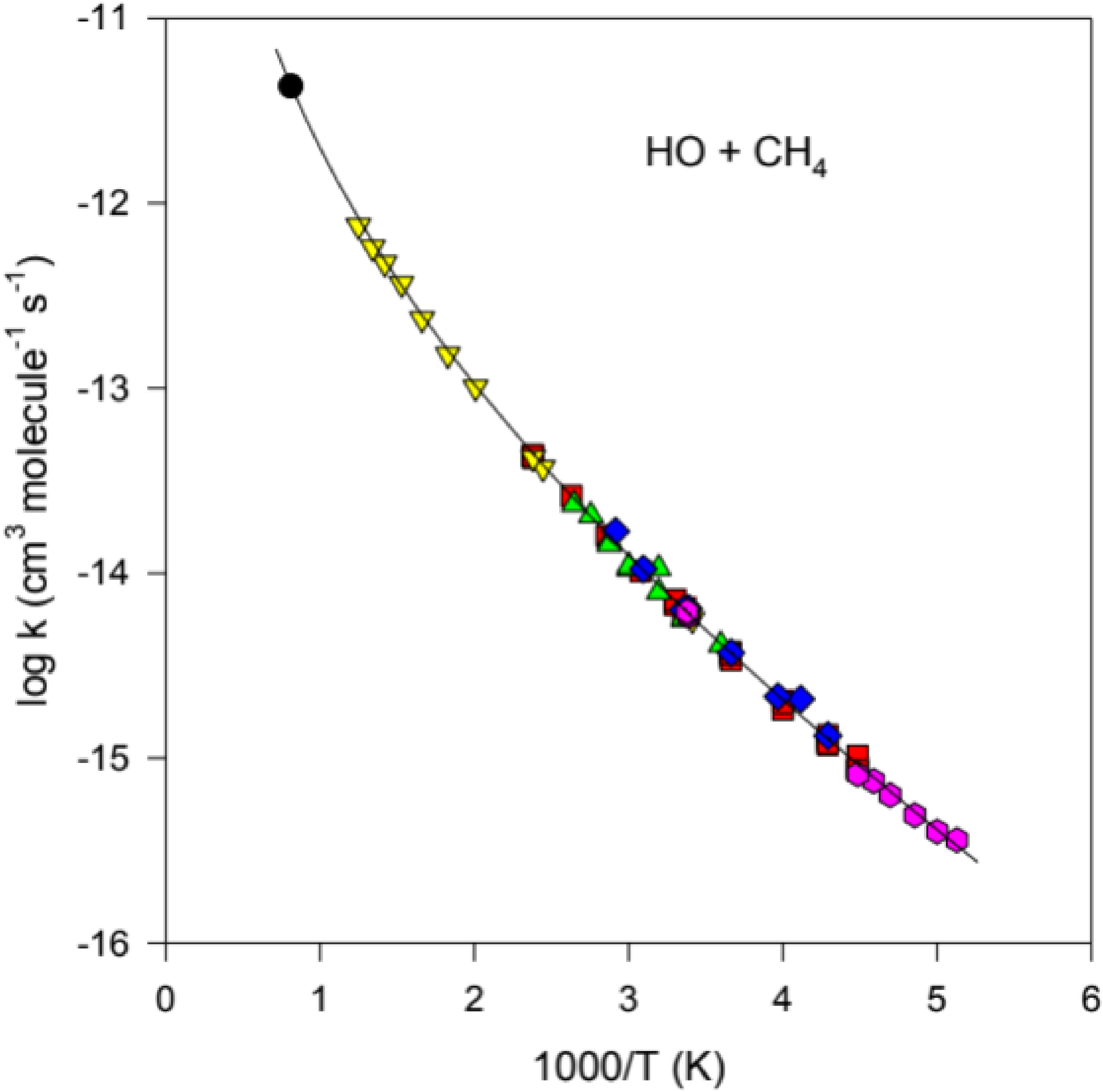


Figure 2.4 — Arrhenius plot for the reaction between methane and the hydroxyl radical in the gas phase.

Fig 2.4 demonstrates that the reaction OH + CH_4 proceeds with a significant activation energy barrier. As temperature is decreased there will be a lower fraction of reactants that can overcome this barrier, and hence the observed rate of reaction will decrease.

2.3 Ter-molecular rate coefficients

A more complex but fairly common type of reaction of interest in the atmosphere is the reaction between two compounds in the presence of a third body:

$$A + B + M \xrightarrow{k_{obs(T)}} \text{products}$$

Strictly speaking, most reactions require a third body (often referred to as M) to remove excess energy from the initial reaction. Consider the reaction between two iodine atoms. When the two atoms come together to form an iodine molecule, energy equal to the bond dissociation energy of I_2 is released into the nascent molecule. This is enough energy to then rupture the I–I bond and reform the iodine atoms. However, if the nascent I_2 molecule (I_2^*) collides with a third body, the encounter can transfer some of the vibrational and rotational energy of the I_2 molecule to translational (and other) excitation of the third body:

$$I + I \underset{k_b}{\overset{k_f}{\rightleftharpoons}} I_2^* \xrightarrow{M} I_2$$

Under atmospheric conditions, M is usually N_2 or O_2, and to a first approximation can be considered as the sum of their concentrations ($[M] = [N_2] + [O_2]$). The rate coefficients for ter-molecular reactions can be experimentally determined as a function of temperature and pressure to derive the low-pressure limit ($k_{0,T}$), which shows pressure dependence, and the high-pressure limit ($k_{\infty,T}$), which doesn't. To predict the observed rate constant as a function of temperature and pressure, we combine these two limits using the Troe equation (a modification of the Lindemann–Hinshelwood expressions), which yields a pseudo-bi-molecular rate coefficient (units molecules^{-1} cm^3 s^{-1}):

$$k_{obs(M,T)} = \left(\frac{k_{0,T}[M]}{1+\frac{k_{0,T}[M]}{k_{\infty,T}}}\right) F\left\{1 + \left[\log_{10}\left(\frac{k_{0,T}[M]}{k_{\infty,T}}\right)\right]^2\right\}^{-1}$$

Exercise 4 — Relating the bi-molecular and ter-molecular reactions

In Kinetics of Chemical Reactions at 1A, we were told that whenever two molecules collide they bring with them enough energy to break the chemical bond that they try to form. This somewhat strange fact can be seen here in more detail.

Consider two iodine atoms colliding:

$$I + I \xrightarrow{k_1} I_2^*$$

(Eq 1)

$$I_2^* \xrightarrow{k_d} I + I$$

(Eq 2)

$$I_2^* + M \xrightarrow{k_{col}} I_2 + M^*$$

(Eq 3)

The only reaction producing our product (I_2) is Eq 3, so:

$$\frac{d[I_2]}{dt} = k_{col}[I_2^*][M]$$

(Eq 4)

I_2^* is an excited-state form of I_2, so it's logical that it should be short-lived — put it into steady state:

$$\frac{d[I_2^*]}{dt} = 0 = k_1[I]^2 - k_{col}[I_2^*][M] - k_d[I_2^*]$$

$$[I_2^*] = \frac{k_1[I]^2}{k_{col}[M] + k_d}$$

(Eq 5)

Substituting Eq 5 into Eq 4:

$$\frac{d[I_2]}{dt} = k_{col}[M] \cdot \frac{k_1[I]^2}{k_{col}[M] + k_d}$$

(Eq 6)

Eq 6 shows the formation of I_2 behaving as a third-order (ter-molecular) process (second order in I, first order in M), whereas Eq 4 shows it as a second-order (bi-molecular) process (first order in M and I_2^*). In general, most (all) reactions that appear bi-molecular are formally ter-molecular, and this example extends to practically every association reaction (two species reacting by colliding together).

At low pressure ($[M]$ small, so $k_{col}[M] < k_d$):

$$\frac{d[I_2]}{dt} \cong [M][I]^2 \frac{k_1 k_{col}}{k_d}$$

(Eq 7)

At high pressure ($[M]$ large, so $k_{col}[M] > k_d$):

$$\frac{d[I_2]}{dt} \cong k_1[I]^2$$

(Eq 8)

Try it yourself

Open `notebooks/02-chemical-kinetics.ipynb` to:

- Reproduce Fig 2.1 (first-order decay for a range of k) and confirm $\tau = 1/k$.
- Solve the $A \to B \to C$ chain reaction numerically for both $(k_1, k_2) = (1, 2)$ and $(1, 10)$ s⁻¹, and check the steady-state approximation for $[B]$ against the full numerical solution.
- Solve Exercise 3 (the $A \rightleftharpoons B$ system) both analytically (Eq 9) and numerically, and quantify how the error in the steady-state approximation depends on k_f/k_b.
- Plot an Arrhenius line from A and E_a, and fit A/E_a back out from a synthetic $k(T)$ dataset — the same exercise you'd do with real IUPAC/JPL kinetics data.
- Explore the low-/high-pressure limits of a ter-molecular reaction using the Lindemann–Hinshelwood form, and see where the Troe falloff curve sits between Eq 7 and Eq 8.

Next: Module 3 — Atmospheric Photochemistry

Atmospheric Photochemistry

Module 3 · Atmospheric Photochemistry

Learning aims By the end of this module you should be able to: 1. Write down and explain each term in the photolysis rate coefficient integral, $J = \int \phi F \sigma d\lambda$. 2. Use the Beer–Lambert law and optical depth to describe how photon flux and photolysis rates vary with altitude and solar zenith angle. 3. Describe the absorption and photolysis behaviour of O_2, O_3, NO_2 and N_2O/CFCs, and explain why the troposphere and stratosphere have such different photochemical environments.

The sun's energy drives atmospheric chemistry. Assuming that the Sun can be represented as a blackbody emitter, Wien's displacement law ($\lambda_{max} = b/T$) can be used to predict the peak wavelength of solar radiation reaching the top of the atmosphere (TOA). Figure 3.1 shows a profile of solar radiation at the TOA and at the Earth's surface. Strong absorption occurs in a number of wavelength intervals. Whilst significant amounts of the Sun's energy are absorbed in the IR bands of H_2O and CO_2, Fig 3.1 shows that it is absorption by O_3 and O_2 that dominate in the visible and near-UV regions. It is the absorption of radiation by these gases that triggers the cascade of reactions we will focus on in this course. One of the most important quantities we will derive is the photolysis coefficient — a first-order rate constant — which can be used to determine the effective lifetime of a gas in the atmosphere with respect to the Sun's energy.

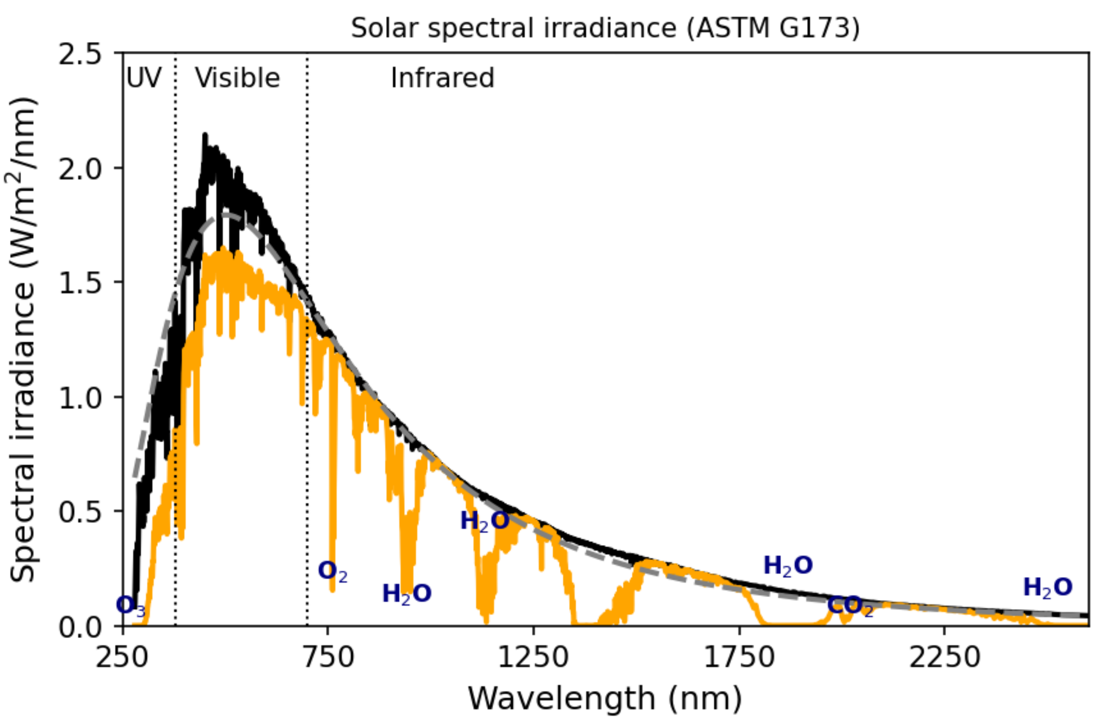


Figure 3.1 — Solar spectral irradiance as a function of wavelength at the TOA and at the surface.

3.1 Calculation of photolysis rate coefficients

Atmospheric chemists commonly use the symbol k for a reaction rate coefficient involving molecular "collisions," and the symbol J for a reaction rate coefficient for a reaction involving the absorption of photons resulting in molecular dissociation. We call this process photolysis (or photodecomposition). The rate coefficient of photolysis of a constituent i (J_i, units s^{-1}) can be written:

$$J_{(i)} = \int_{\lambda=0}^{\infty} \phi_{(i,\lambda)} F_{(\lambda)} \sigma_{(i,\lambda)} d\lambda$$

where $\phi_{(i,\lambda)}$ is the quantum yield (molecule photon^{-1}) for photolysis (the number of reactant molecules decomposed per quantum of radiation absorbed; sometimes defined per reaction pathway), $\sigma_{(i,\lambda)}$ is the wavelength-dependent absorption cross-section (usually cm^2 molecule^{-1}) of species i, and $F_{(\lambda)}$ is the incident photon intensity at a given wavelength (photons cm^{-2} s^{-1} nm^{-1}).

The absorption cross-section is defined by the Beer–Lambert law, which describes the attenuation of light by a homogeneous absorbing system:

$$I = I_0 \,\text{exp}(-\sigma n l)$$

where I_0 and I are the incident and transmitted light intensities, l is the absorption path length (cm), n is the concentration of absorber (molecule cm^{-3}), and σ is the absorption cross-section (cm^2 per molecule).

Given that the source of radiation of interest to us is the sun, we can use the Beer–Lambert law to determine the flux of photons at a specific wavelength as a function of altitude (z):

$$F_\lambda(z) = F_\lambda(\text{TOA}) \,\text{exp}(-\text{O.D.})$$

where $F_\lambda(\text{TOA})$ is the intensity of radiation (flux of photons) at the top of the atmosphere. The optical depth (O.D.) can be calculated as:

$$\text{O.D.}(\lambda) = \sum_i \sigma_{i,\lambda} \times \text{column}_i$$

— where we saw in Exercise 2 (Module 1) how to calculate the column of a well-mixed gas. In the cases we will consider, the relevant distance for the column is from the TOA to our point of interest.

If the sun is at zenith angle θ, the path length is increased by a factor $\sec(\theta)$. The optical depth thus increases rapidly as the zenith angle approaches 90°. This may be important near sunrise or sunset, or at high latitudes, and can result in a marked reduction in photolysis J values.

In principle the O.D. depends on all absorbing species at the wavelength of interest. In practice, for this course most of the time we will need only consider that O_2 (λ < 240 nm) and O_3 absorb solar radiation significantly, and only these two need be considered when calculating $F_\lambda(z)$.

3.2 Altitude dependence of photolysis

Figure 3.2 shows the altitude at which the intensity of solar radiation is reduced by $1/e$ from its TOA value, as a function of wavelength. There is a lot of variation in this plot, but in general we can note: (i) as we move to longer λ, radiation penetrates closer to the surface before being attenuated; (ii) beyond 320 nm, there is no difference between the photon flux at the TOA and at the surface.

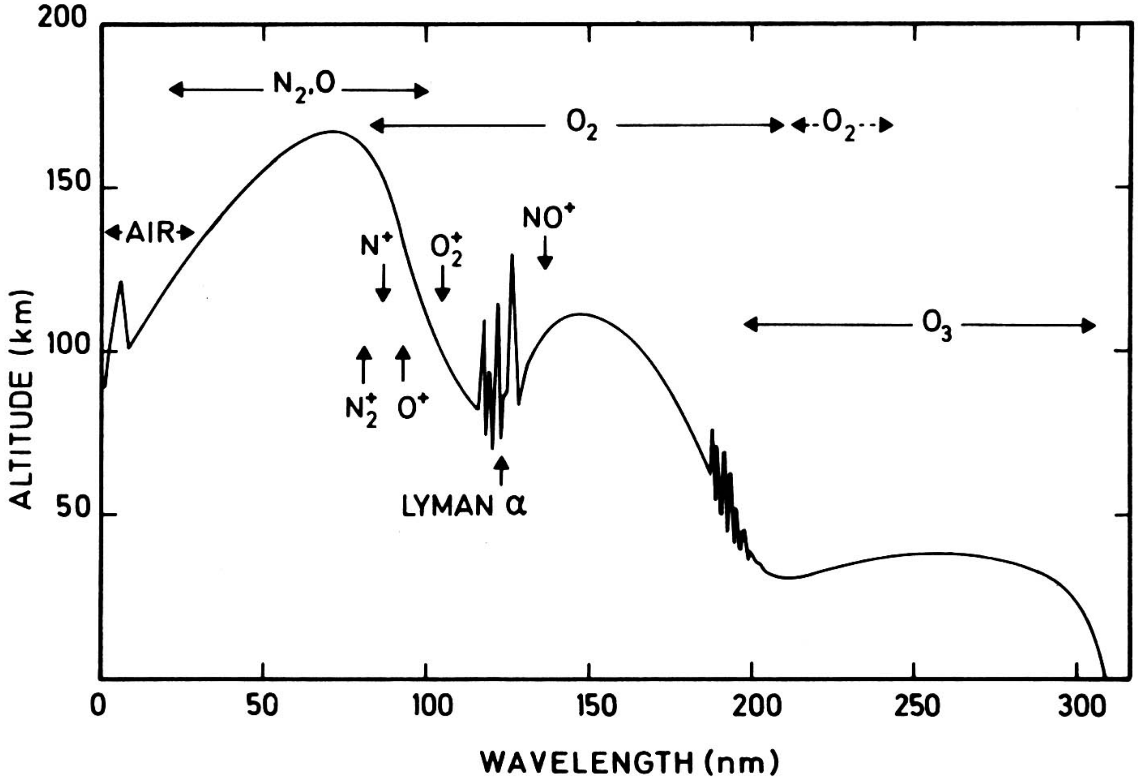


Figure 3.2 — Altitude at which attenuation reaches 1/e. Shows how the depth of penetration of radiation varies with wavelength. Based on Thomas 1980 DOI:10.1098/rsta.1980.0167

The fine structure in Fig 3.2 comes from the fine structure in the absorption cross-sections of the atmospheric absorbers. Briefly: Fig 3.2 shows that little radiation of $\lambda < 300$ nm penetrates below the tropopause. This means the photochemistry of the troposphere is very different from that of the stratosphere.

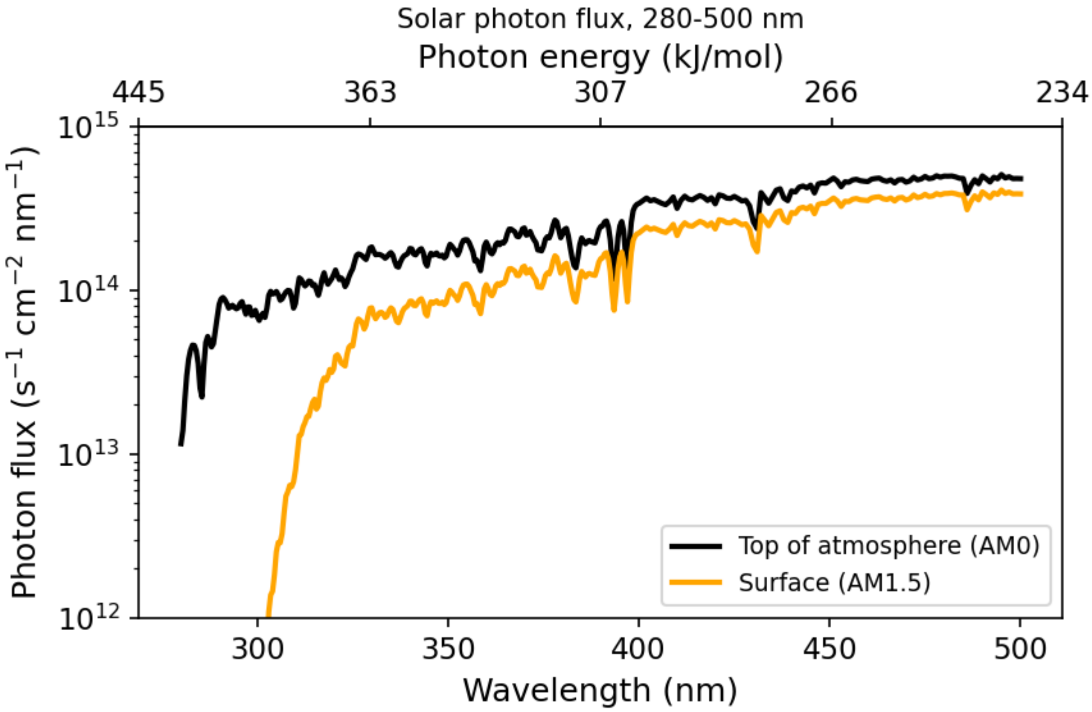


Figure 3.3 — Difference between photon flux at the top of atmosphere and the surface.

These altitude dependencies of photon fluxes matter when we come to calculate photolysis rates in different parts of the atmosphere — Fig 3.3 shows that at 290 nm there is over a factor-of-1000 difference in photon flux between the top of the atmosphere and the surface, so processes requiring this wavelength proceed much more slowly in the troposphere than the stratosphere.

J for O_2 ($\lambda < 250$ nm) varies strongly with altitude, as the relevant photons are absorbed and removed on the way down. In contrast, J for NO_2 + hv → NO + O ($\lambda < 400$ nm) varies little with altitude, and it is this photolysis reaction that leads to ozone production in the troposphere (covered in more detail later).

3.3 Photolysis of O_2

The presence of O_2 has a profound effect on the photochemistry of the atmosphere. O_2 has likely been present in our atmosphere for the last 2 billion years, with its current mixing ratio roughly constant over the last 500 million years. O_2 is produced through photosynthesis and is involved in a number of biogeochemical cycles giving rise to long lifetimes in the lower atmosphere (thousands of years). However, in the stratosphere O_2 can absorb short-λ radiation and photo-dissociate. The absorption cross-section for O_2 consists of characteristic features, many named after the scientists who discovered them. The cross-section at $\lambda < 150$ nm is large but not important to O_2 photolysis in the stratosphere owing to a lack of photons at these wavelengths. The major absorption in the stratosphere is in the Herzberg continuum (220–260 nm), $A^3\Sigma_u^+ \leftarrow X^3\Sigma_g^-$.

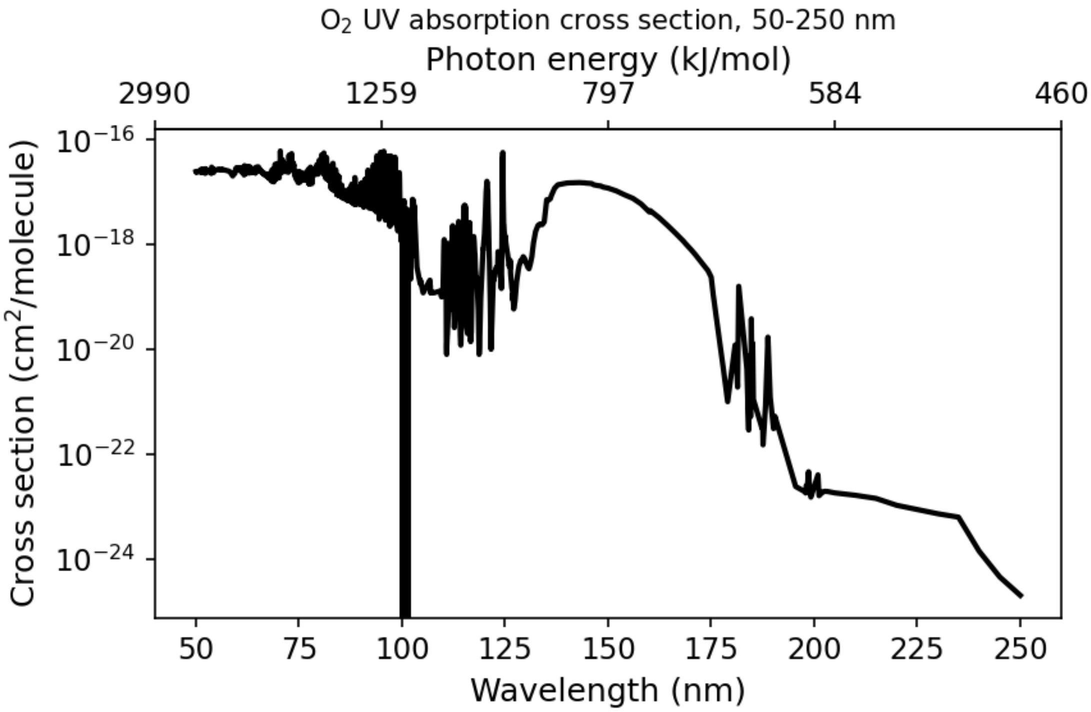


Figure 3.4 — Molecular oxygen absorption cross-sections.

Assuming unit quantum yield for photolysis of O_2 ($O_2 + hv \rightarrow O(^3P) + O(^3P)$), we can combine the wavelength-dependent flux profile with the absorption cross-section to calculate the photolysis frequency J_{O_2}; the reciprocal gives the lifetime/timescale for photolysis. The lifetime of O_2 with respect to photolysis varies over 5 orders of magnitude across the altitude range 20–100 km. The longer-wavelength Herzberg absorption is relatively more important at low altitudes, while the shorter wavelengths (Schumann–Runge Bands, Schumann–Runge Continuum) are only important at high altitudes (photons at these wavelengths are absorbed high up).

3.4 O_3 photolysis

Ozone is central to the chemistry of the atmosphere! The presence of ozone in a layer in the stratosphere (the ozone layer) has helped provide an environment suitable for the evolution of life at the Earth's surface. Yet ozone itself is toxic to most plants and animal life. Hence the phrase "good up high, bad nearby" is apt for ozone.

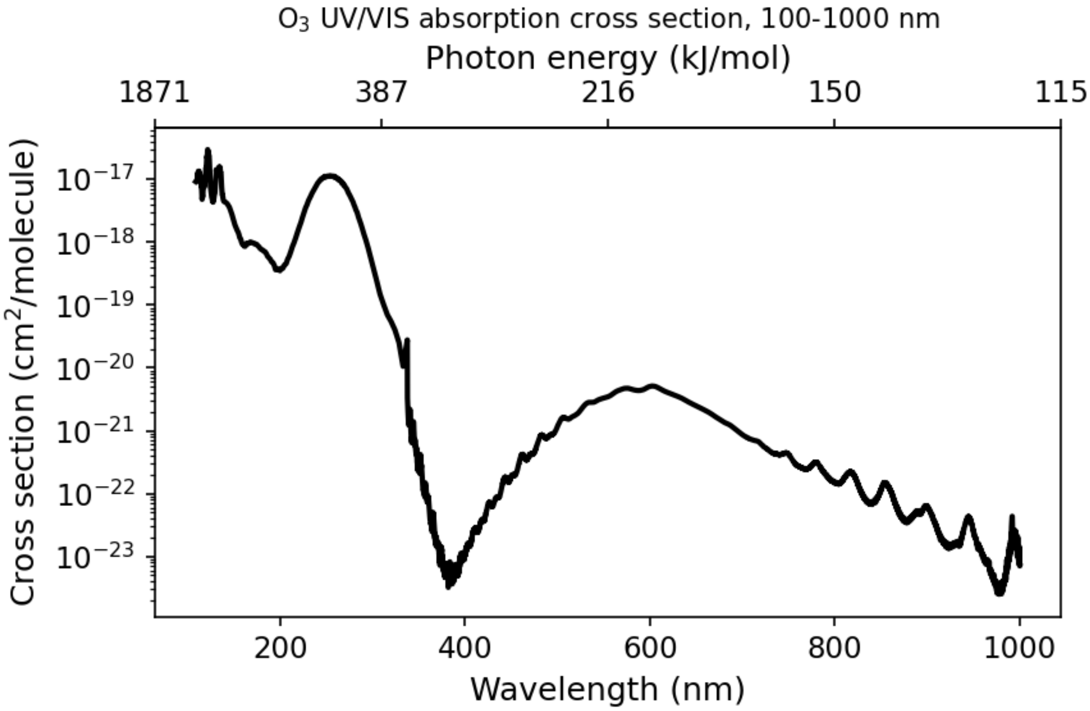


Figure 3.5 — Molecular ozone absorption cross-sections.

O_3 is relatively weakly bound compared to O_2 and can be photolysed over a wide range of wavelengths, with a theoretical threshold at 1180 nm. Absorption occurs in the so-called Hartley, Huggins and Chappuis bands. The precise photolysis products depend on the photon energy. The most important processes are photolysis to yield electronically excited oxygen atoms (rate constant $J_{O(^1D)}$) and photolysis to ground-state atomic oxygen ($J_{O(^3P)}$):

$$O_3 + h\nu \xrightarrow{J_{O(^1D)}} O_2(^1\Delta) + O(^1D) \qquad \lambda < 310 \text{ nm (actually longer!)}$$

$$O_3 + h\nu \xrightarrow{J_{O(^3P)}} O_2(^3\Sigma) + O(^3P) \qquad \lambda < 1180 \text{ nm}$$

The exact products that are formed depend on the energy from the photons. Calculations of these are given below where the values correspond to the maximum limits of the photons (in nm):

	$O_2(^3\Sigma_g^-)$	$O_2(^1\Delta_g)$	$O_2(^1\Sigma_g^+)$	$O_2(^3\Sigma_u^+)$	$O_2(^3\Sigma_u^-)$
$O(^3P)$	1180	590	460	230	170
$O(^1D)$	410	310	260	167	150
$O(^1S)$	234	196	179	129	108

Using the absorption cross-section data together with the altitude-dependent spectral irradiance, we can calculate photolysis frequencies for O_3, just as for O_2. Photolysis of O_3 is much more rapid than photolysis of O_2: lifetimes of O_3 with respect to photolysis vary from ~8 hours to ~3 minutes over the altitude range 10–50 km.

Of more interest to our understanding of atmospheric chemistry is the reaction channel producing excited oxygen atoms, $J_{O(^1D)}$. Here we must follow the same approach as before, but rather than assuming unit quantum yield we need the wavelength (and possibly temperature) dependence of $\phi_{(\lambda)}$: at small λ the quantum yield is ~ 1.0, whilst at $\lambda > 310$ nm it drops to ~ 0.0.

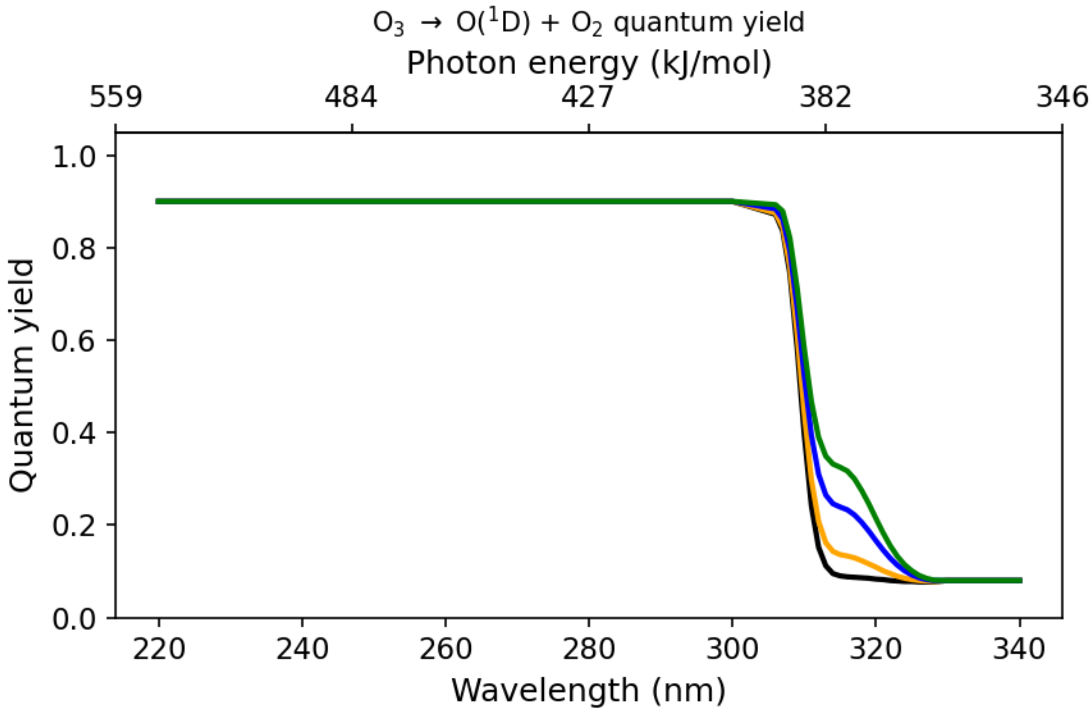


Figure 3.8 — Wavelength dependence of the O(¹D) quantum yield.

We will see later that O(¹D), while present in very low concentrations, is extremely important for atmospheric chemistry. It can actually be produced at somewhat longer wavelengths than 310 nm, possible explanations being: 'hot band' absorption by O_3, or possibly $O_3 \rightarrow O(^1D) + O_2(^3\Sigma)$ (spin-forbidden).

The very low concentrations of O(¹D) (order 1–100 molecules cm^{-3}) occur because it is collisionally quenched back to the ground electronic state very rapidly:

$$O(^1D) + O_2 \rightarrow O_2 + O(^3P) \qquad k_1 = 4.2 \times 10^{-11} \text{ molecules}^{-1}\text{cm}^3\text{s}^{-1}$$

$$O(^1D) + N_2 \rightarrow N_2 + O(^3P) \qquad k_2 = 2.8 \times 10^{-11} \text{ molecules}^{-1}\text{cm}^3\text{s}^{-1}$$

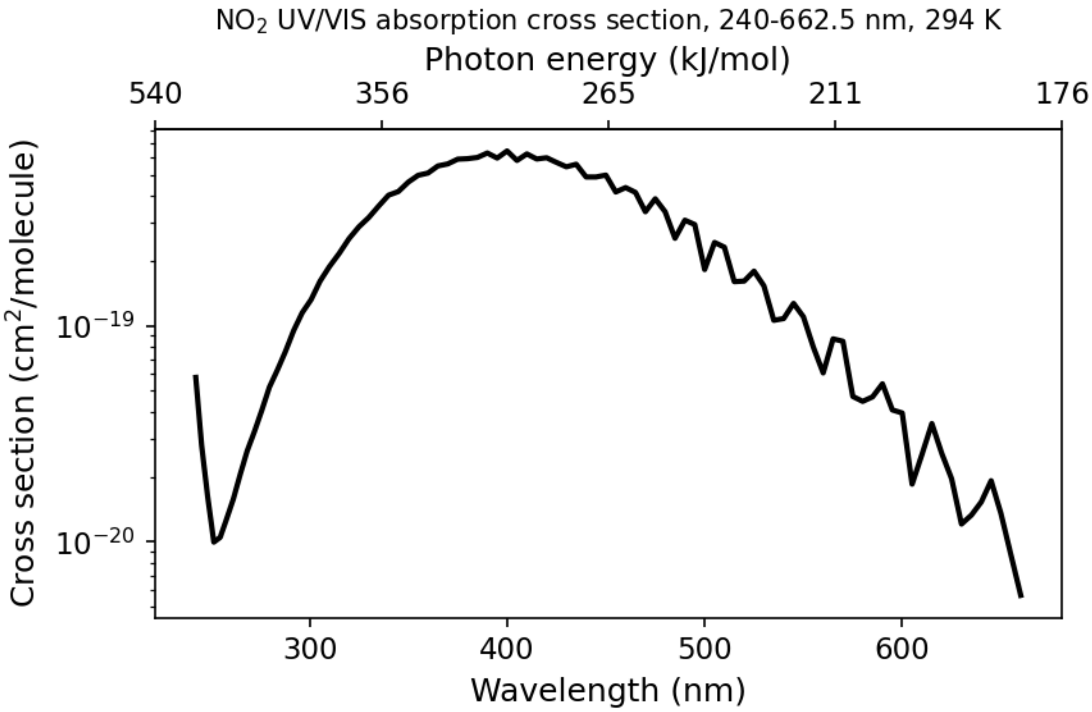
3.5 Photolysis of NO_2

Nitrogen dioxide was likely first discovered by Priestley in the late 18th century. At room temperature it's a brownish gas, and — as you might imagine — this means it has a stonking great absorption cross-section in the visible spectrum.

Photolysis of NO_2 proceeds as:

$$NO_2 + h\nu \xrightarrow{J_{NO_2}} NO(^2\Pi) + O(^3P)$$

The photolysis products can go on to form O_3 (through reaction with O_2 in a ter-molecular process). Indeed, photolysis of NO_2 is the main source of O_3 production in the troposphere (where O_2 cannot be photolysed).



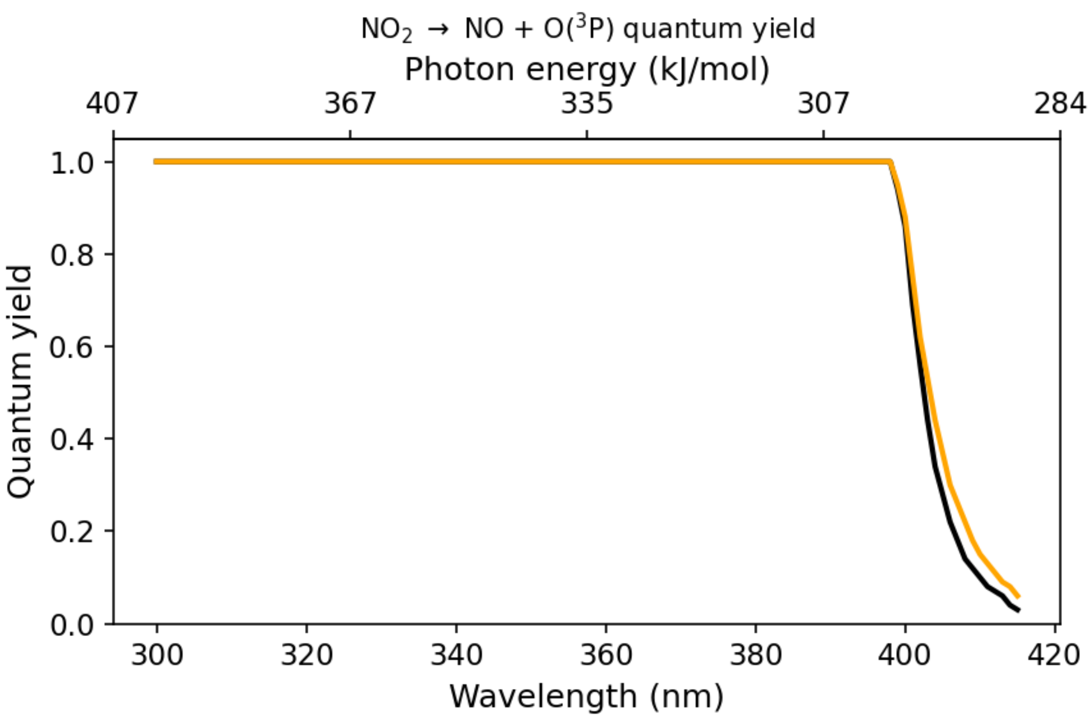


Figure 3.9 — NO_2 absorption cross-section and quantum yield.

As Fig 3.9 shows, the cross-section remains relatively large even at long wavelengths (where the atmosphere is otherwise practically transparent), resulting in large photolysis frequencies even near the surface:

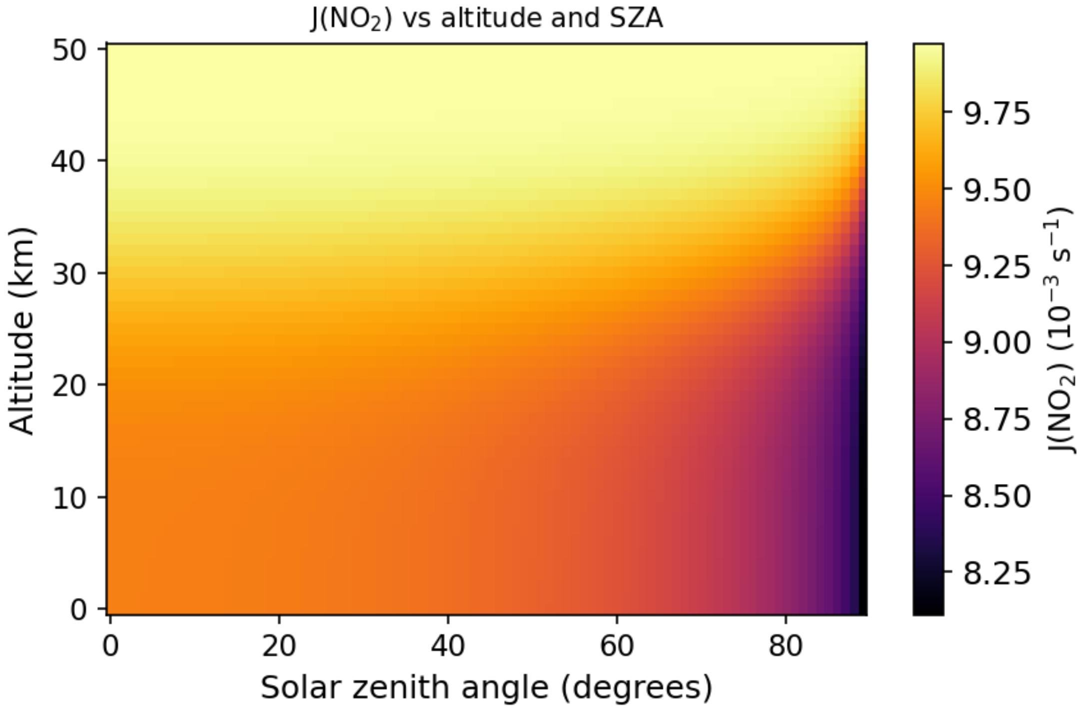


Figure 3.10 — NO_2 photolysis rates as a function of solar zenith angle and altitude.

NO_2 photolysis happens so rapidly that NO_2 can usually be thought of as being in steady state (i.e. the first-order rate of loss — photolysis — is large enough to keep its concentration low and steady) during daylight hours. Fig 3.10 shows that even at high solar zenith angles photolysis remains rapid ($\Delta\tau$ ~ 5% between midday and 6 am). This is an important aspect of NO_2 chemistry, and one that will let us simplify complex reaction chains later in the course.

3.6 Photolysis of N_2O and other compounds

Photolysis of N_2O almost exclusively proceeds as:

$$N_2O + h\nu \xrightarrow{J_{N_2O}} N_2 + O(^1D)$$

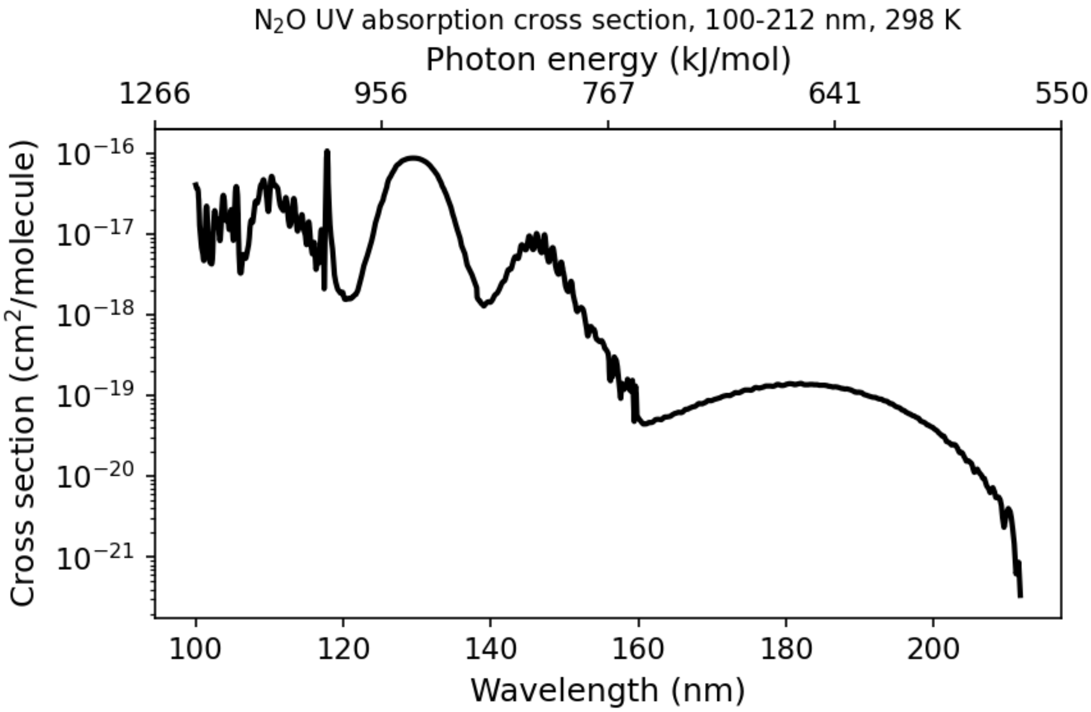


Figure 3.11 — N_2O absorption cross-sections as a function of wavelength (and energy).

N_2O absorbs strongly at short λ (< 220 nm). The main things to note: at the longer wavelengths available in the troposphere there is no photolysis of N_2O, and N_2O can contribute significantly to the optical depth at low wavelengths.

CFCs (chlorofluorocarbons) contain carbon, fluorine and chlorine and are entirely man-made. PFCs (perfluorinated carbons) are largely man-made, but the simplest PFC (CF_4) has a natural source from the oxidation of certain volcanic rocks. PFCs are very stable — in fact the absorption cross-section of CF_4 is so small that under most stratospheric conditions its lifetime is millions of years, so we won't worry about PFCs for now (although they are strong greenhouse gases, so don't think they aren't important!).

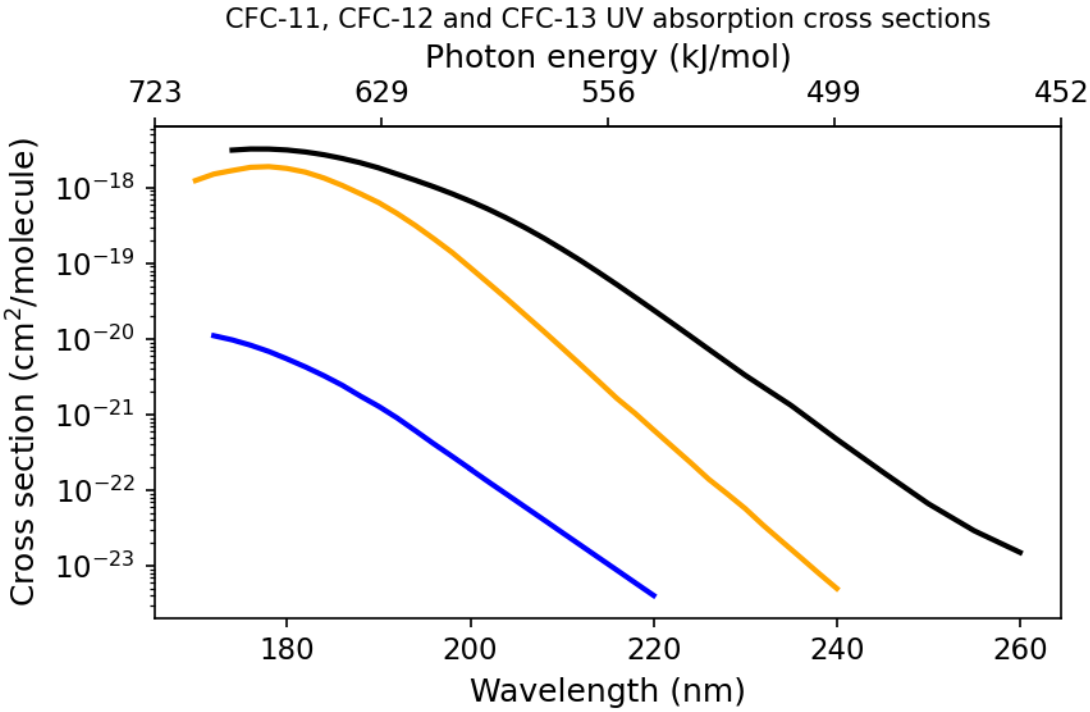


Figure 3.12 — CFC absorption cross-sections as a function of wavelength (and energy).

The naming convention for CFCs: the rightmost digit in the name gives the number of fluorine atoms (n_F); the next digit to the left gives the number of hydrogen atoms plus 1 ($n_H + 1$); and the next gives the number of carbon atoms minus 1 ($n_C - 1$), with zero values omitted. For example, CFC-11 has $n_F = 1, n_H = 0$ (so digit = 1), $n_C = 1$ (so digit = 0), giving CFC-11 = CCl_3F.

CFC absorption cross-sections are fairly large at very short λ and almost completely featureless (compare O_3 and NO_2). There is a huge range in CFC lifetimes as a function of altitude — tropospheric lifetimes are many thousands of years, but upper-stratospheric lifetimes are only a few days.

Important breakdown products from the photolysis of Cl- and Br-containing compounds in the stratosphere are the halogen nitrates. Their absorption cross-sections differ greatly at $\lambda > 260$ nm, with $BrONO_2$ having a sufficiently large cross-section at $\lambda > 320$ nm that its lifetime is very short.

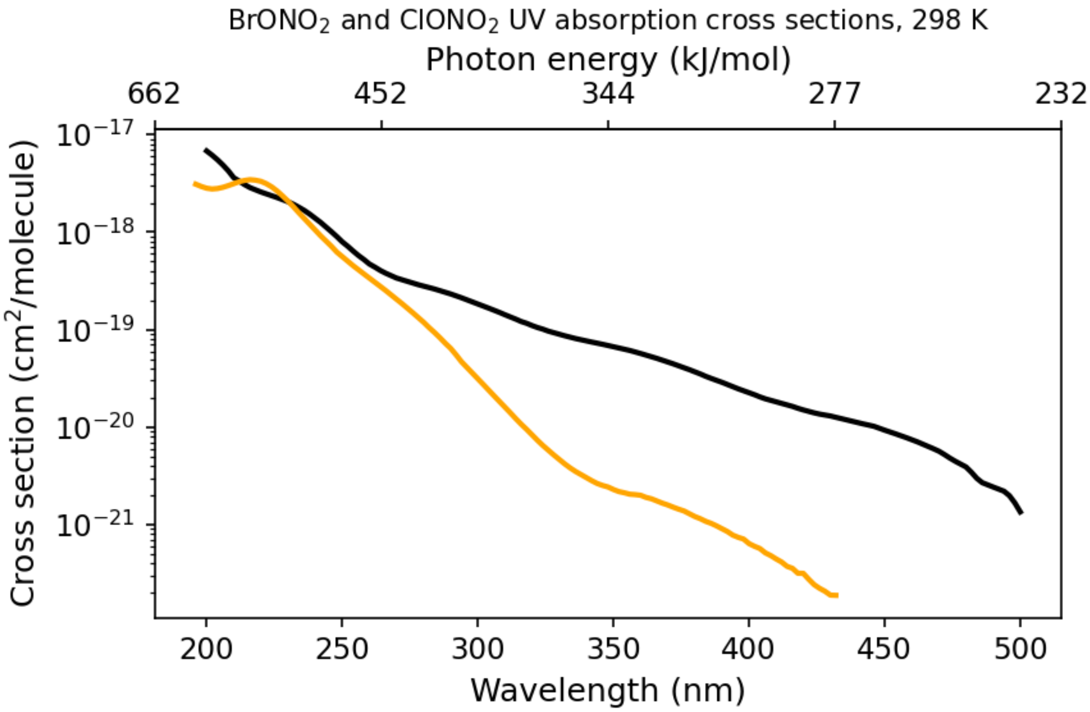


Figure 3.13 — Halogen nitrate absorption cross-sections as a function of wavelength (and energy).

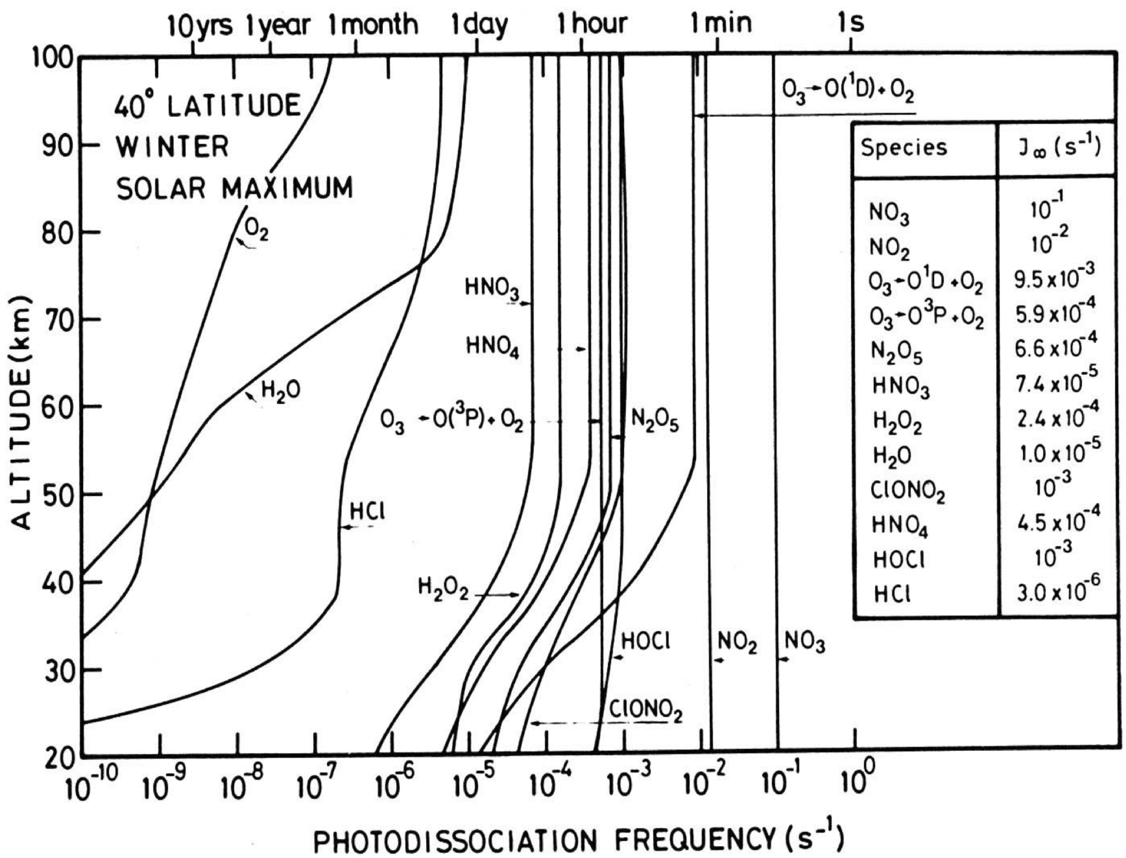


Figure 3.14 — Overview of the photolysis coefficients of important trace gases in the stratosphere.

Finally, Figure 3.14 gives an important overview of the photolysis coefficients (J-rates) of many trace gases in the stratosphere. This is a key figure we can come back to as it provides an important justification for the use of steady state. Essentially, during daylight hours we can see that species which have lifetimes with respect to photolysis shorter than 1 hour are very good candidates to apply steady state to.

3.7 Scattering and albedo

So far when discussing the transmission of radiation through the atmosphere we have considered the role of atmospheric gases absorbing solar radiation. In addition to this we also need to consider scattering of radiation. Scattering occurs when incident radiation is redirected from its original path. The amount of scattering that happens is a complex function that depends on the wavelength of the radiation and the size and type of particles and gases present. Three types of scattering take place:

- Rayleigh scattering – λ » dp (size of particle); light scattered isotropically; gives rise to blue sky.
- Mie scattering – λ ~ dp; scattering strongest along direction of incident beam.
- Non-selective scattering – λ « dp; gives rise to white appearance of fogs and clouds.

If the atmosphere is considered to be multiple scattering, we see a significant enhancement in the ratio of the incident radiation to that at the top of the atmosphere ($I(z)/I(\infty)$). These enhancements as a function of altitude and wavelength are shown in Figure 3.15 The scattering allows photons to traverse a region more than once, increasing the apparent photon flux proportionately and hence the photolysis rates for species. Since the degree of scattering is proportional to air density, only those wavelengths which penetrate to the lower atmosphere show enhancement.

If surface albedo (i.e. reflectivity) is also included, the enhancement is even more marked. The albedo of the Earth varies from ~100% (fresh snow and ice) to ~5% (forest), with the average, known as the planetary albedo, being ~35%. Figure 3.15 shows how different wavelengths of light are affected by not only scattering but also by the presence of a reflective lower surface (albedo 50%). As with the impacts of enhancements from scattering, our main concern is that increases in the apparent flux of photons as a result of changes in albedo will affect the photolysis frequencies we calculate.

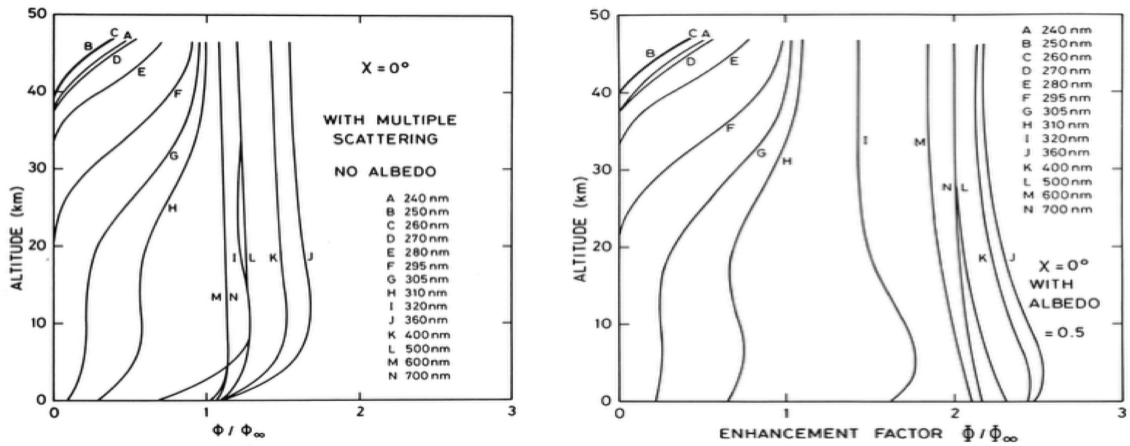


Figure 3.15 — (left) Enhancement ratios as a function of altitude and wavelength including the effect of multiple scattering; (right) Enhancement ratios including multiple scattering and surface albedo of 0.5.

Try it yourself

Open `notebooks/03-photolysis-rates.ipynb` to:

- Implement the Beer–Lambert / optical-depth calculation and reproduce the qualitative shape of Fig 3.2 (attenuation depth vs. wavelength) for a simplified O_2/O_3 atmosphere.
- Compute a toy photolysis coefficient $J_i = \int \phi F \sigma d\lambda$ numerically given synthetic $\sigma(\lambda)$, $\phi(\lambda)$ and $F(\lambda)$ data, and convert to a photolysis lifetime.
- Explore how J scales with solar zenith angle via the sec(θ) airmass factor, and reproduce the qualitative NO_2 vs. O_2 altitude-dependence contrast described in §3.2.
- Plot the $O(^1D)$ quantum-yield step function (Fig 3.8) and see how a sharp cutoff wavelength propagates into a photolysis rate integral.

Next: Module 4 — Stratospheric Ozone Chemistry: the Chapman Mechanism

Stratospheric Ozone Chemistry: the Chapman Mechanism and Catalytic Cycles

Module 4 · Stratospheric Ozone Chemistry

Learning aims By the end of this module you should be able to: 1. Write down the Chapman reactions and derive the steady-state Chapman ozone concentration. 2. Explain why the Chapman mechanism alone overestimates stratospheric ozone, and what this implies about additional loss processes. 3. Describe the general form of a catalytic ozone-destruction cycle and derive the odd-oxygen loss rate for the NOx, HOx, ClOx and BrOx families. 4. Explain the role of reservoir species in moderating the efficiency of each catalytic family.

4.1 The Chapman Reactions

The discovery of O_3 in the atmosphere is usually attributed to Christian Friedrich Schönbein (1799–1868). Schönbein noticed a strange odour associated with electrical discharges — for those of you who have ever operated an "old" photocopier, you may recognize this odour.

Schönbein quantified the presence of O_3 in air using pieces of parchment soaked in a solution of KI, which would change colour on exposure to O_3. These early measurements are our only record of O_3 levels in the atmosphere at the time, and as you can imagine represent far-from-ideal data sets. Interestingly, Schönbein's method has continued to this day, albeit with modifications, in the form of the KI electrochemical cell.

It wasn't until the work of Gordon Dobson (1889–1975), who pioneered spectroscopic measurements of O_3 using the Dobson spectrophotometer during the 1920s, that we had a good picture of the distribution and abundance of O_3 in the upper atmosphere.

These early observations provided the first challenge for atmospheric chemists: to explain the occurrence of a thick layer of O_3 in the stratosphere. In 1930, Sidney Chapman (1888–1970) proposed a series of reactions to explain this distribution of stratospheric ozone.

Reaction	Rate coefficient	Eq.
$O_2 + h\nu \rightarrow O + O$	$J_2 \sim 5 \times 10^{-10}$ s^{-1}	4.1
$O + O_2 + M \rightarrow O_3 + M$	$k_3 = 6 \times 10^{-34}(300/T)^2$ cm^6 s^{-1}	4.2
$O_3 + h\nu \rightarrow O + O_2$	$J_3 \sim 2 \times 10^{-3}$ s^{-1}	4.3
$O + O_3 \rightarrow O_2 + O_2$	$k_4 = 1.0 \times 10^{-11} \exp(-2100/T)$ cm^3 s^{-1}	4.4

Notice that the photolysis of oxygen is much slower than the photolysis of ozone ($J_2 \ll J_3$; values given are for ~40 km). Reactions 4.2 and 4.3 have very short time constants for the conversion of O to O_3 and vice versa, and establish a steady state much more rapidly than reactions 4.1 and 4.4.

Exercise 5 — Time constants for inter-conversion of O and O_3

As in Exercise 3, we found that the time scale for interconversion of two species is $(k_f + k_b)^{-1}$ (both must be first-order or pseudo-first-order rate constants).

O and O_3 are linked by interconversion reactions:

$$O\,(+O_2 + M) \xrightarrow{k_f} O_3 \qquad O_3\,(+h\nu) \xrightarrow{k_b} O\,(+O_2)$$

Here $k_f = k_{O+O_2}[M][O_2]$ (pseudo-first-order for reaction 4.2) and $k_b = J_3$.

Focusing on the upper/middle stratosphere (40 km, $T = 225$ K), from Module 1's Exercise 1 we have $[M]_{40\text{km}} = 2.4 \times 10^{19}$ exp$(-40/7) = 7.9 \times 10^{16}$ cm^{-3} ($[O_2] = 0.2[M]$). The interconversion timescale is:

$$\left(7.9 \times 10^{16} \times 0.2 \times 7.9 \times 10^{16} \times 6 \times 10^{-34}(300/225)^2 + 2 \times 10^{-3}\right)^{-1} = 0.75 \text{ seconds! i.e. fast!}$$

Reactions 4.2 and 4.3 only interconvert the "odd oxygen" species O and O_3 — i.e. they conserve odd oxygen ($[O_x] = [O] + [O_3]$). In defining O_x we've invoked the idea of a chemical family: a collection of compounds connected via fast interconversion reactions, in steady state with each other. Since odd oxygen is only formed by reaction 4.1 and only removed by reaction 4.4:

$$\frac{d[O_x]}{dt} = 2J_2[O_2] - 2k_4[O][O_3]$$

Without invoking the O↔O_3 steady state we'd instead have the more complex system:

$$\frac{d[O]}{dt} = 2J_2[O_2] - k_3[O][O_2][M] + J_3[O_3] - k_4[O][O_3]$$

$$\frac{d[O_3]}{dt} = k_3[O][O_2][M] - J_3[O_3] - k_4[O][O_3]$$

$$\frac{d([O] + [O_3])}{dt} = 2J_2[O_2] - 2k_4[O][O_3]$$

The use of odd oxygen thus leads, as we'll see again later, to both conceptual and mathematical simplification.

Note on oxygen

Note that we will always consider $[O_2]$ to be unaffected by this chemistry ($d[O_2]/dt = 0$ and $[O_2] \approx 0.2[M]$) unless otherwise stated.

The timescale for steady state between reactions 4.1 and 4.4 varies strongly with altitude: ~hours at 40 km, but many years at 20 km. Invoking steady state in the upper stratosphere is a good approximation; in the lower stratosphere it would be poor, since solar intensity, temperature, and atmospheric transport all vary too rapidly to allow steady state to be established.

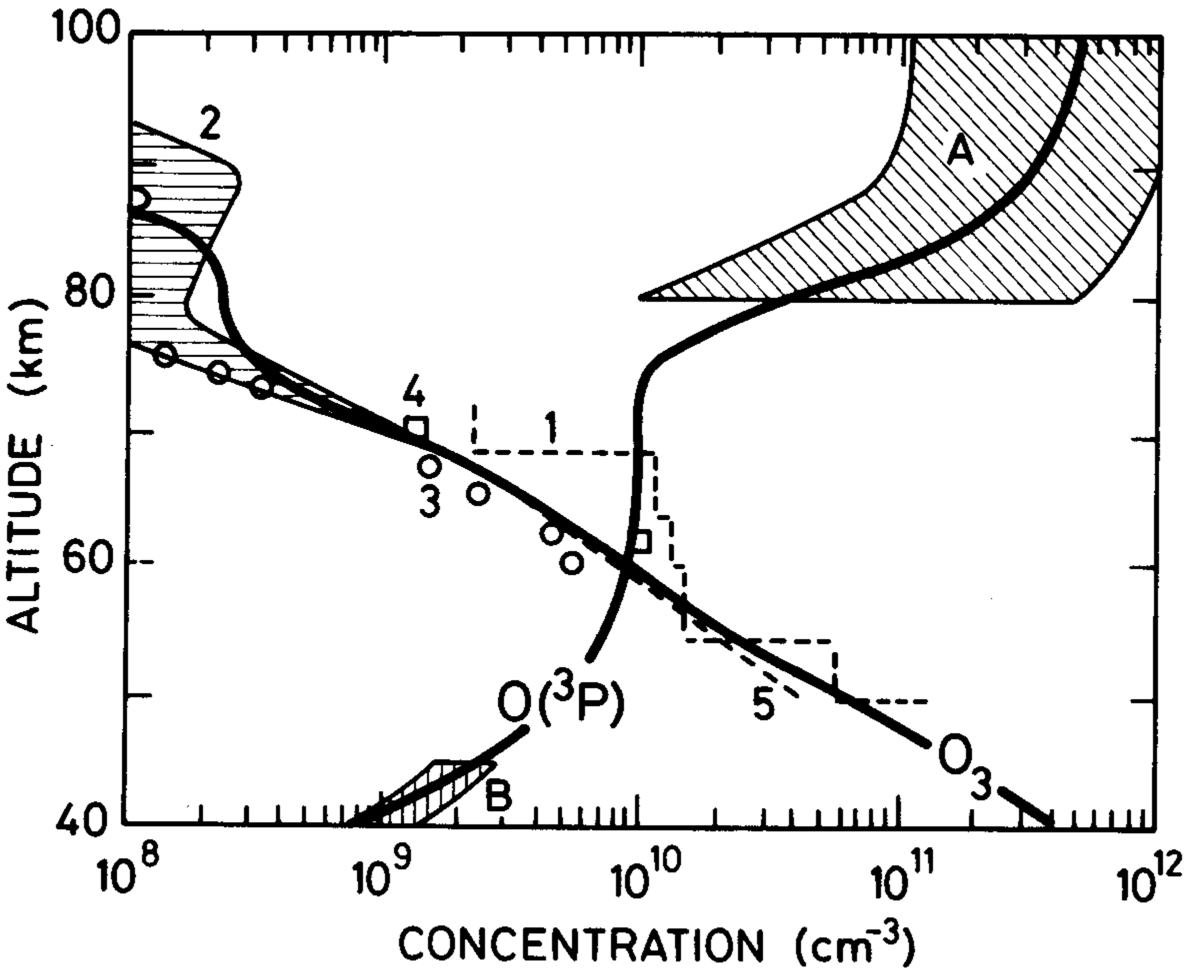


Figure 4.1 — Observed and calculated altitude profiles of $O(^3P)$ and O_3.

From the steady state between O and O_3, [O] decreases very rapidly with decreasing altitude (both [M] and $[O_2]$ are proportional to pressure), so reaction cycles involving O become very inefficient at low altitudes (Fig 4.1).

Exercise 6 — Steady-state distribution of ozone

This calculation only works where chemical production and loss rates balance. At steady state:

$$\frac{d[O_x]}{dt} = 2J_2[O_2] - 2k_4[O][O_3] = 0 \quad \Rightarrow \quad [O_3] = \frac{2J_2[O_2]}{2k_4[O]}$$

(Eq 2)

The problem is we don't yet know [O]. Using the O↔O_3 interconversion (reactions 4.2/4.3):

$$\frac{[O_3]}{[O]} = \frac{k_3[O_2][M]}{J_3}$$

(Eq 3)

Rearranging Eq 3 for [O] and substituting into Eq 2:

$$[O_3]^2 = \frac{2J_2[O_2]k_3[O_2][M]}{2k_4J_3} \quad \Rightarrow \quad [O_3] = \sqrt{\frac{J_2[O_2]k_3[O_2][M]}{k_4J_3}}$$

(Eq 5)

Now $[O_3]$ can be calculated purely from rate constants and altitude (via $[O_2]$ and [M]).

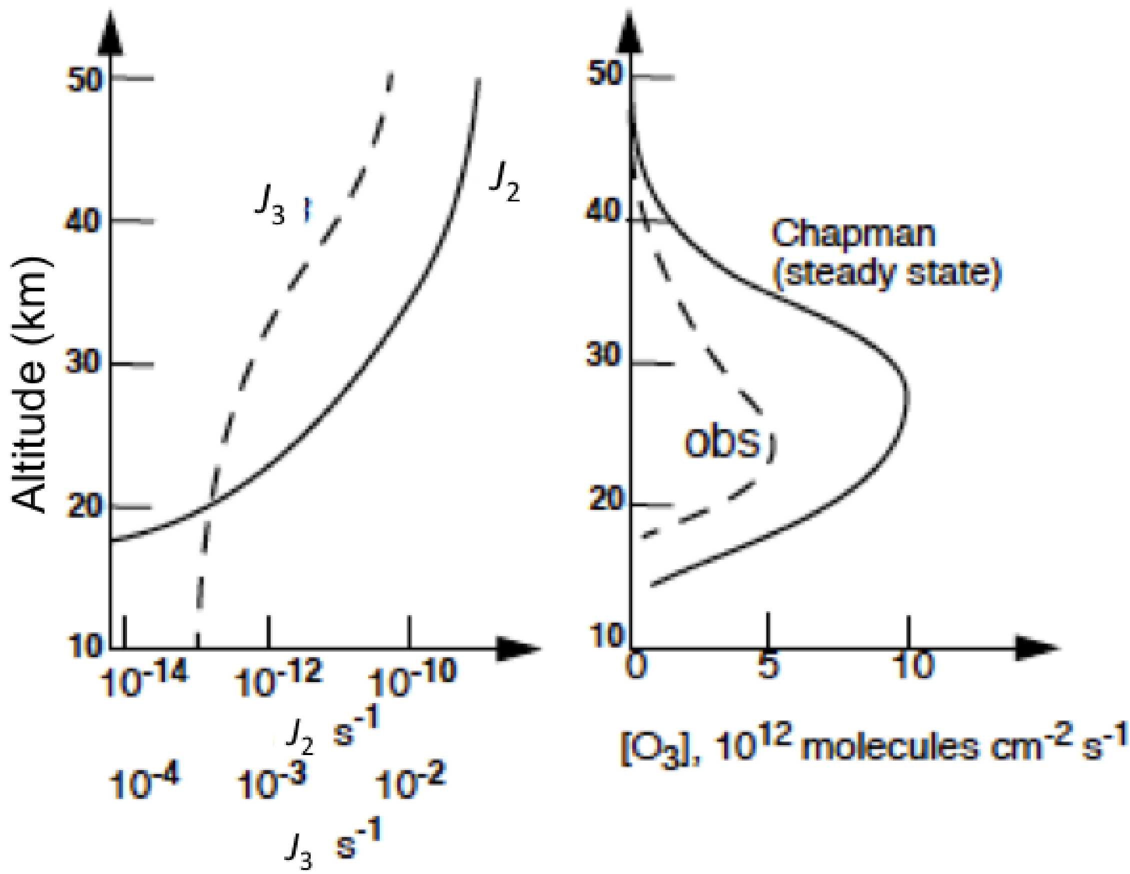


Figure 4.2 — Measured and calculated (Chapman mechanism) distributions of O_3 mixing ratios vs. altitude.

For many years it was thought that Chapman's model could adequately explain the distribution of stratospheric ozone. However, with improved laboratory and atmospheric measurements, it became apparent that reaction 4.4 only removes about 25% of the odd oxygen produced by oxygen photolysis. Calculations based on just the Chapman reactions seriously overestimate stratospheric ozone concentrations (Fig 4.2).

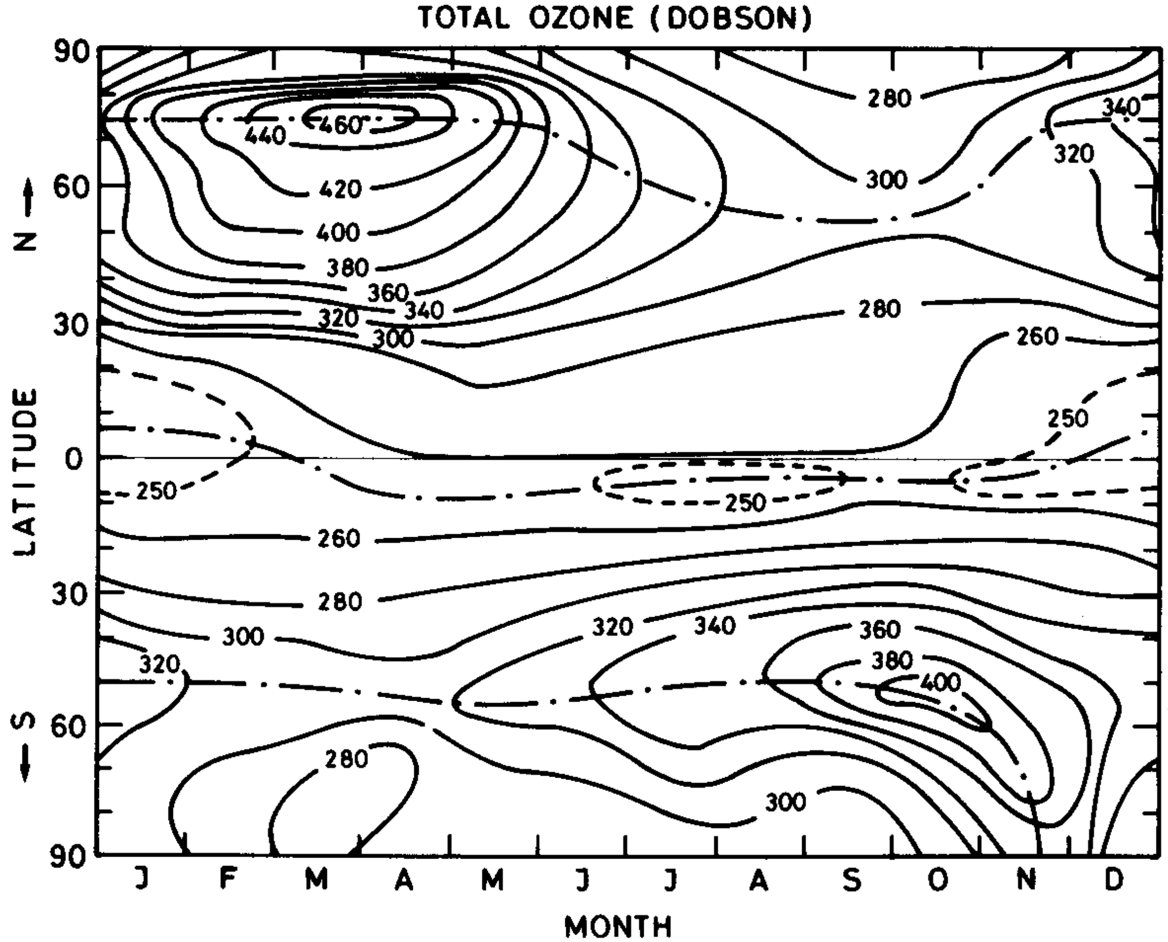


Figure 4.3 — Variation of total ozone as a function of latitude and season, showing highest values at high latitudes just after the polar night. Photochemical theory alone would predict highest ozone over the tropics; the observed distribution is strong evidence for the role of large-scale transport (the winds) in shaping the ozone distribution.

Observations of the total ozone column helped meteorologists and atmospheric physicists show there is a global overturning circulation, known after its discoverers as the Brewer–Dobson Circulation (BDC), which takes air from the tropics and redistributes it poleward.

4.2 Catalytic cycles

Reaction 4.4 has an unexpectedly high activation energy ($E_a \sim 17.5$ kJ mol^{-1}) for such an exothermic reaction ($\Delta H \sim -390$ kJ mol^{-1}). It was realised that, at stratospheric temperatures (200–290 K), odd oxygen could be removed efficiently by catalytic cycles which achieve the same net result as reaction 4.4, without net loss of the catalytic species X or XO:

$$X + O_3 \rightarrow XO + O_2$$

$$XO + O \rightarrow X + O_2$$

net: $O + O_3 \rightarrow O_2 + O_2$ (i.e. reaction 4.4)

The rate-determining step is usually the reaction involving O. For efficiency we assume each step must be exothermic, which constrains the X–O bond energy to lie in the range $107 < D(X\text{–}O) < 498$ kJ mol^{-1}. Suitable candidates for X present in the stratosphere include H, OH, NO, Cl and Br.

4.3 The oxides of nitrogen, NOx

4.3.1 Sources

NO and NO_2 ($NO+NO_2$ = NOx) are present in the stratosphere at ~10 ppb (compare ~10 ppm of O_3). The main source of NOx in the stratosphere is nitrous oxide (N_2O), produced at the Earth's surface by denitrifying bacteria. N_2O is well mixed in the troposphere (~335 ppb), has a very long tropospheric lifetime, and is transported to the stratosphere where its main fate is photolysis (λ < ~220 nm):

$$N_2O + h\nu \rightarrow N_2 + O(^1D)$$

About 1% is instead converted to NO via reaction with electronically excited oxygen atoms:

$$O_3 + h\nu \rightarrow O_2(^1\Delta) + O(^1D) \qquad \lambda < 310 \text{ nm}$$

$$N_2O + O(^1D) \rightarrow NO + NO \qquad k \sim 10^{-10} \text{ cm}^3\text{s}^{-1}$$

$$\rightarrow N_2 + O_2$$

N_2O mixing ratios are increasing at ~0.8 ppb per year. Nitrogen oxides are also emitted directly into the stratosphere by high-flying aircraft.

> Key point: Emissions of NOx into the troposphere from power stations, motor vehicles, etc. are very large and important for the troposphere, but little of this reaches the stratosphere, since conversion to HNO_3 and rainout is rapid.

The stratospheric residence time is sufficiently long that NOx from aviation is calculated to cause a significant reduction of stratospheric ozone, especially from a hypothetical fleet of advanced supersonic aircraft projected to fly even higher in the stratosphere.

4.3.2 Catalytic cycle

With X = NO:

$$\begin{aligned} NO + O_3 &\rightarrow NO_2 + O_2 \quad (k_a) \\ NO_2 + O &\rightarrow NO + O_2 \quad (k_b) \end{aligned}$$

This cycle is responsible for about 50% of odd oxygen removal from the stratosphere, despite competing reactions — the most important being NO_2 photolysis, which is very rapid even at low altitudes (Module 3). NO_2 photolysis produces a null cycle in which odd oxygen is conserved:

$$\begin{aligned} NO_2 + h\nu &\rightarrow NO + O \quad (J_c) \\ O + O_2 + M &\rightarrow O_3 + M \\ NO + O_3 &\rightarrow NO_2 + O_2 \end{aligned}$$

Steady state between NO and NO_2 is rapidly established (~100 s). Taking the rate of change of NO:

$$\frac{d[NO]}{dt} = -k_a[NO][O_3] + k_b[NO_2][O] + J_c[NO_2] = 0$$

$$\Rightarrow \quad [NO] = \frac{k_b[NO_2][O] + J_c[NO_2]}{k_a[O_3]}$$

(I)

The rate of odd oxygen destruction by NOx:

$$\frac{d[O_x]}{dt} = -k_a[NO][O_3] - k_b[NO_2][O] + J_c[NO_2]$$

(II)

Substituting [NO] from (I) into (II):

$$\left.\frac{d[O_x]}{dt}\right|_{NO_x} = -2k_b[NO_2][O]$$

4.3.3 Reservoirs and sinks

A more comprehensive picture of stratospheric NOy chemistry (NOy = sum of all nitrogen oxides excluding N_2O) also involves NO_3, N_2O_5, HNO_4, HNO_3 (and $ClONO_2$ — §4.5).

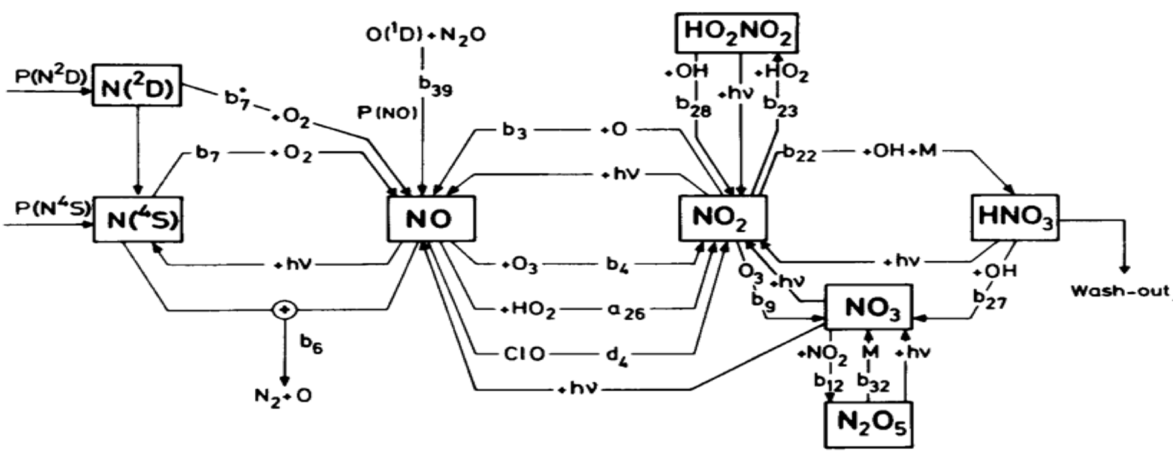


Figure 4.4 — Key sources, sinks and species involved in the chemistry of NOy in the stratosphere.

N_2O_5, HNO_3 and $ClONO_2$ are important reservoirs for NOx (and HOx, ClOx) — states where otherwise-reactive O_3-destroying radicals are held in less reactive form. For example, HNO_3 is formed by:

$$OH + NO_2 + M \rightarrow HNO_3 + M \tag{4.5}$$

and destroyed by

$$HNO_3 + h\nu \rightarrow OH + NO_2 \tag{1}$$
$$HNO_3 + OH \rightarrow H_2O + NO_3 \tag{4.6}$$

At higher latitudes HNO_3 can become the major nitrogen oxide species. HNO_3 is transported from the stratosphere into the troposphere, where it is efficiently rained out — the major sink of stratospheric nitrogen oxides.

Typical (modelled) distributions of NOy species are shown in (Fig 4.5).

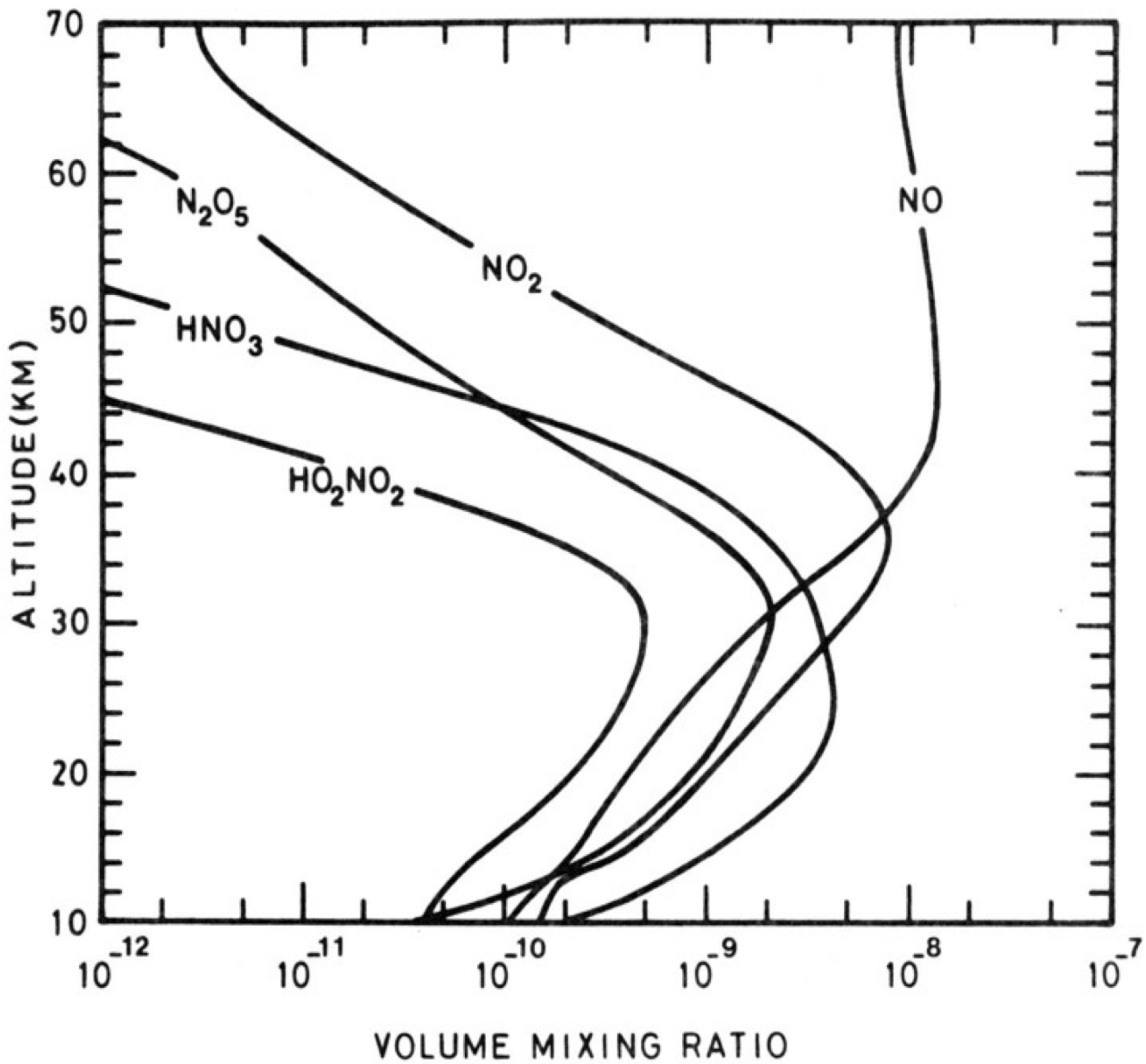


Figure 4.5 — Vertical profile of key NOy species in the stratosphere.

Reactions 4.5 and 4.6 are both favoured by low temperatures. For the termolecular reaction 4.5, this is obvious; the temperature (and pressure!) dependence for HNO_3 + OH (4.6) is more surprising — it clearly doesn't proceed by simple H-abstraction, but instead via a short-lived intermediate. Both reactions are very important in the very low stratosphere.

Climate change predictions indicate the troposphere will warm ('global warming') while the stratosphere cools. A cooler lower stratosphere would favour formation of the reservoir species (reaction 4.5) and water vapour from HOx (reaction 4.6). The resulting reduction in HOx radicals is predicted to lead to an increase in lower-stratospheric ozone later this century (with consequences also for the troposphere, since the stratosphere is a source of ozone to it).

4.3.4 Diurnal variation of the nitrogen oxides

Balloon-borne tunable diode laser spectroscopy (TDLS) measurements have confirmed the basic chemistry above, including the night-time decay of NO_2.

During sunlight hours, NO and NO_2 are in rapid equilibrium via:

$$NO + O_3 \rightarrow NO_2 + O_2 \quad (2)$$
$$O + NO_2 \rightarrow NO + O_2 \quad (3)$$
$$NO_2 + h\nu \rightarrow NO + O \quad (4)$$

At sunset, photolysis ceases and [O] decreases rapidly, so [NO_2] rises rapidly. Thereafter [NO_2] falls via:

$$NO_2 + O_3 \rightarrow NO_3 + O_2 \qquad k = 8.5 \times 10^{-13}\,\mathrm{exp}(-2450/T)\ \mathrm{cm^3 s^{-1}}$$

During the day NO_3 is photolysed very rapidly (~seconds), but at night it's removed by the fast termolecular reaction:

$$NO_2 + NO_3 + M \rightarrow N_2O_5 + M$$

The slope of the night-time NO_2 decay (Fig 4.6) is thus set by [O_3] and temperature (since at high temperatures $N_2O_5 \rightarrow NO_2 + NO_3$).

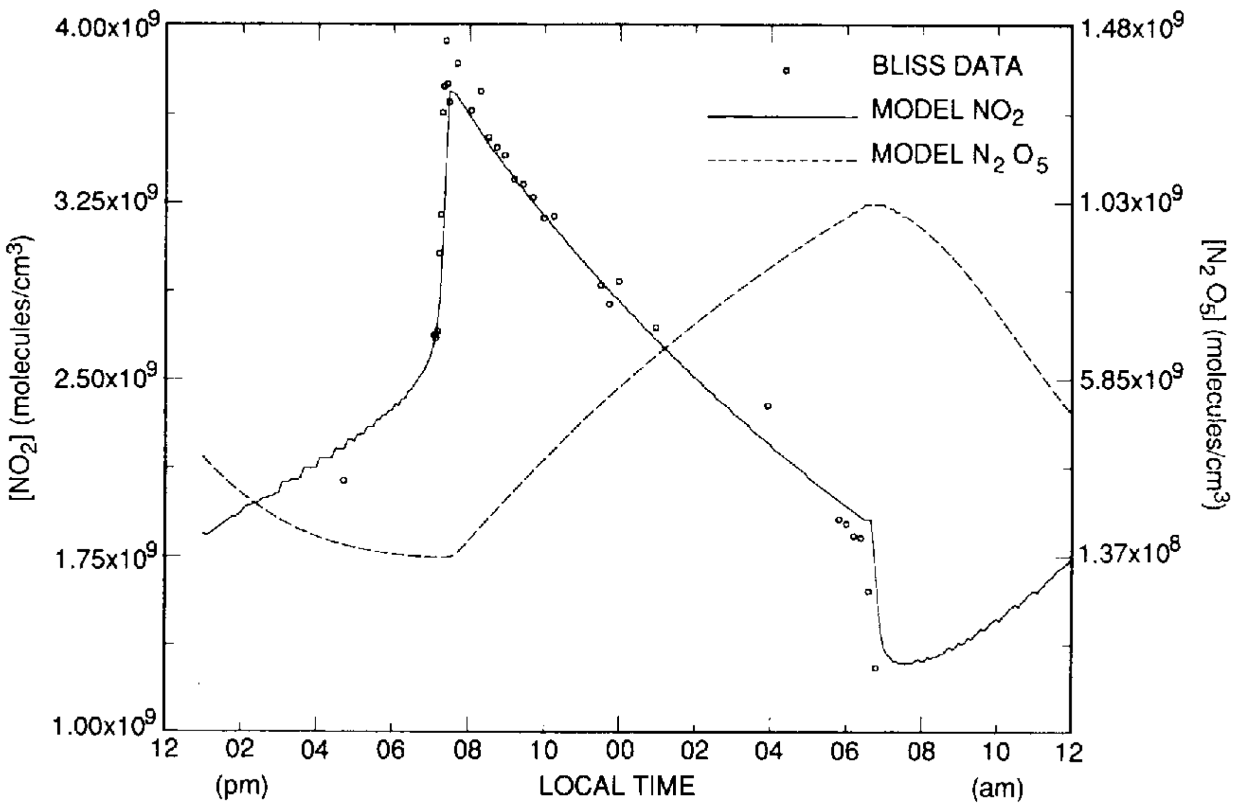


Figure 4.6 — Measured and modelled diurnal variation of NO_2.

Fig 4.6 shows [NO_2] falling at dawn with the onset of photolysis; the daytime increase in [NO_2] reflects the slow (~hours) photolysis of N_2O_5. Global measurements of NO_2, N_2O and HNO_3 have also been obtained from satellite instruments working in the infrared.

4.4 The oxides of hydrogen, HOx

4.4.1 Sources

The hydroxyl radical is one of the most important atmospheric species, in both the troposphere and the middle atmosphere. As in the troposphere, the main middle-atmosphere source of OH is:

$$O(^1D) + H_2O \rightarrow OH + OH$$

Because air enters the stratosphere at the cold equatorial tropopause, the stratosphere is extremely dry (H_2O mixing ratios 2–6 ppm). CH_4 is also transported to the stratosphere, where it is destroyed by reaction with OH and $O(^1D)$:

$$O(^1D) + CH_4 \rightarrow OH + CH_3 \qquad OH + CH_4 \rightarrow H_2O + CH_3$$

— followed by oxidation of CH_3 to H_2O and hydrogen radicals. In the upper stratosphere CH_4 mixing ratios are low and hydrogen is present mainly as H_2O, with the relationship $\Delta H_2O \sim 2\Delta CH_4$ holding. CH_4 is thus an additional source of H_2O and hence HOx, as well as playing other roles.

Major CH_4 sources are anaerobic fermentation in wetlands and paddy fields, ruminants, termites and biomass burning. Atmospheric CH_4 has been increasing (~1% yr^{-1}, followed by a period of little change during the 2000s, then a significant increase in recent years — poorly understood), leading to the expectation that stratospheric HOx concentrations will increase.

4.4.2 Catalytic cycles

H, OH and HO_2 (collectively HOx) are closely coupled on a timescale of seconds by interconversion reactions. The three-body reaction

$$H + O_2 + M \rightarrow HO_2 + M \qquad (k = 6 \times 10^{-32}(300/T)^2 \text{ cm}^6\text{s}^{-1})$$

is 100 times faster than O + O_2 + M (4.2), keeping the HO_2:H ratio very high in the stratosphere. At lower pressures in the upper stratosphere/mesosphere, H atom concentrations can rise, enabling:

$$H + O_3 \rightarrow OH + O_2$$
$$OH + O \rightarrow H + O_2$$

More important in the stratosphere itself:

$$OH + O_3 \rightarrow HO_2 + O_2 \quad (k_a)$$
$$HO_2 + O \rightarrow OH + O_2 \quad (k_b)$$

with, as for NOx, a null cycle:

$$OH + O_3 \rightarrow HO_2 + O_2$$
$$HO_2 + NO \rightarrow NO_2 + OH \quad (k_c)$$
$$NO_2 + h\nu \rightarrow NO + O$$
$$O + O_2 + M \rightarrow O_3 + M$$

The steady state between OH and HO_2 is rapidly established. By the same procedure as §4.3.2:

$$\left.\frac{d[O_x]}{dt}\right|_{HO_x} = -2k_b[HO_2][O]$$

In the lower stratosphere, as [O] falls, this cycle becomes inefficient and an alternative cycle not involving O becomes important:

$$OH + O_3 \rightarrow HO_2 + O_2$$
$$HO_2 + O_3 \rightarrow OH + O_2 + O_2$$
$$\text{net: } 2O_3 \rightarrow 3O_2$$

4.4.3 Reservoirs and sinks

A comprehensive description of stratospheric HOx chemistry is shown below:

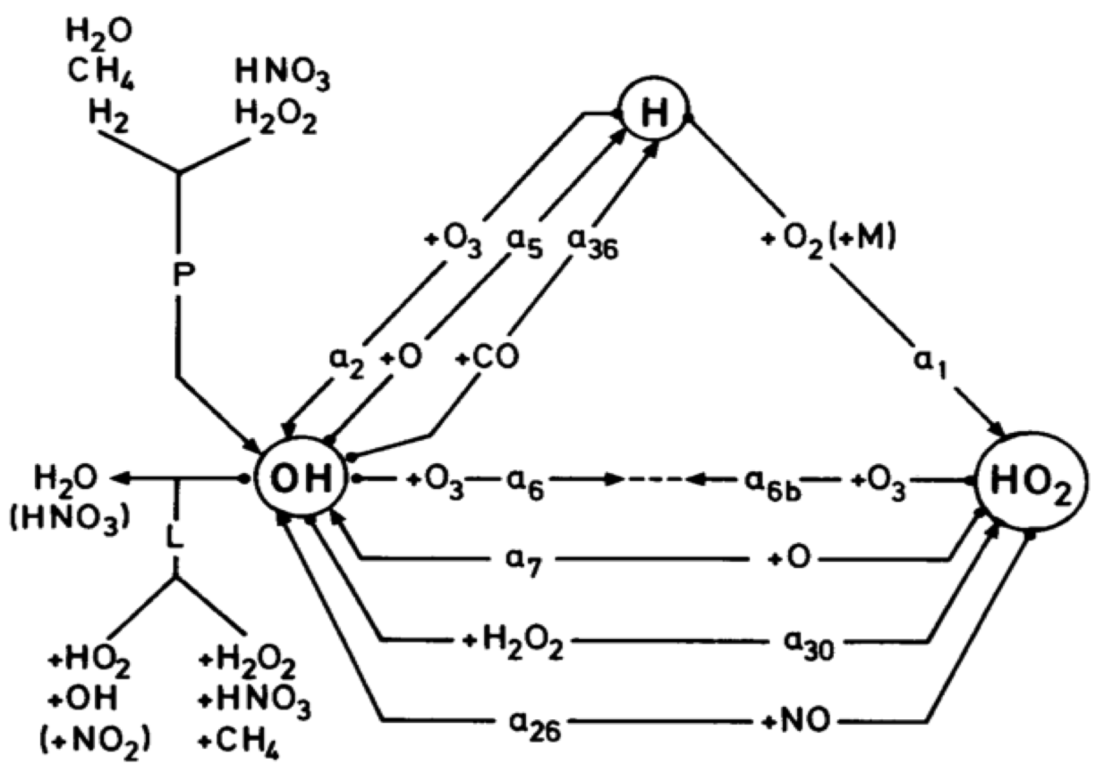


Figure 4.7 — Reaction fluxes of HOx species.

Hydrogen radicals are removed from the stratosphere by reformation of H_2O and reactions producing the temporary reservoirs HNO_3 and H_2O_2:

$$OH + NO_2 + M \rightarrow HNO_3 + M \tag{5}$$
$$HO_2 + OH \rightarrow H_2O + O_2 \tag{6}$$
$$HO_2 + HO_2 \rightarrow H_2O_2 + O_2 \tag{7}$$

Typical modelled distributions of HOx species in the stratosphere are shown in Fig 4.8.

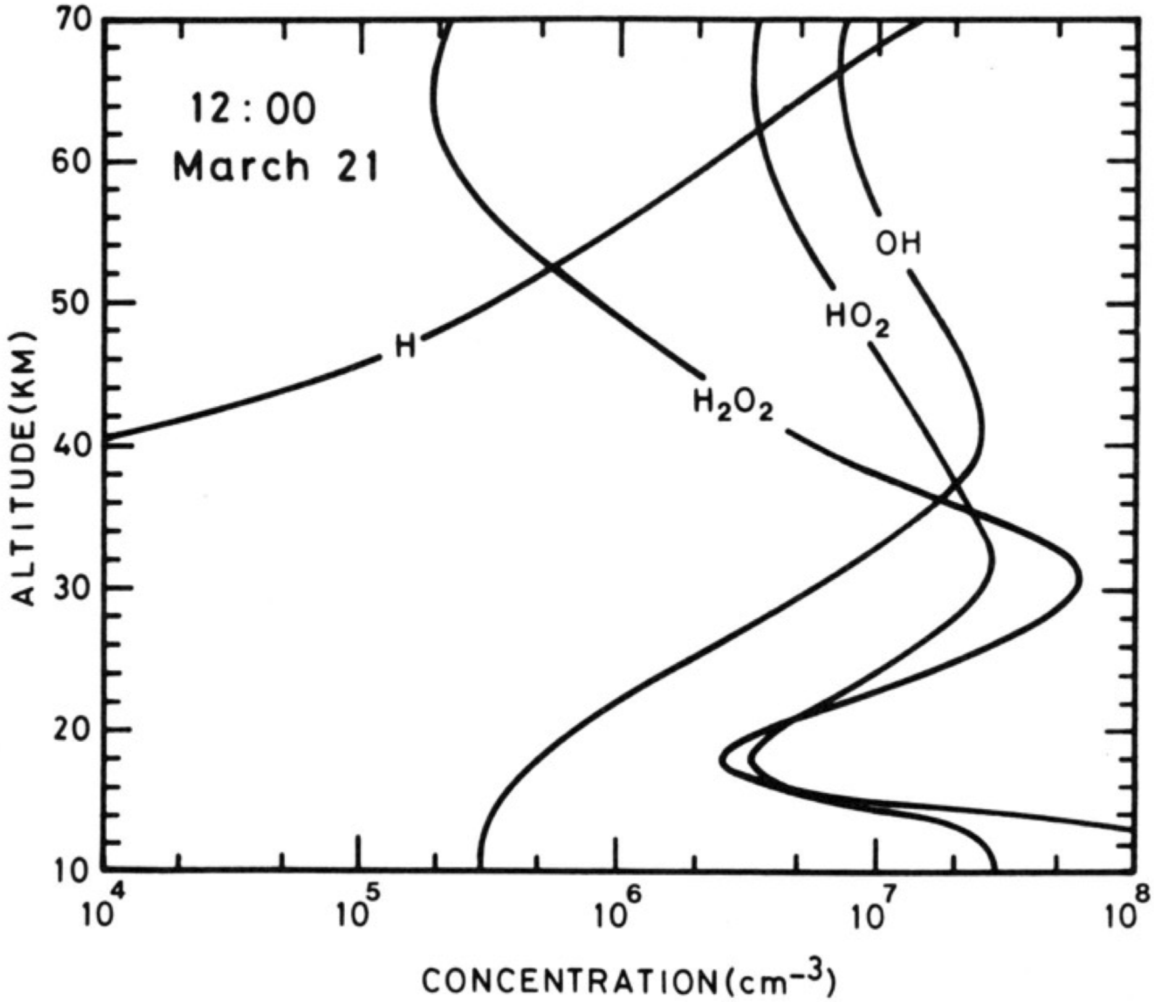


Figure 4.8 — Vertical profile of HOx species in the stratosphere.

4.5 The oxides of chlorine, ClOx

4.5.1 Sources

The major natural source of chlorine to the stratosphere is CH_3Cl (~0.6 ppb). Stratospheric chlorine is currently over 3 ppb, with the increase due largely to manmade CF_2Cl_2 and $CFCl_3$, used as aerosol propellants, refrigerants and foam-blowing agents since the 1960s. Chemically inert in the troposphere, these CFCs are transported effectively to the stratosphere, where they are broken down by photolysis and reaction with $O(^1D)$, releasing chlorine atoms. The troposphere acts as a reservoir for these species, giving them lifetimes of 50–100 years (the long atmospheric lifetime reflects the time needed to cycle up to the very low pressures where local loss is more rapid). Other species contribute to the total chlorine budget (e.g. CCl_4, CH_3CCl_3).

As a consequence of the large stratospheric O_3 losses observed in polar regions, driven by Cl-compounds (see Part III), the use and release of all these compounds is now regulated by the Montreal Protocol.

The historic and predicted future evolution of these source gases are shown in Fig 4.9.

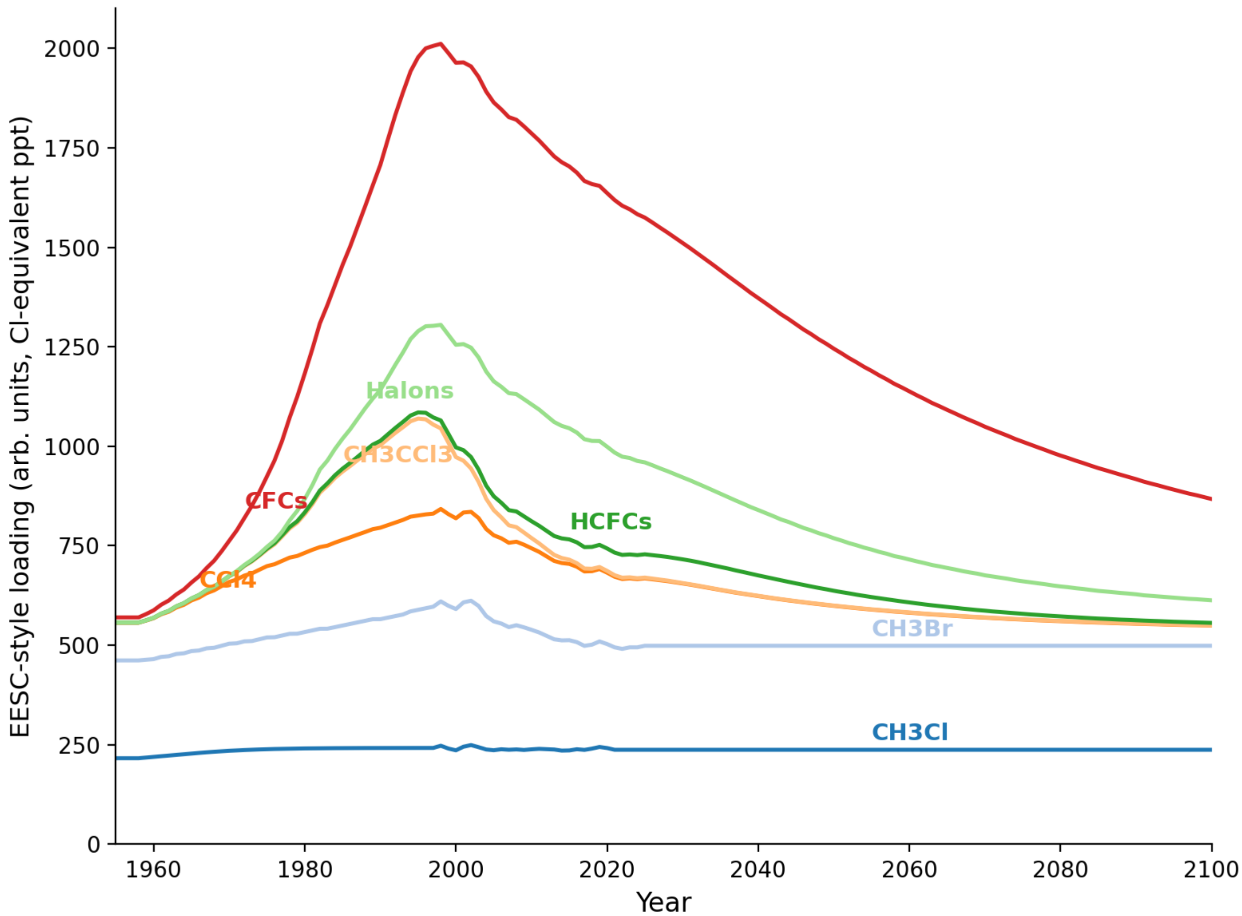


Figure 4.9 — Historic and expected future mixing ratios of ODS shown in effective equivalent stratospheric chlorine (EESC).

Because of their long lifetimes, these species will continue to play an important, if diminishing, role in ozone chemistry for at least the next 100 years. A group of substitute compounds, the HCFCs, are also regulated — being shorter-lived (they have a C–H bond, so react with OH in the troposphere), meaning the chlorine they carry is less likely to reach the stratosphere. However, many are potent greenhouse gases, as are some of the chlorine- and bromine-free replacements, the HFCs. Because of the long CFC lifetimes, there is little scope for more rapid reduction in total stratospheric chlorine.

4.5.2 Catalytic cycle

With X = Cl:

$$Cl + O_3 \rightarrow ClO + O_2 \qquad (k_a) \tag{8}$$

$$ClO + O \rightarrow Cl + O_2 \qquad (k_b) \tag{9}$$

with, again, a null cycle:

$$
\begin{aligned}
Cl + O_3 &\rightarrow ClO + O_2 && && (10)\\
ClO + NO &\rightarrow NO_2 + Cl && (k_c) && (11)\\
NO_2 + h\nu &\rightarrow NO + O && && (12)\\
O + O_2 + M &\rightarrow O_3 + M && && (13)
\end{aligned}
$$

By the same steady-state procedure as before:

$$\left.\frac{d[O_x]}{dt}\right|_{ClO_x} = -2k_b[ClO][O]$$

4.5.3 Reservoirs and sinks

A more comprehensive description of stratospheric ClOx chemistry is shown below.

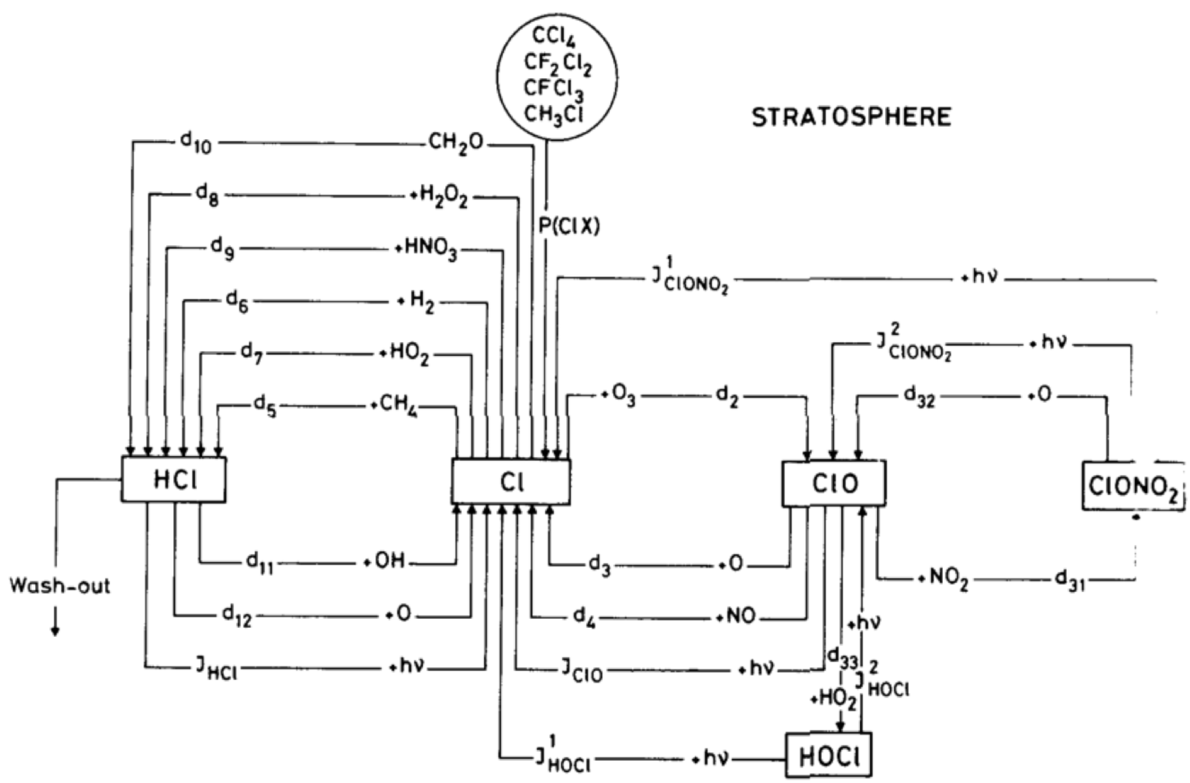


Figure 4.10 — Sources, sinks and reservoirs involved in stratospheric Cly chemistry.

Two important reservoirs for ClOx are HCl and $ClONO_2$, formed by:

$$
\begin{aligned}
Cl + CH_4 &\rightarrow HCl + CH_3 && (14)\\
ClO + NO_2 + M &\rightarrow ClONO_2 + M && (15)
\end{aligned}
$$

and removed by:

$$OH + HCl \rightarrow H_2O + Cl \tag{16}$$
$$ClONO_2 + h\nu \rightarrow Cl + NO_3 \tag{17}$$

Note the contrast with NOx: whereas OH forms the NOx reservoir HNO_3, here OH acts to release Cl from the reservoir HCl. Typical distributions of reactive chlorine species are shown in Fig 4.11.

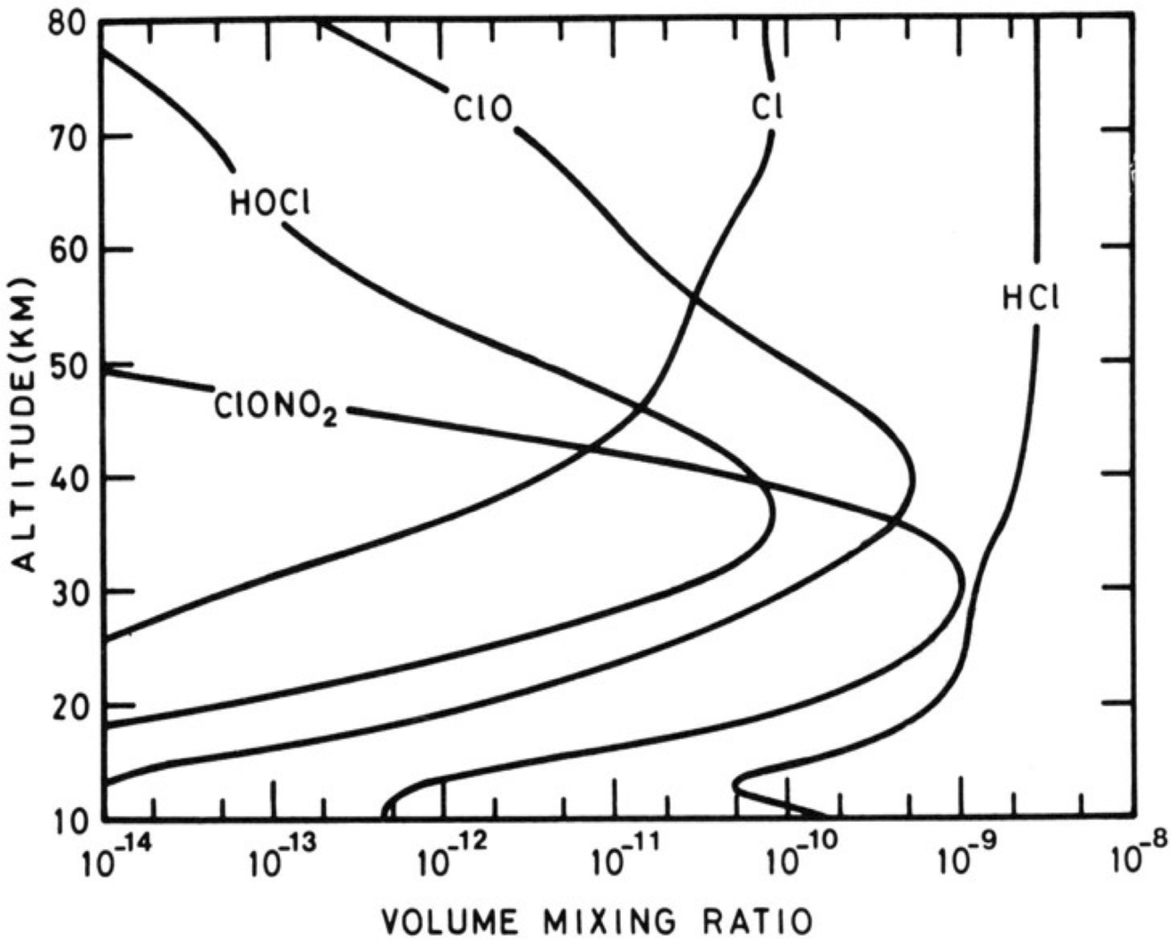


Figure 4.11 — The vertical distribution of main stratospheric Cly species.

4.6 Other halogens

4.6.1 Sources

Catalytic destruction cycles could in principle also involve F, Br and I. The F cycle is unimportant because of the high stability of the HF bond (565 kJ mol^{-1}), and there are probably insufficient I-containing species for I to be a significant O_3 destroyer.

Atmospheric bromine levels are very low (~20 ppt), with the major natural source CH_3Br, and major man-made sources CH_3Br (soil fumigation), CF_3Br, CF_2BrCl and $C_2F_4Br_2$ (fire retardants) — but Br can nonetheless be important.

4.6.2 Catalytic cycle

As with chlorine:

$$Br + O_3 \rightarrow BrO + O_2 \tag{18}$$
$$BrO + O \rightarrow Br + O_2 \tag{19}$$

4.6.3 Reservoirs and sinks

Bromine differs from chlorine in that Br + $CH_4 \rightarrow$ HBr + CH_3 is 74 kJ mol^{-1} endothermic, making the slower reaction with HO_2 the main source of HBr instead:

$$HO_2 + Br \rightarrow HBr + O_2$$

Analogously to chlorine, $BrONO_2$ is formed and destroyed via:

$$BrO + NO_2 + M \rightarrow BrONO_2 + M \tag{20}$$
$$BrONO_2 + h\nu \rightarrow Br + NO_3 \tag{21}$$

However, $BrONO_2$ photolysis occurs at longer wavelengths than for $ClONO_2$, with a larger σ, making it ~20 times faster than for chlorine. The net effect: active BrO, rather than reservoirs, are favoured for BrOx — increasing its O_3 destruction efficiency per atom relative to chlorine.

Typical distributions of reactive bromine species are shown in Fig 4.12.

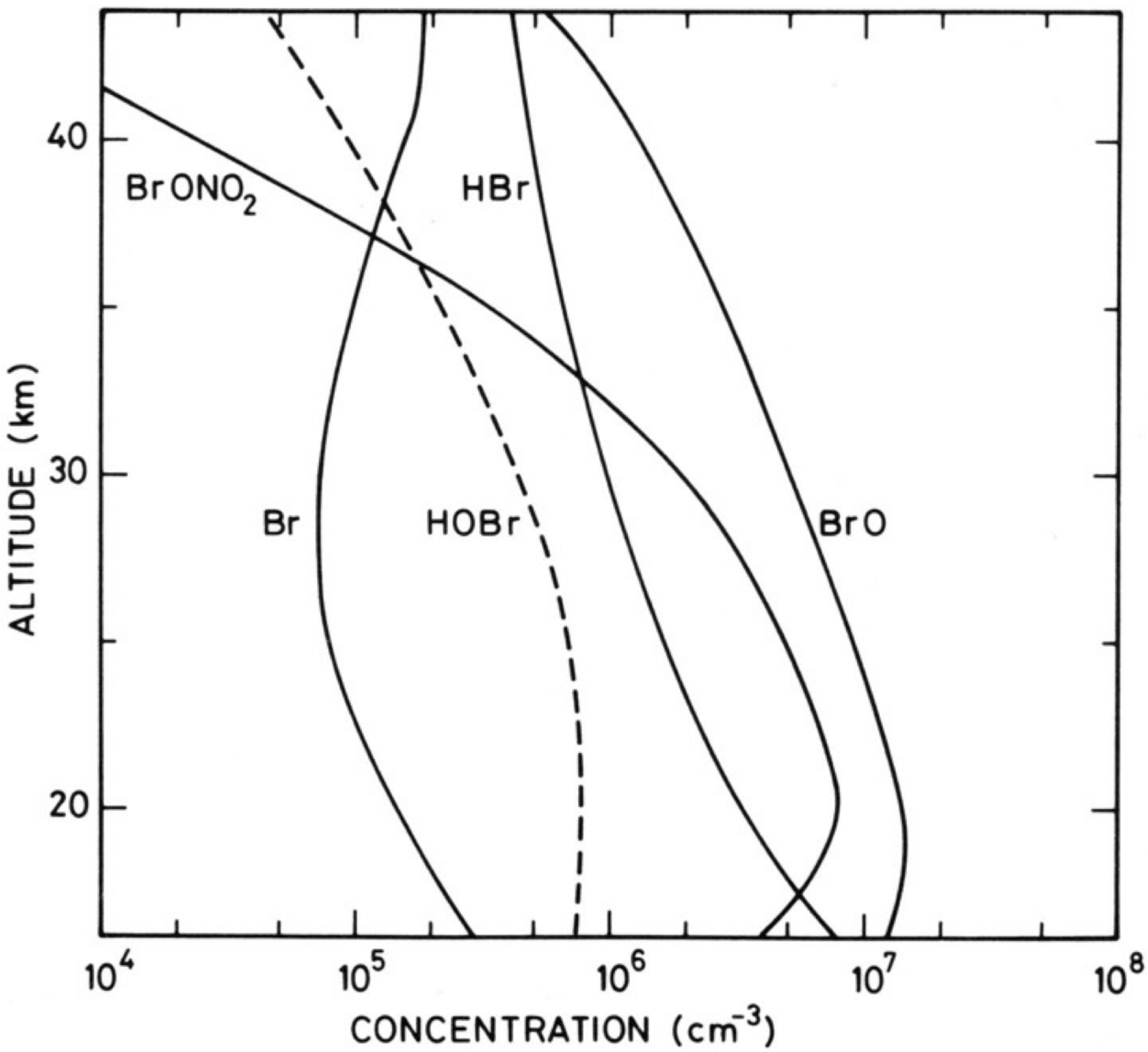


Figure 4.12 — The vertical distribution of main stratospheric Bry species.

4.7 Relative importance of the Ox, NOx, HOx, ClOx and BrOx cycles

The relative importance of the different catalytic destruction cycles is set by: the reaction rates of the different cycles; the stratospheric lifetimes of the various source gases (a longer lifetime implies release of the catalysts at a higher altitude, and further poleward); and the partitioning between reactive and reservoir species. The different chemical cycles contribute to odd oxygen loss in different proportions as a function of altitude — HOx and NOx cycles tend to dominate in different altitude bands, with ClOx/BrOx becoming disproportionately important in the polar lower stratosphere where heterogeneous chemistry (Part III) activates them from their reservoirs.

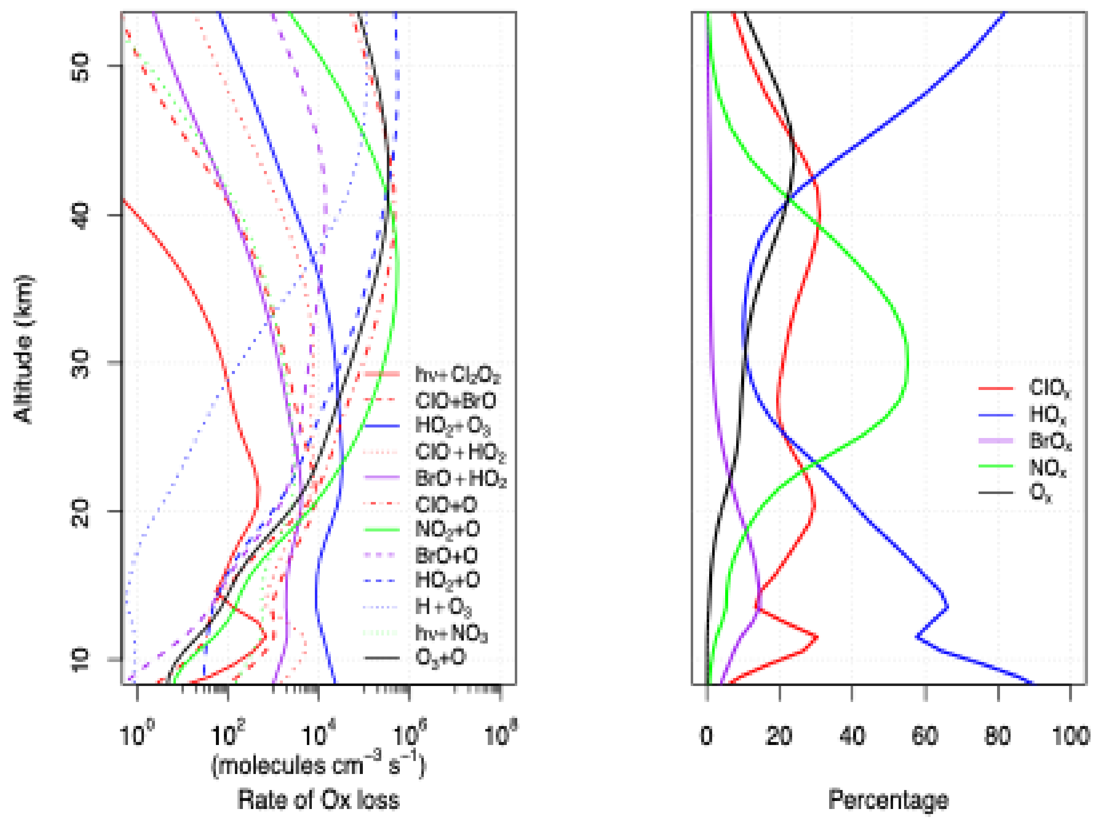


Figure 4.13 — The vertical distribution of the main stratospheric Ox destruction fluxes. On the right the relative role of each family is shown.

Try it yourself

Open `notebooks/04-chapman-mechanism.ipynb` to:

- Solve Exercise 5 ($O \leftrightarrow O_3$ interconversion timescale) as a function of altitude, and confirm it becomes "fast" only in the upper stratosphere.
- Implement Exercise 6's steady-state Chapman $[O_3]$ formula (Eq 5) and reproduce the qualitative shape of Fig 4.2 — including the ~overestimate relative to the real atmosphere once you compare against a fixed "observed" profile.
- Build a generic catalytic-cycle odd-oxygen-loss calculator, $d[O_x]/dt|_X = -2k_b[XO][O]$, and compare the relative importance of NOx-, HOx- and ClOx-driven loss for representative concentrations at a chosen altitude.
- Explore how reservoir partitioning (e.g. the $HNO_3 \rightleftharpoons NOx$ or $HCl \rightleftharpoons ClOx$ balance) changes the effective catalytic efficiency of a family, even when the intrinsic k_a, k_b are fixed.

Next: Module 5 — Model Predictions of Changes in Global O_3

Model Predictions of Changes in Global O_3

Module 5 · Model Predictions of Changes in Global O_3

Learning aims By the end of this module you should be able to: 1. Write down the multi-family odd-oxygen budget equation combining Ox, NOx, HOx, ClOx and BrOx loss terms. 2. Explain why gas-phase catalytic cycles predict ozone loss concentrated in the upper stratosphere. 3. Explain why a large local percentage change in ozone does not necessarily translate into a large column change, and why this means gas-phase chemistry alone cannot explain the Antarctic ozone hole.

On the basis of the previous discussion, the rate of change of ozone (odd oxygen) can be written as:

$$
\begin{aligned}
\frac{d([O]+[O_3])}{dt} &= 2J_a[O_2] && (22)\\
&- 2k_a[O][O_3] && (23)\\
&- 2k_b[O][NO_2] && (24)\\
&- 2k_c[O][HO_2] && (25)\\
&- 2k_d[O][ClO] && (26)\\
&- 2k_e[O][BrO] && (27)\\
&- \text{etc.} && (28)
\end{aligned}
$$

It was realised in the 1970s that an increase in the concentration of the radical catalysts (e.g. NOx emitted directly into the stratosphere by aircraft; ClOx following degradation of CFCs; HOx either directly from aviation or through a changing climate) would change the balance in this equation, leading to lower ozone. Fig 4.13 (Module 4) showed a simplified version of the relative importance of the various catalytic Ox destruction cycles as a function of altitude.

In the case of the CFCs, peak depletion was predicted at about 40 km — the altitude at which ClO was predicted to peak. Chlorine-driven loss was predicted to be small in the low stratosphere, where [O] is low and ClOx was thought to be mostly locked up in the reservoirs $ClONO_2$ and HCl. Model calculations of the change in O_3 for a change in ClOx from around 1 ppb to 3 ppb (somewhat below present-day values) show reductions of greater than 20% at ~40 km at high latitudes Fig 5.1.

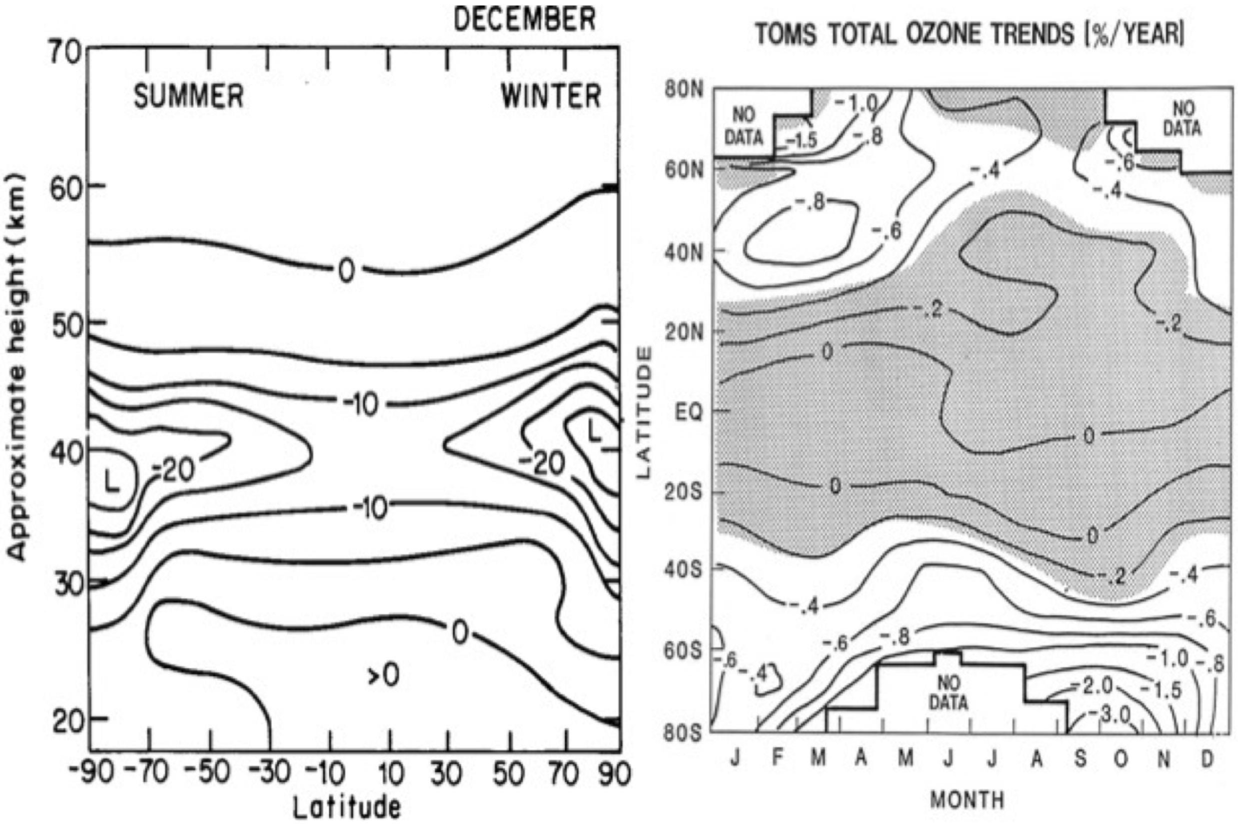


Figure 5.1 — Modelled % change in O_3 for a change in ClOx from around 1 to 3 ppb (left panel) and observed rate of change in ozone column (%/yr) (right panel).

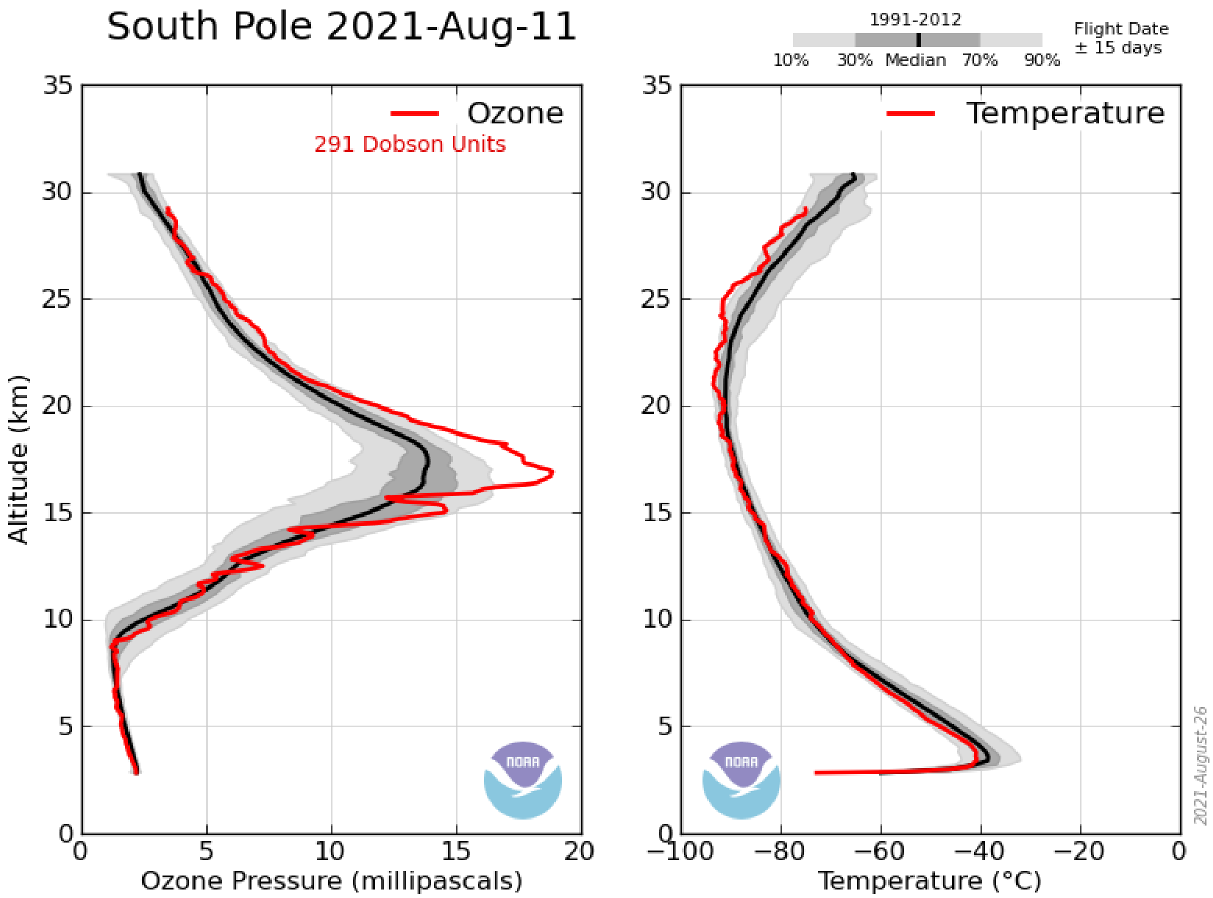
Why the high-latitude enhancement? As CH_4 mixing ratios fall toward high latitudes (because the source of methane is the tropical troposphere), the reaction converting ClOx to HCl slows:

$$Cl + CH_4 \rightarrow HCl + CH_3$$

allowing more ClOx to remain in active form — leading to greater O_3 loss.

Ozone depletion has indeed been detected in the upper stratosphere in line with this picture. But do these calculations also explain the observed loss of ozone in Antarctica, first reported in 1985, where the total ozone column falls from >300 DU to ~100 DU in a six-week period each springtime?

The answer is ‘No!’ Despite the large percentage change in local O_3 calculated above, the change in the total column O_3 from upper-stratospheric loss alone is small.



South Pole 2021-Aug-11
1991-2012
10%
30%
Median
70%
90%
Flight Date
± 15 days
Ozone
291 Dobson Units
Temperature
Altitude (km)
Ozone Pressure (millipascals)
Temperature (°C)
2021-August-26

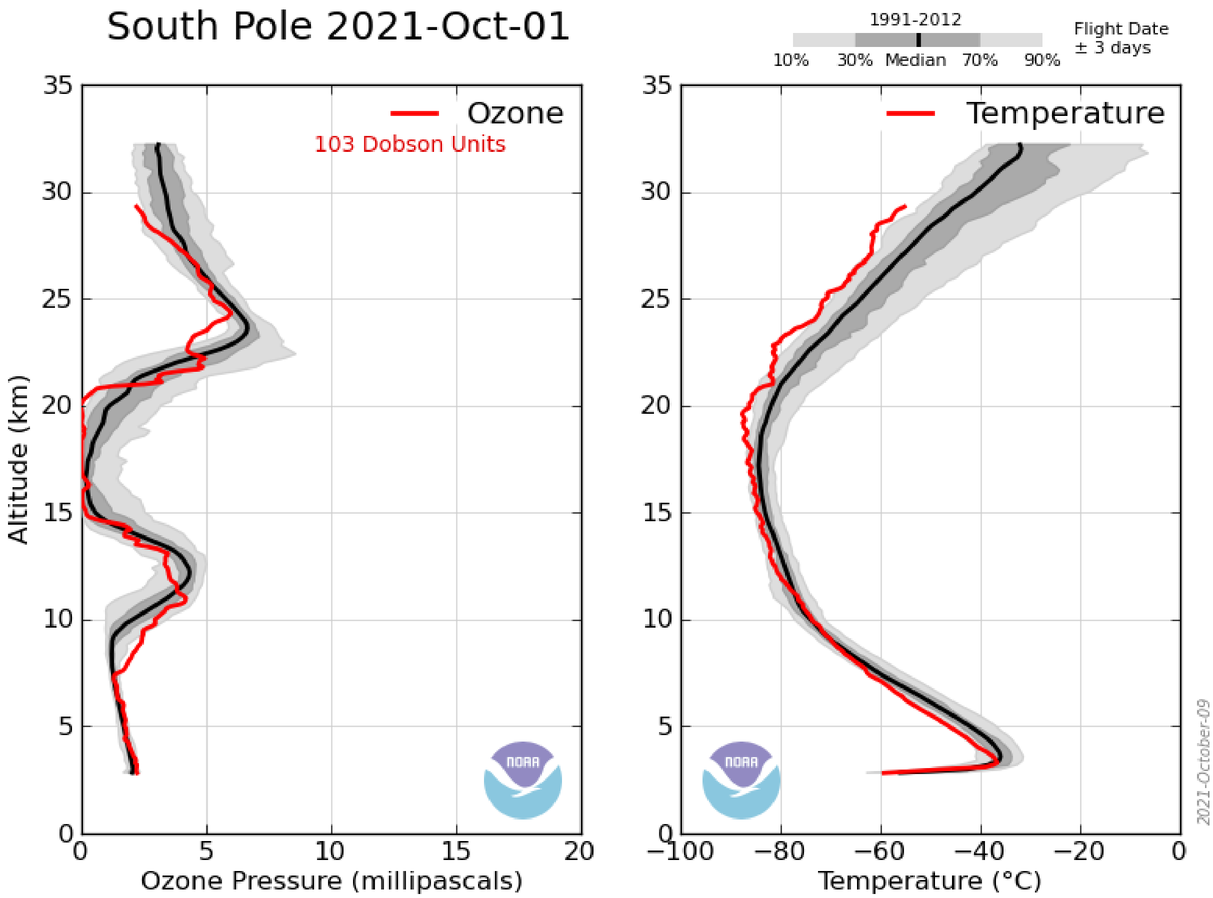


Figure 5.2 — Observed (ozone-sonde) vertical profiles of temperature and ozone. Note the difference between the two panels after 6 weeks — the springtime Antarctic ozone hole is a lower-stratospheric phenomenon, not the upper-stratospheric signature predicted by gas-phase catalytic cycles alone.

So the large observed change in the Antarctic is not explained by the upper-stratospheric losses calculated from gas-phase catalytic cycles. Our theory of catalytic cycles is correct — it explains ozone behaviour away from polar regions (e.g. the observed loss in the upper stratosphere) — but it is incomplete. Explaining the Antarctic ozone hole requires additional chemistry (heterogeneous reactions on polar stratospheric cloud surfaces, which activate ClOx from its reservoirs far more efficiently at low temperatures) — the subject of further courses.

One of the key questions current research aims to answer is how changes in stratospheric ozone are affected by, and will in turn affect, climate change.

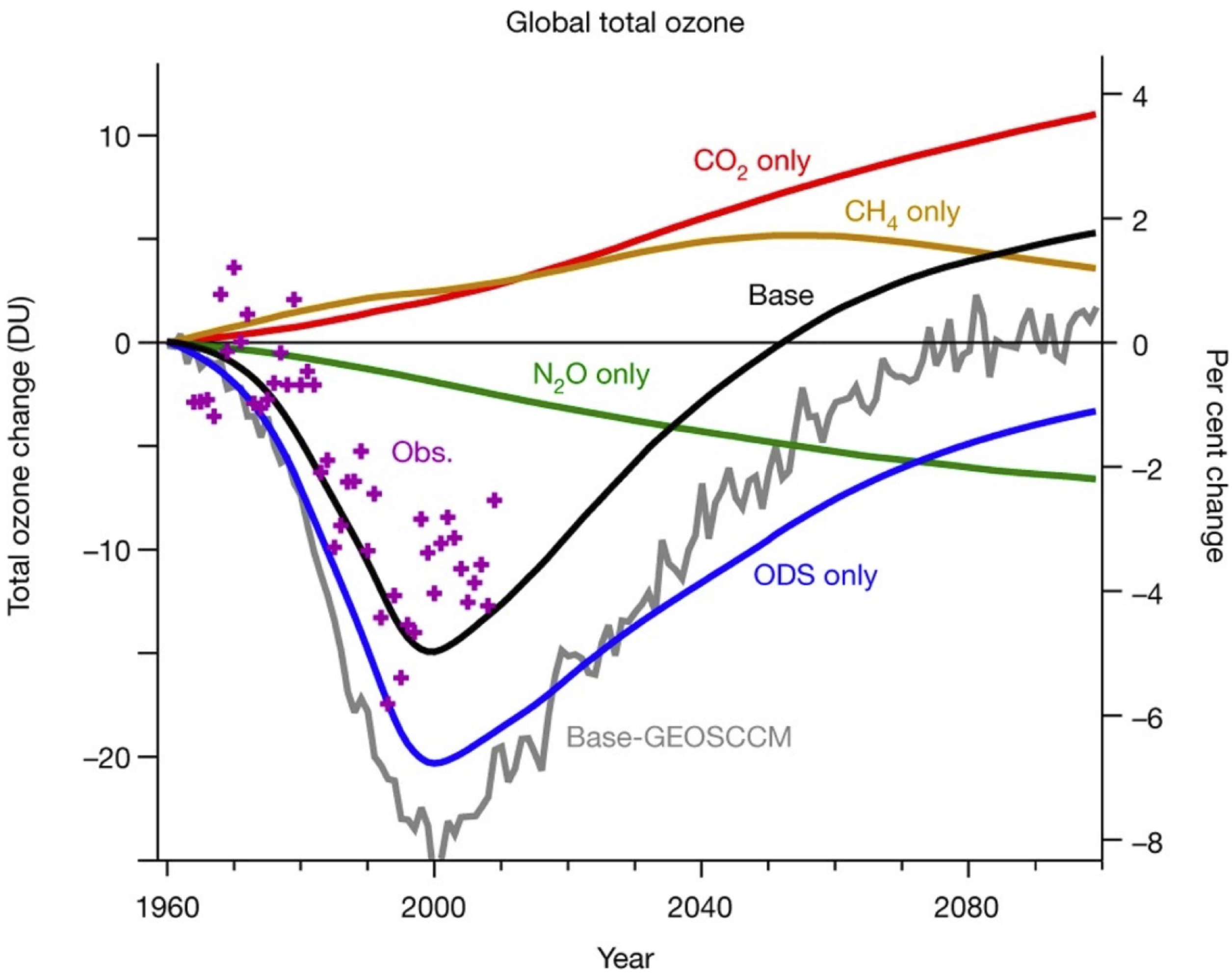


Figure 5.3 — Observed and modelled ozone from Chemistry Climate Modelling Initiative models. The observations show good skill of the models and the future scenarios highlight how different emissions/climate pathways will impact ozone. Taken from WMO Global Ozone Research and Monitoring Project Report No. 55)

Try it yourself

Open `notebooks/05-ozone-loss-antarctic.ipynb` to:

- Build the multi-family odd-oxygen budget equation from this module as a simple weighted-sum model, and explore how sensitive total loss is to each catalytic term.
- Reproduce, schematically, why a large local percentage change at 40 km barely moves the total column — by integrating a percentage perturbation applied only above some altitude against a realistic ozone profile, versus the same perturbation applied throughout the whole column.
- Compare that column-integral behaviour to what a lower-stratospheric perturbation (like the real ozone hole) would do to the total column, to see why the altitude at which loss occurs matters so much.

Next: Module 5b — Measurement Techniques for Stratospheric Composition.

Measurement Techniques for Stratospheric Composition

Module 5b · Measurement Techniques for Stratospheric Composition

Learning aims By the end of this module you should be able to:

1. Explain the principle behind Dobson-spectrophotometer ozone measurements and derive the two-wavelength ratio method.
2. Distinguish in situ from remote measurement strategies, and nadir from limb-sounding satellite geometries, in terms of their spatial/temporal coverage trade-offs.
3. Describe the physical basis of absorption and emission spectroscopy for atmospheric composition, including how pressure broadening enables vertical-profile retrieval.
4. Outline the operating principles of the main in situ techniques used for O_3, H_2O, ClO/BrO and OH, and explain why each technique suits its target species.

5b.1 Measurement strategies: in situ vs remote

A variety of methods are used to measure atmospheric composition. These divide into two broad categories.

In situ (local) measurements are obtained only in the vicinity of the sampling instrument, from a variety of platforms — aircraft, balloons, the ground.

- Advantages: high spatial/temporal resolution.
- Disadvantages: limited coverage (altitude, latitude, longitude).
- Methods include: grab sampling of long-lived species for later analysis; gas chromatography; mass spectrometry; electrochemical methods; chemiluminescence; and spectroscopic techniques including resonance fluorescence and tunable diode laser spectroscopy (TDLS).

Remote (non-local) measurements can be made some distance from the sampling instrument, from satellites, aircraft, balloons or the ground.

- Advantages: good spatial and temporal coverage (satellites).
- Disadvantages: poor spatial resolution.
- Methods: spectroscopic, spanning microwave to UV, using either emission or absorption.

5b.2 Remote measurement: principles and geometries

Remote measurements rely on the fact that many molecules of interest have absorption (and thus emission) features in the earth's atmosphere. Measurements are made over a wide range of wavelengths, from microwave through to the ultraviolet; the technique used depends on the wavelength of the spectral feature of the molecule of interest.

Solar energy input at the top of the earth's atmosphere is characteristic of that from a black body at ~6000 K, while outgoing infrared emission from the atmosphere is broadly characteristic of a black body at 245 K.

Over this wavelength range, a wide range of atmospheric constituents have absorption features.

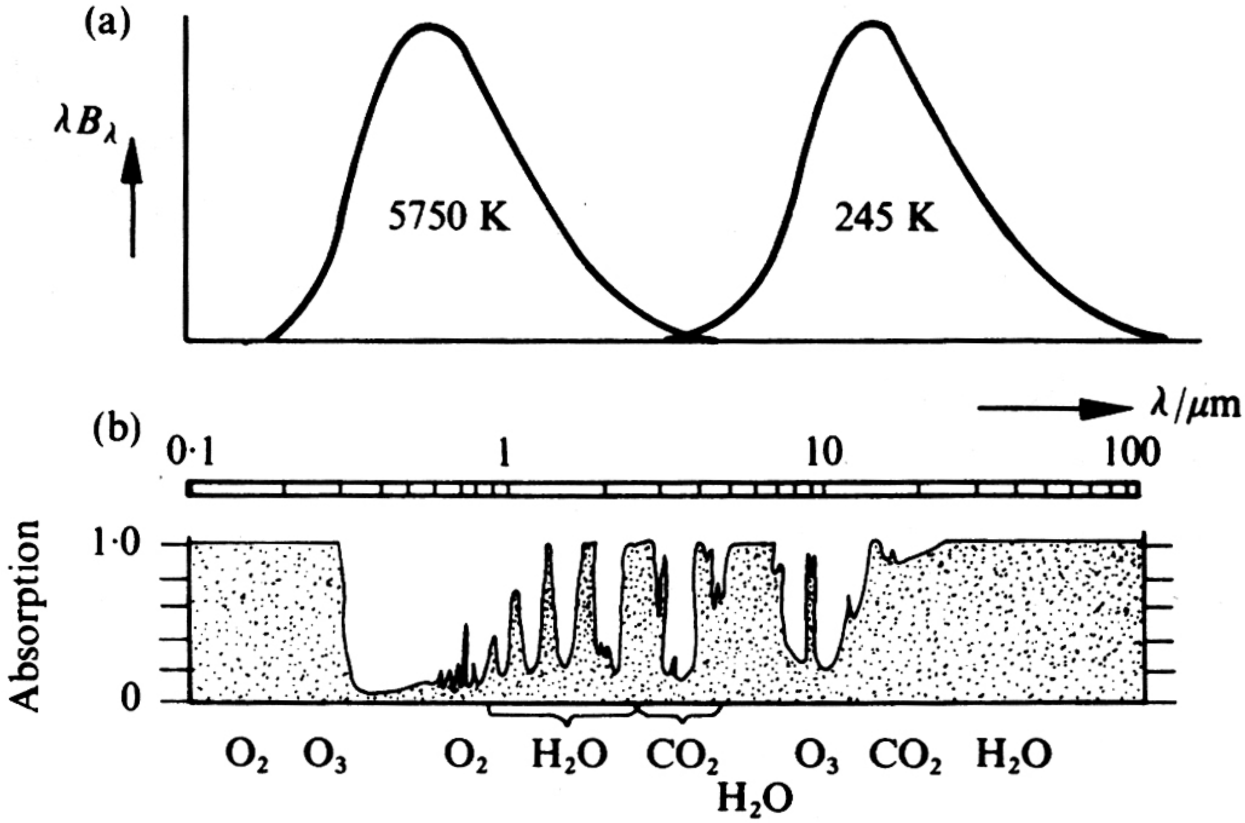


Figure 5b.1 — Black-body curves corresponding to solar and mean atmospheric temperatures (top) and absorption by atmospheric gases (bottom) as a function of wavelength. (Houghton)

Although, particularly in the IR, the absorption spectrum of the atmosphere is very complex — with overlapping absorption features due to the vibration–rotation bands of many molecules — given sufficient spectral resolution, transitions of individual molecules can be resolved.

The spectroscopic signal detected by an instrument, $I_{\nu,o}$, depends on two broad factors, depending on the observational geometry, as seen in the basic radiative transfer equation. One term depends on the (external) source, $I_{\nu,s}$, and the transmission of radiation from the source through the atmosphere. A further term describes the contribution from emission (and subsequent absorption) along the atmospheric path to the instrument:

$$I_{\nu,o} = I_{\nu,s}\,\mathrm{exp}(-\tau') + \int B_\nu(T(\tau))\,\mathrm{exp}(-\tau)\,d\tau \qquad \text{(5b.1)}$$

where $d\tau = \sum_i n_i \sigma_i \, dz$ is the optical depth at ν.

Usually only one of these terms is important. In UV spectroscopy, the sun is usually the source: there is absorption by atmospheric constituents, but no emission in the UV along the atmospheric path, so only the first term matters. Measurements of atmospheric emission are often made in the infrared or microwave, where the radiance measured depends on the emission and subsequent absorption along the path.

For composition measurements in emission, satellite measurements usually view the limb (with "cold space" as the source, which is negligible) rather than the nadir (downward-looking, with the earth's surface as the source).

Absorption spectroscopy. In the ultraviolet and visible, absorption methods are generally used, in which the absorption by the atmosphere of radiation from a distant source (usually the sun) is detected. Knowledge of the absorption characteristics of the species of interest allows the quantity of absorber between source and detector to be deduced. Measurements using this technique can be made from satellites, balloons, and the ground.

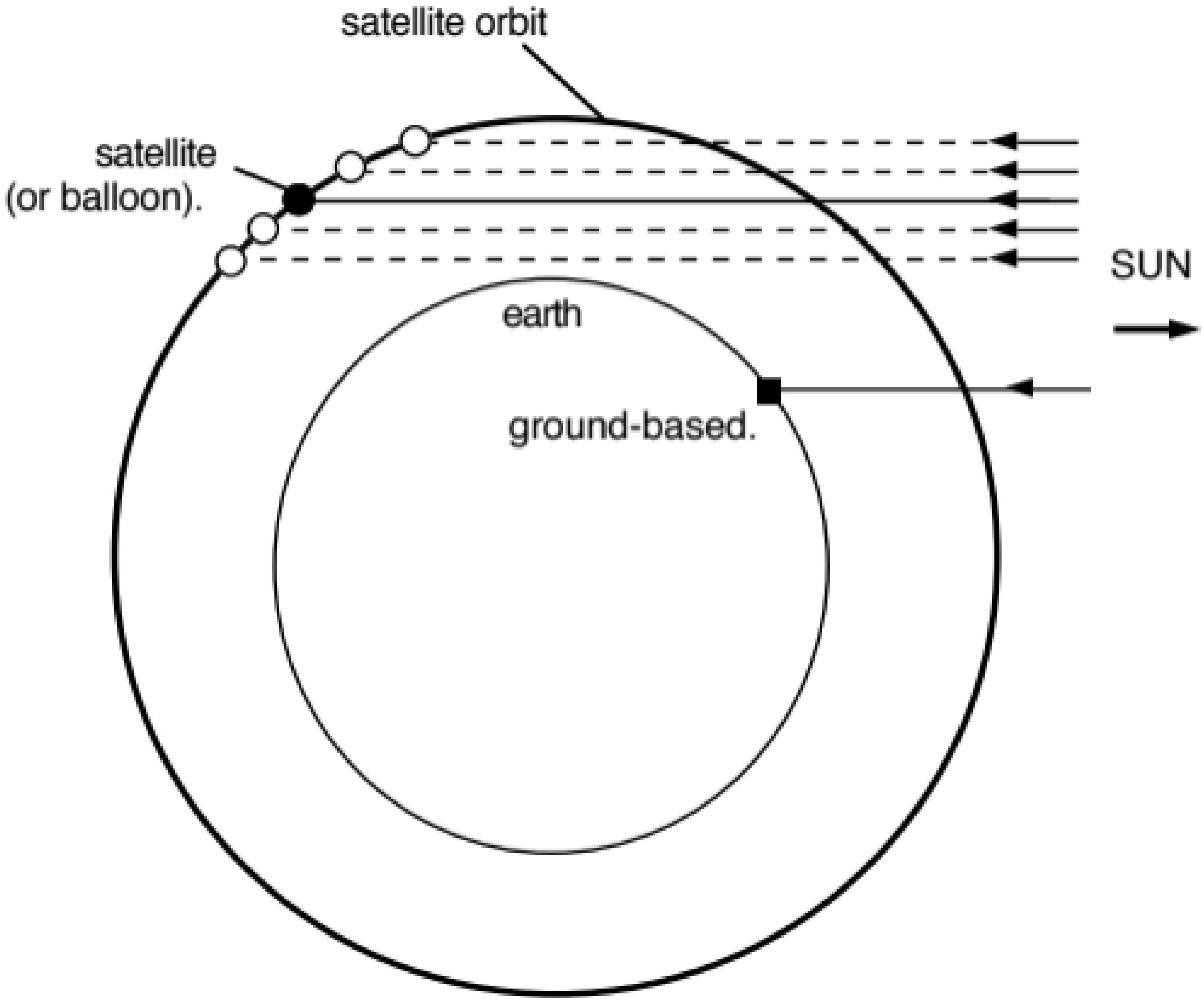


Figure 5b.2 — Schematic of absorption measurements.

Emission spectroscopy. In the atmosphere, peak energy is in the infrared. In this region, emission methods — in which emission from molecules of interest is detected — can be employed. As the energy emitted from a region depends on both the absorption properties of the gas and the temperature, the latter must also be measured. Using interferometry, simultaneous measurements of a wide range of constituents are possible.

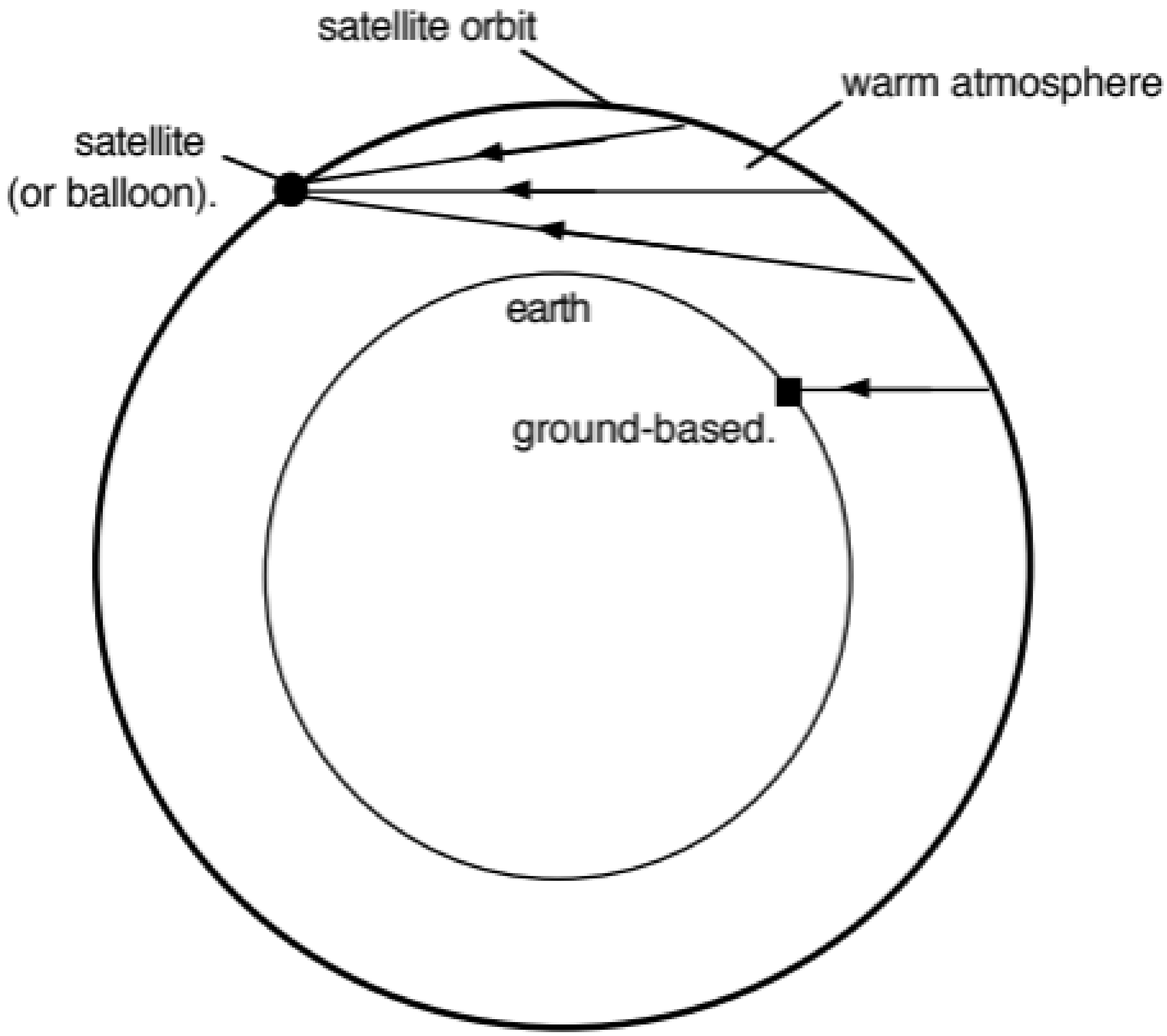


Figure 5b.3 — Schematic of emission measurements.

Different signals — reflecting the relative importance of the terms in Eq. 5b.1, and the different viewing geometries — are shown below.

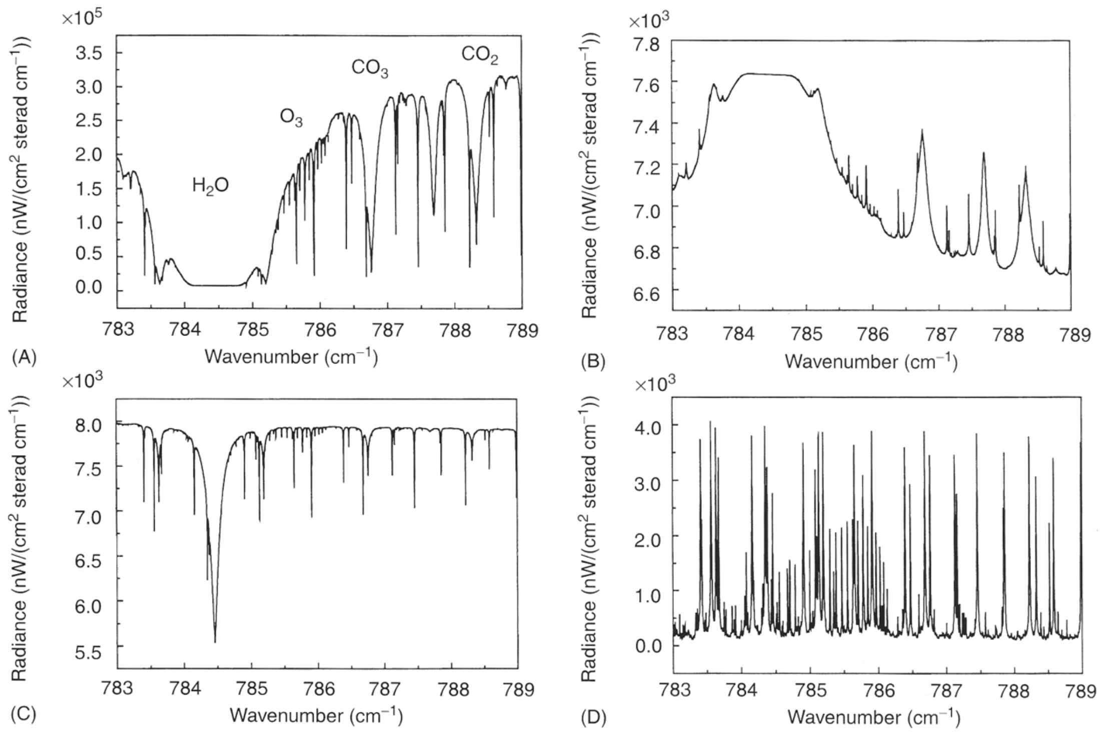


Figure 2 Calculated atmospheric spectra in the same spectral interval for various observational modes. (A) A ground-based solar absorption spectrum. Some of the most striking features are labeled. All narrow absorption lines belong to O_3; the strong H_2O line around 784.5 cm^{-1} is completely opaque. (B) The spectrum emitted by the atmosphere as observed from the ground. The spectral features now appear in emission and the O_3 signatures are strongly damped. The saturated H_2O line around 784.5 cm^{-1} reaches the limiting black body radiance of the atmospheric temperature near ground. (C) A nadir spectrum according to the view from a satellite. The warmer ground acts as a background source, so the spectral features appear in absorption. Note the changed ordinate scale and the poor contrast between lines and continuum in (B) and (C) as compared to (A). This occurs because the temperatures involved are not so different as in case (A), where the hot Sun outshines the atmospheric emission. (D) A limb spectrum for a tangent height of 20 km. The emission of stratospheric O_3 dominates, whereas the strong H_2O feature seen in (A)–(C) is not detectable, owing to the low amount of H_2O in the stratosphere.

Figure 5b.4 — Calculated spectra for different observational modes.

5b.3 Specific examples of remote sensing

Dobson spectrophotometer

As we've seen, ozone absorbs strongly at wavelengths shorter than ~310 nm (the Hartley band, see Fig 3.5), preventing it reaching the earth's surface. However, at longer wavelengths (in the Huggins band) ozone absorption is much weaker, and at these wavelengths a significant amount of solar radiation does reach the earth's surface. Dobson exploited the fact that if you know the absorption cross-section of ozone in the Huggins band, and measure the attenuation of the atmosphere, the total number of molecules of ozone between the measuring instrument and the top of the atmosphere can be determined.

The Beer–Lambert law states that the attenuation of an absorbing path with absorption cross-section σ_1 (cm²) and absorber amount U (cm⁻²) is given by:

$$\text{attenuation} = \exp\{-\sigma_1 U\}$$

Thus if T_1 is the solar intensity at the top of the atmosphere, the irradiance reaching the surface, S_1, will be:

$$S_1 = T_1 \exp\{-\sigma_1 U\}$$

In practice, Dobson found it much easier to measure the relative intensity at two nearby wavelengths. To do this he inserted a mechanical attenuator at one wavelength, adjusting the attenuation until the intensity at the two wavelengths was identical. Pairs of wavelengths were typically ~20 nm apart (e.g. 305 nm and 325 nm), with an order of magnitude difference in cross-section (see Huggins band, Fig 3.5). The relative intensity at the two wavelengths is thus:

$$S_1/S_2 = (T_1/T_2)\,\mathrm{exp}\{-(\sigma_1 - \sigma_2)U\}$$

The ratio of solar flux at the top of the atmosphere T_1/T_2 is known to be constant and can be measured, allowing U to be deduced. Of course, U is the ozone amount in the slant line of sight between the observer and the sun. To deduce the vertical column of ozone, account has to be made of the solar zenith angle, θ. In a horizontally stratified atmosphere the vertical column is related to that observed at an angle θ by:

$$U_{\mathrm{vert}} = U_{\mathrm{slant}}\cos(\theta)$$

In fact there is additional attenuation in the atmosphere due both to Rayleigh scattering and scattering and absorption by cloud particles. The derivation can be modified to account for these, and once the difference at two wavelengths is taken, the residual effect of scattering is small and fairly easily accounted for.

The Dobson measurement uses pairs of wavelengths. For other species, measurements often match absorption features over a broader spectral range — e.g. Differential Optical Absorption Spectroscopy (DOAS), a fitting procedure that nevertheless gives excellent selectivity for species with banded structure similar to the ozone Huggins bands (NO_2, NO_3, OClO, BrO, etc.).

Ground-based methods: microwave emission spectrometry

In the IR and microwave, the absorption cross-section σ cannot be assumed independent of pressure and temperature (i.e. altitude), because transitions are pressure- and Doppler-broadened. The absorption of an atmospheric path then depends on the distribution of the absorbing gas along that path. For emission (rather than absorption) there is an additional factor: the radiant energy emitted from a layer depends on temperature via the Planck function, so the emitted energy depends on both the absorber distribution and the temperature profile.

Several ground-based techniques exploit these facts to obtain the vertical profile of constituents. In the microwave, transitions are usually pure rotational, and their shape is dominated by pressure broadening:

$$\sigma_L = \frac{S\,\alpha\,P\,U}{\pi\,[(\nu - \nu_0)^2 + (\alpha P)^2]}$$

where ν is frequency (ν_0 the line centre), S is the transition strength, α is the Lorentz width at atmospheric pressure, P is the ambient pressure (bar), and U is the absorber amount. The width at half height is thus αP — proportional to pressure — while the area under the line equals the transition strength S times the absorber amount U.

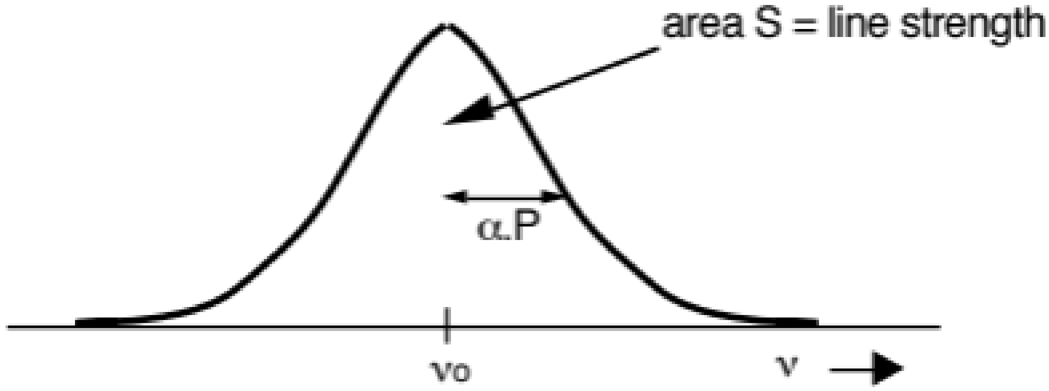


Figure 5b.5 — Pressure-broadened line shape.

Atmospheric pressure falls approximately exponentially with altitude, so an emission line's width at high altitude is much narrower than one from low altitude. When atmospheric absorption is weak, the energy reaching the surface closely approximates the sum of that emitted from all altitudes.

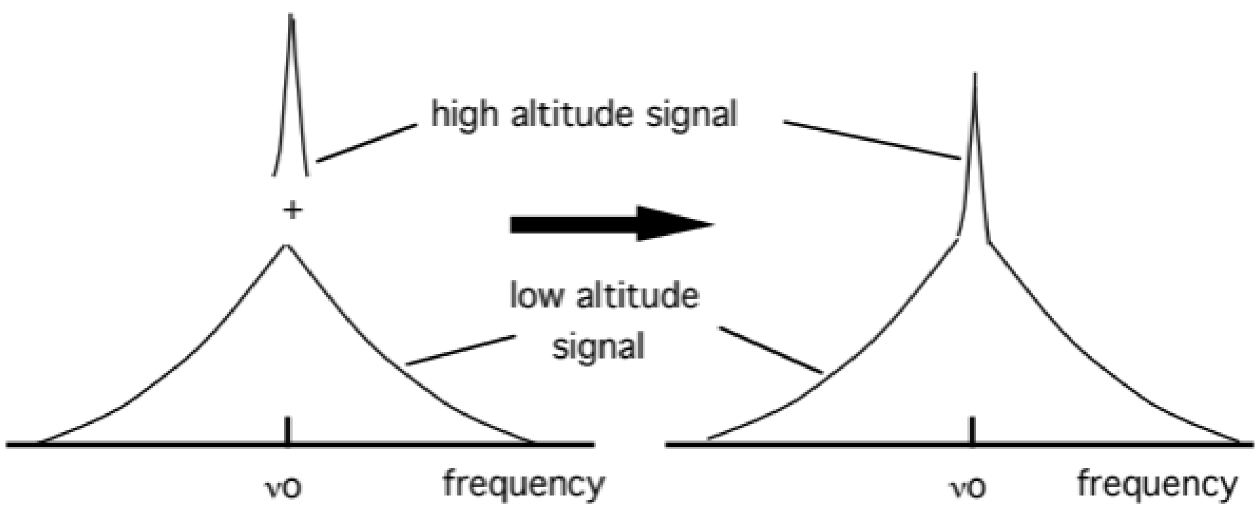


Figure 5b.6 — Schematic of the signal seen by a microwave radiometer.

This effect is clearly visible in data showing the normalised emission spectrum of the $J = 15/2 \rightarrow 13/2$ rotational transition of ClO at 278 GHz, measured with a ground-based heterodyne microwave receiver in Antarctica.

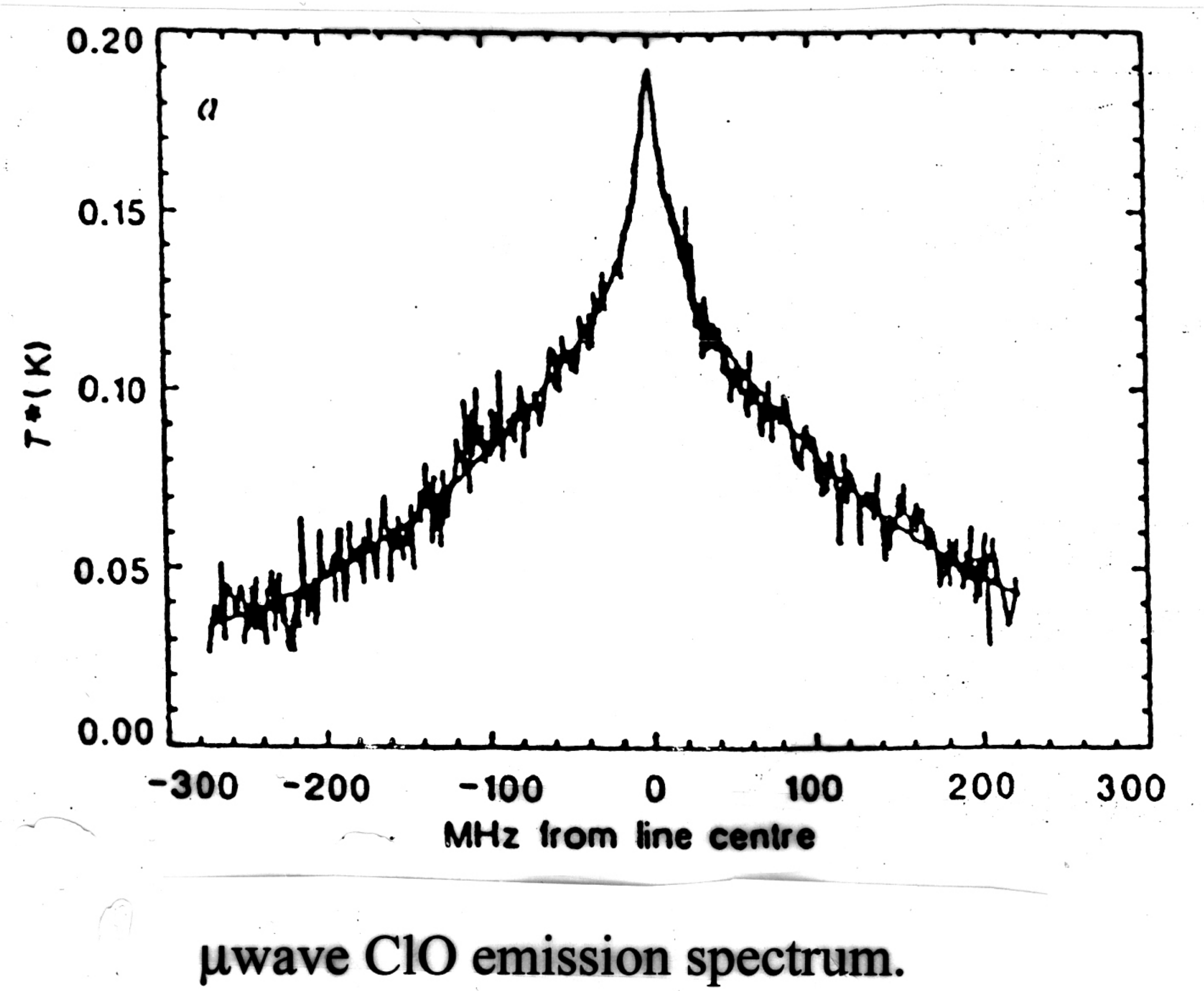
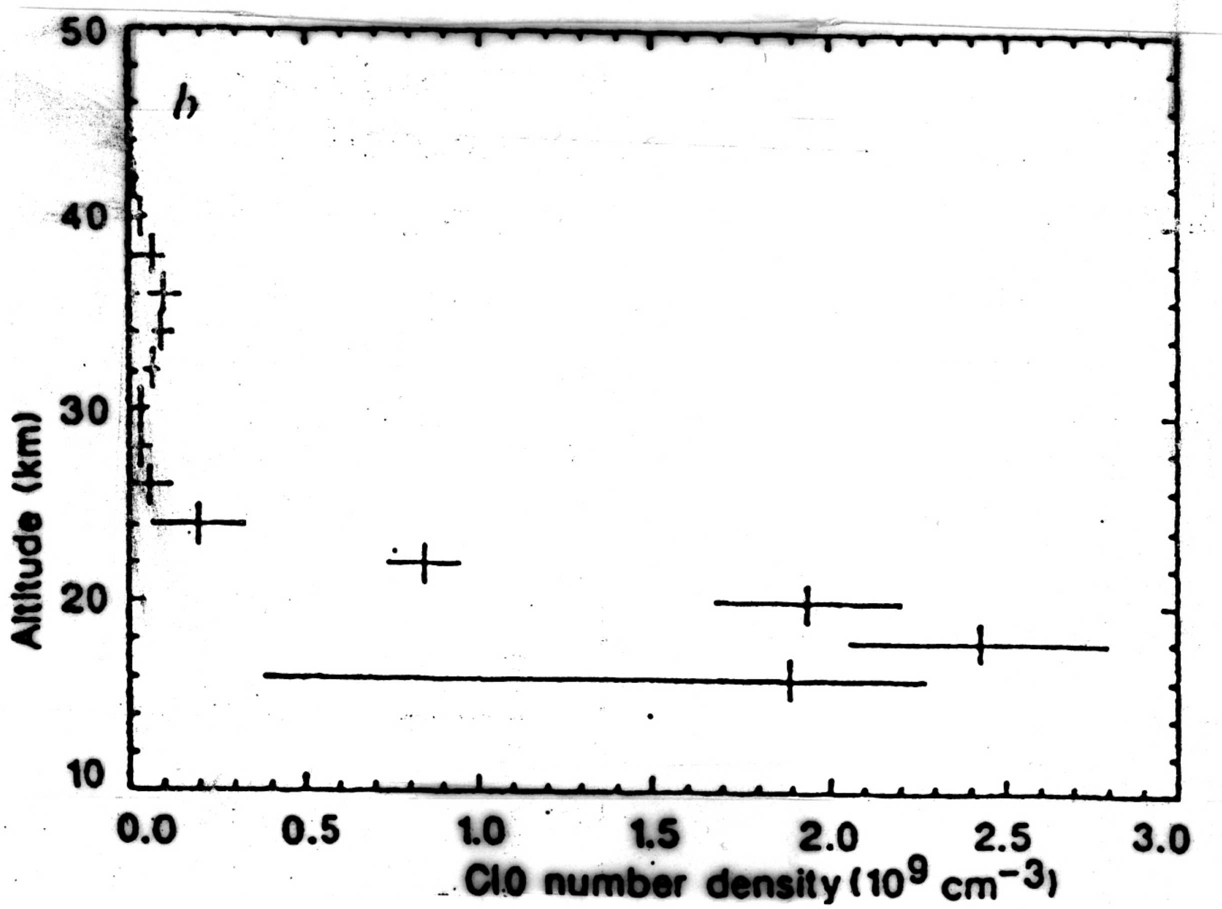


Figure 5b.7 — Ground-based microwave emission spectrum of the ClO 278 GHz transition.

The spectrum consists of a broad feature (~200 MHz wide) plus a narrow feature (~20 MHz wide), corresponding to atmospheric pressures of ~80 mbar and ~8 mbar — i.e. ~18 km and ~35 km respectively. The relative sizes of the two peaks show that a relatively larger proportion of the ClO resides at the lower altitude. A least-squares fit to the data produces the vertical profile below.



Corresponding ClO profile.

Figure 5b.8 — Retrieved vertical profile of ClO.

This measurement was particularly important because it demonstrated that high concentrations of ClO existed in the low Antarctic stratosphere, linking chlorine to the ozone loss found there. A very important series of ClO measurements has also been made using a satellite-borne microwave emission spectrometer, MLS.

Satellite measurements of vertical profiles

Satellites fall into two classes defined by their orbits: geostationary (geosynchronous) orbiters, and polar (low Earth) orbiters (LEO). Although costly to develop and launch, satellite measurements have advantages over ground-based techniques — most obviously, for satellites orbiting pole to pole, near-global coverage is obtained relatively quickly.

Two basic viewing geometries are used:

- Nadir (vertical) sounding — the instrument views the atmosphere vertically. This gives good horizontal resolution (a few km) but generally poor vertical resolution. Nadir sounding has been used very successfully for temperature sounding (using CO_2 emission), but is less suitable for measuring minor constituents.

• Limb sounding — the instrument views the limb of the earth. This improves vertical resolution at the expense of horizontal resolution.

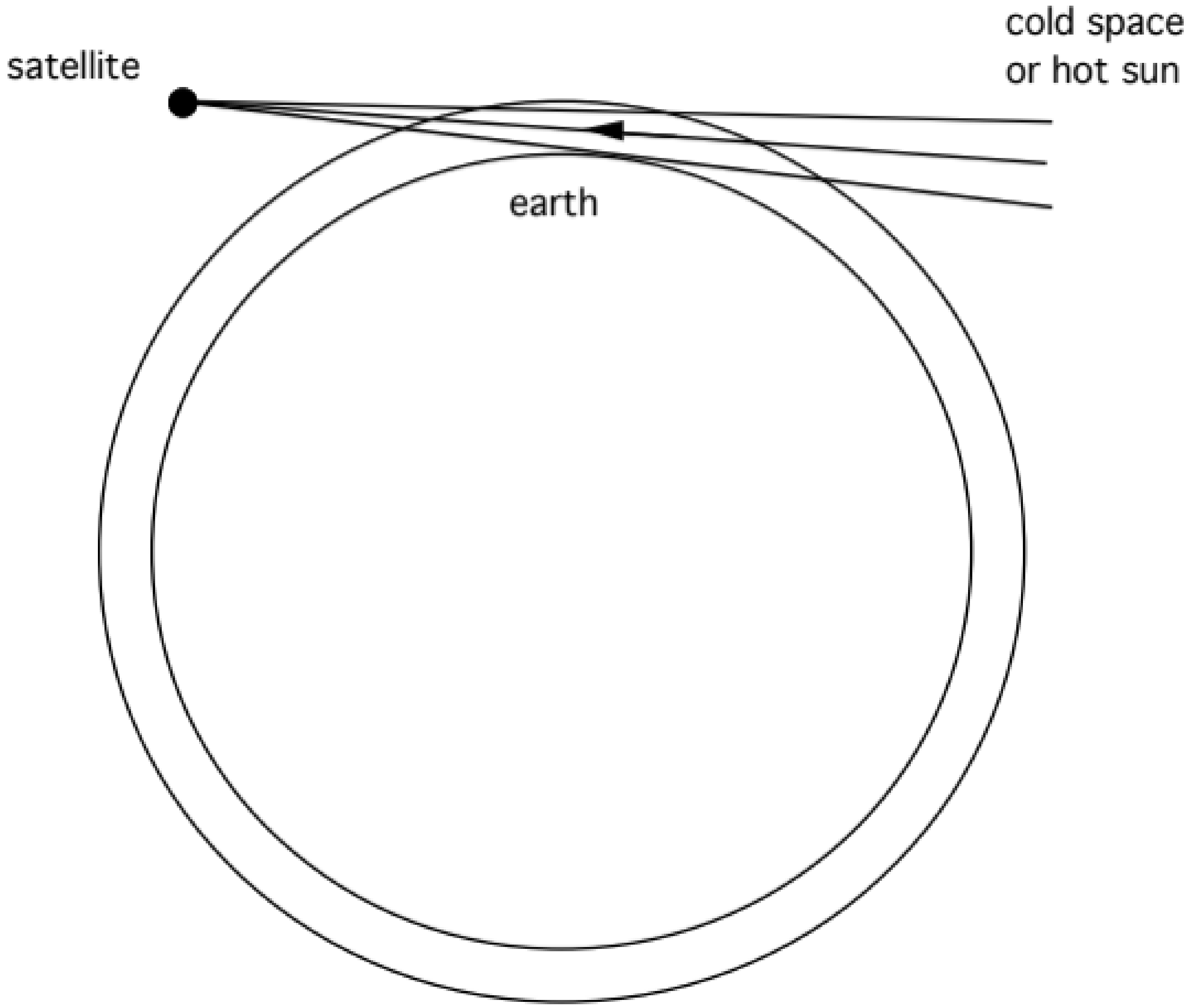


Figure 5b.9 — Schematic of limb sounding.

An important advantage of limb sounding is that the absorbing path sampled is ~70 times the vertical path. The extra path length increases the absorption of more weakly absorbing or less abundant species, permitting their detection. Using limb sounding, observations are possible by measuring thermal emission from the atmosphere (IR and microwave), or by measuring the absorption of solar radiation by the atmosphere (solar occultation).

Profile measurements are made by scanning the instrument field of view across the limb of the earth; for occultation this is done during sunrise and sunset. The profile is built up as follows: the concentration in the topmost shell (shell 1) is determined first, with the line of sight at viewing angle 1. The scan then steps down to viewing angle 2 (following the sun, for solar occultation); the line of sight now passes through shells 1 and 2, but since the concentration in shell 1 is already known, that in shell 2 can be determined. Successive steps to lower altitudes build up a full profile.

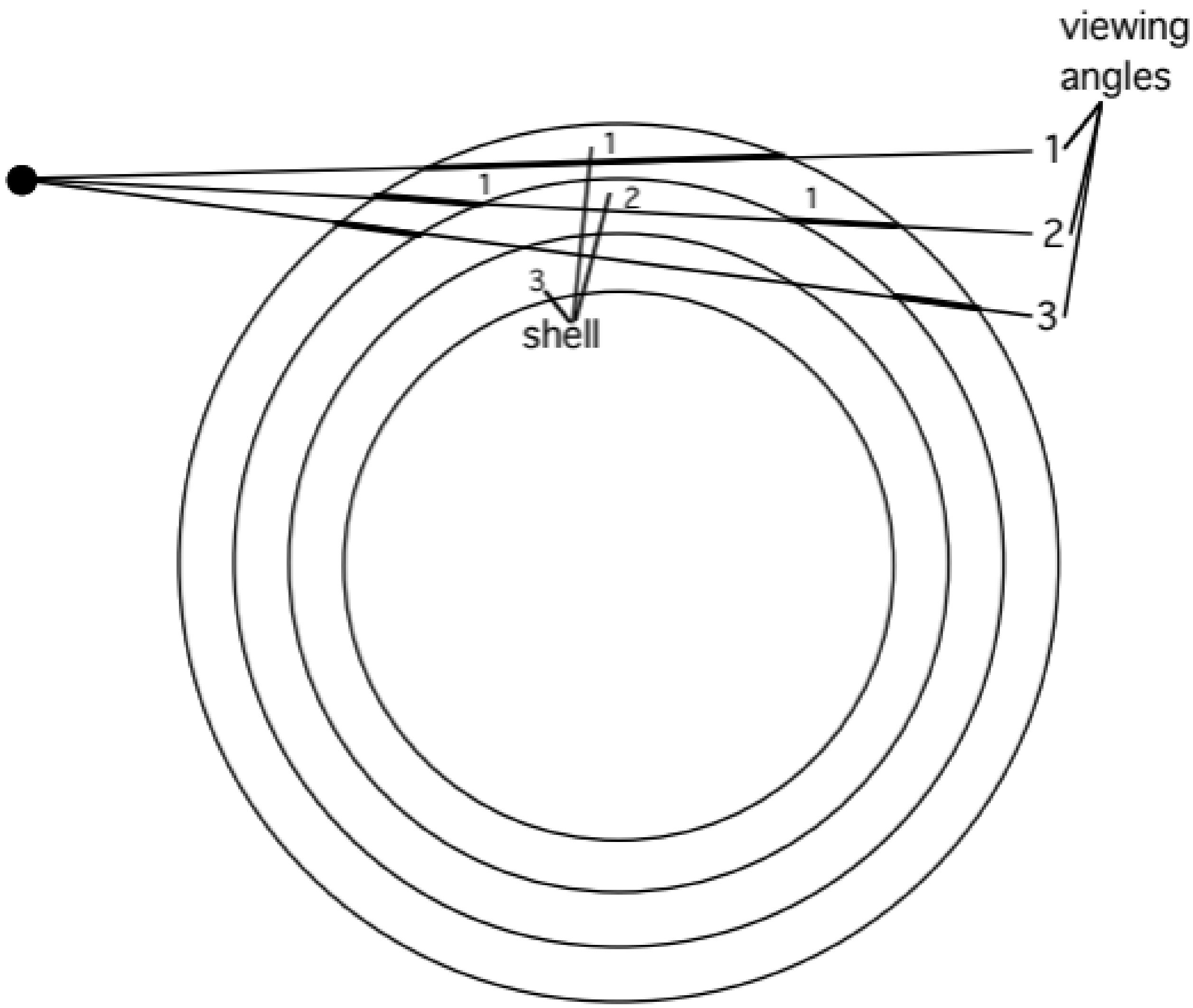


Figure 5b.10 — Schematic of the limb-sounding retrieval.

5b.4 In situ measurements

Ozone measurements: UV absorption

O_3 is routinely measured using its absorption in the Hartley band (see Fig 3.5) around 250 nm (a Hg resonance lamp provides a source at 253.7 nm), where ozone absorbs strongly. The absorption occurs in a cell in which the light path is folded many times. The light beam is split (to allow measurement of I_0 and I_{tr}), and the Beer–Lambert law is used to determine absolute concentrations of ozone (down to ~1 ppbv).

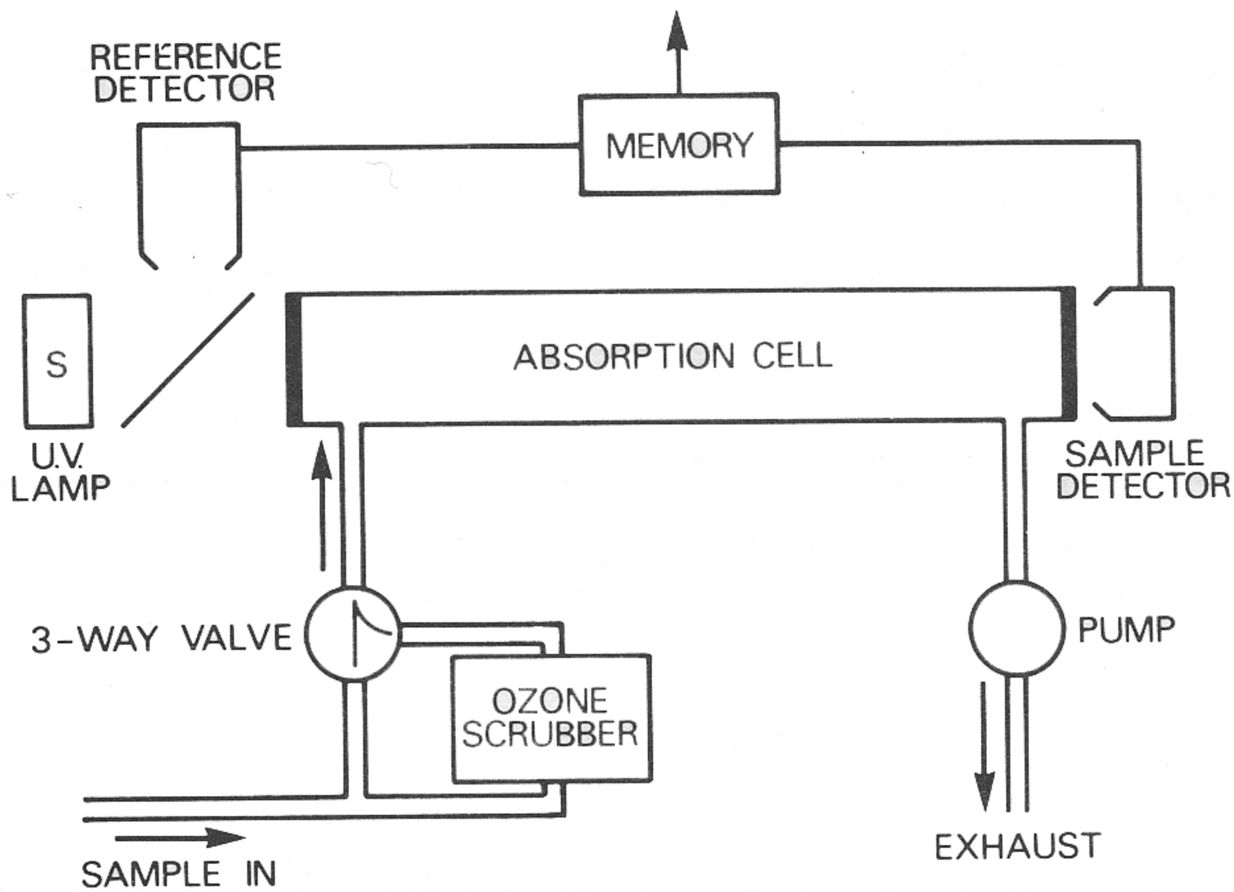


Figure 5b.11 — Schematic of a UV absorption ozone measurement.

Ozone electrochemical concentration cell (ECC) sonde. A very widely used method for O_3 is the ECC sonde — a simple, compact, inexpensive device for measuring O_3 profiles with good vertical resolution up to about 30 km. The throw-away sonde, weighing a few kg, is carried by a small balloon. It consists of an electrochemical cell containing potassium iodide in solution; ozone reacts with the KI, forming iodine:

$$2\text{KI} + \text{O}_3 + \text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{I}_2 + \text{O}_2$$

This simple device has been used to show the extent of ozone losses in the Antarctic.

Chemiluminescence. A variety of measurements are possible by the chemiluminescence technique: a known amount of chemical is injected into the air sample, reacting with the species of interest to produce an excited product; photons emitted as the excited product de-excites are then detected. Two commonly made measurements are O_3 (reacting it with ethylene, C_2H_4) and NO (by reaction with injected O_3). An extension passes the sample over a heated gold catalyst in the presence of added CO, reducing all available reactive nitrogen compounds to NO, which is then measured as before — giving a measurement of total reactive nitrogen in the air sample.

H_2O measurements by fluorescence

Knowledge of water concentrations is crucial, since reaction of H_2O with excited atomic oxygen (O^1D) is the major stratospheric source of the hydroxyl radical OH. Very sensitive measurements of H_2O have been made with the Lyman-α fluorescence hygrometer, which uses a Lyman-α light source at 121.6 nm to dissociate H_2O and form an excited OH molecule:

$$\mathrm{H_2O} + h\nu \rightarrow \mathrm{OH}^* + \mathrm{O} \qquad k_1\ (\lambda = 121.6\ \mathrm{nm}) \tag{5b.2}$$

The excited OH molecule can then radiate:

$$\mathrm{OH}^* \rightarrow \mathrm{OH} + h\nu \qquad A\ (\lambda = 309\ \mathrm{nm}) \tag{5b.3}$$

or be quenched by collision:

$$\mathrm{OH}^* + M \rightarrow \mathrm{OH} + M \qquad k_q[M] \tag{5b.4}$$

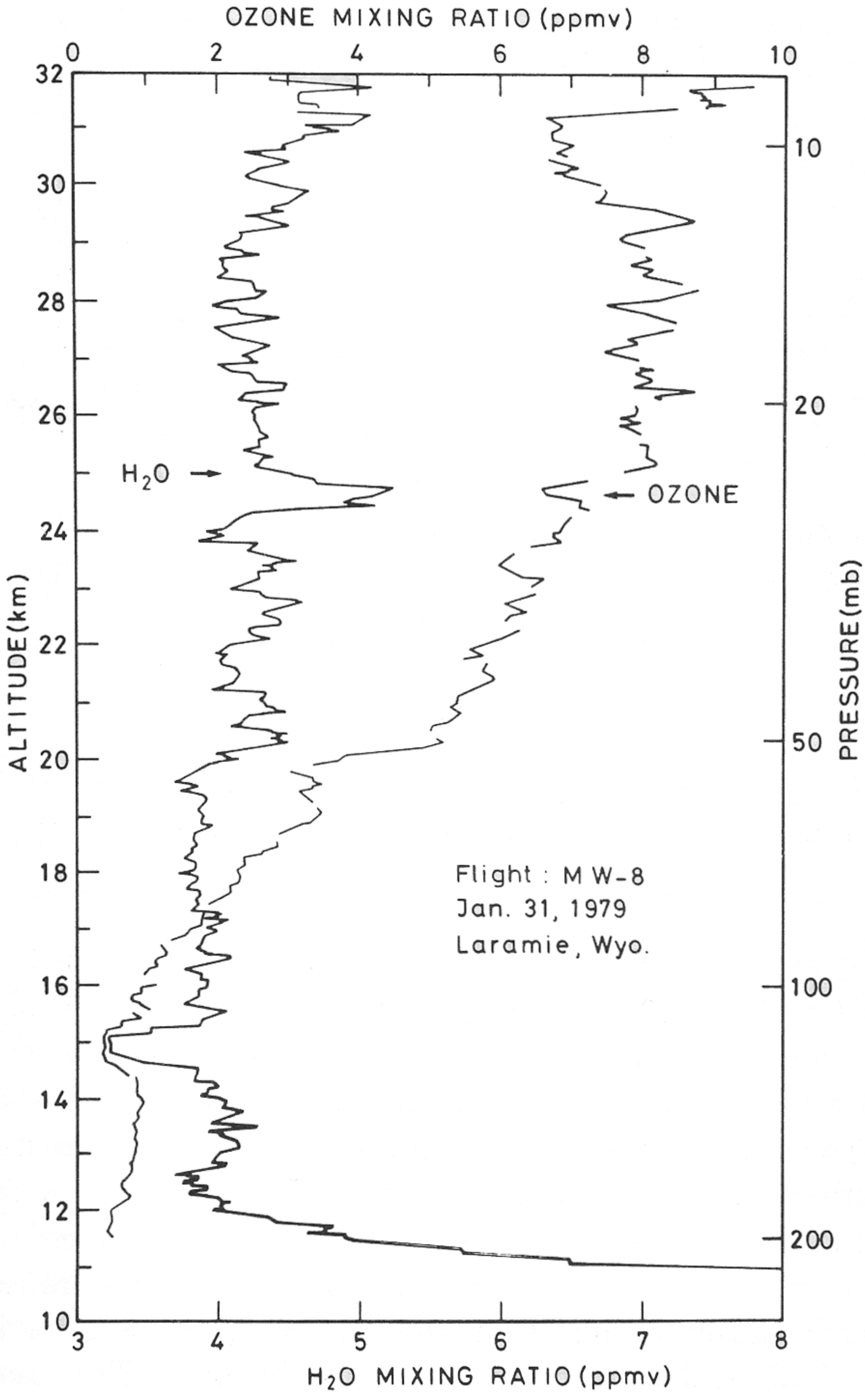
The rate of fluorescence from the sampled air is then proportional to:

$$RF = \frac{[\mathrm{H_2O}]}{A + k_q[M]} \tag{5b.5}$$

A photomultiplier detects the resonantly scattered photons; calibration is necessary to determine various instrumental factors. Equation 5b.5 can be approximated as:

$$RF \approx \frac{[\mathrm{H_2O}]}{[M]\, k_q} \tag{5b.6}$$

showing that the measurement is, in principle, one of mixing ratio, not concentration. This makes the instrument sufficiently sensitive to measure in the upper stratosphere, where — although the absolute water concentration is much lower — $[M]$, and hence the quenching, is also much reduced.



OZONE MIXING RATIO (ppmv)
0
2
4
6
8
10
32
30
28
26
24
22
20
18
16
14
12
10
ALTITUDE (km)
10
20
50
100
200
PRESSURE (mb)
H_2O
OZONE
Flight : M W-8
Jan. 31, 1979
Laramie, Wyo.
3
4
5
6
7
8
H_2O MIXING RATIO (ppmv)

Figure

5b.12 — H_2O profile measured using a Lyman-α fluorescence hygrometer.

The measurements above were made on a balloon-borne instrument. The structure in the H_2O profile is consistent with the long lifetime of H_2O in the upper stratosphere and can be used to identify how long air has spent there; the gradual increase with altitude is due to methane oxidation.

Tunable diode laser spectroscopy (TDLS)

TDLS is used for absorption measurements in the infrared. It employs a laser light source of very narrow wavelength, tunable over a small range — narrow enough to resolve weak absorptions lying between, e.g., H_2O and CO_2 lines, so that small absorptions due to specific rotational lines in a rovibrational spectrum can be measured with high selectivity. Conversely, it is difficult to measure many species simultaneously (unlike conventional FTIR spectroscopy). Lead-salt lasers have often been used, with tuning usually accomplished by varying the temperature (~3 cm^{-1} per K); folded cells are usually used to increase the absorption path. Successful measurements include NO_2, CH_4, HNO_3, HCHO and SO_2 — for example, the diurnal-cycle measurements of NO_2 discussed in section 4.3 were made by TDLS.

ClO (and other radicals) by resonance fluorescence

Radical concentrations determine the partitioning between different species in the atmosphere and thus control the ozone photochemical balance. However, because of their small concentrations and high reactivities, few measurements of important atmospheric radicals have been made. One successful technique is resonance fluorescence — sometimes preceded by chemical conversion — which, by its nature, is limited to in situ approaches. It involves exciting an allowed electronic transition of the species of interest and detecting any resonantly scattered photons; the number of scattered photons is related to the concentration of scattering molecules by laboratory calibration.

Anderson made measurements of the hydroxyl radical by direct fluorescence using the $A^2\Sigma \leftarrow X^2\Pi$ transition of OH at 308 nm, with a balloon-borne instrument providing profiles up to 40 km.

For ClO, resonance fluorescence cannot be applied directly, since ClO pre-dissociates on a time-scale shorter than that for re-emission of a photon. Instead, ClO is converted to chlorine atoms by injecting nitric oxide:

$$\text{ClO} + \text{NO} \rightarrow \text{Cl} + \text{NO}_2$$

Using the apparatus described, the conversion efficiency to chlorine atoms is around 95%. Resonant scattering from the ${}^2P_{3/2} - {}^2D_{5/2}$ Cl transition at 118.9 nm is then detected in the normal way; BrO is detected similarly.

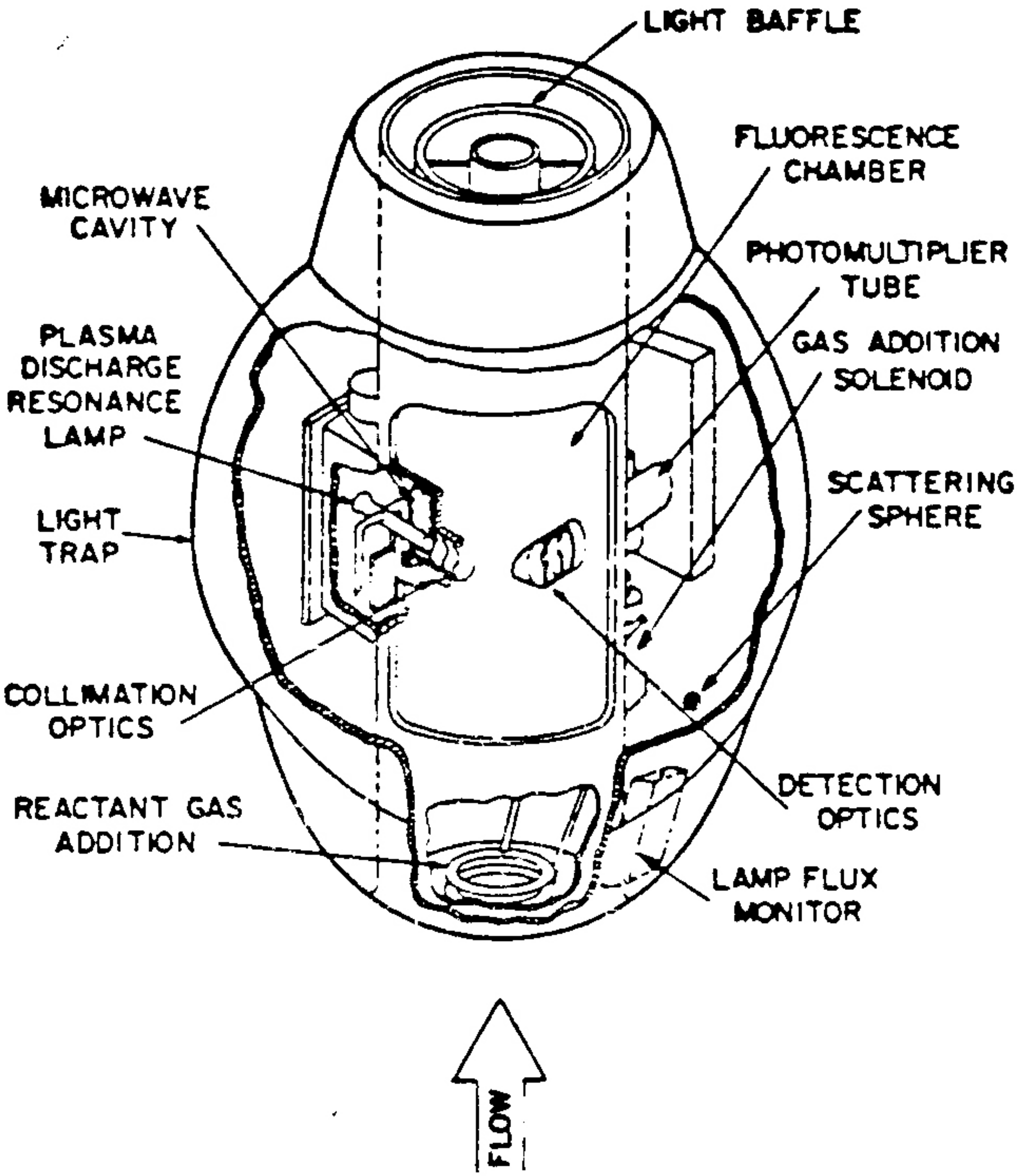


Figure 5b.13 — The balloon-borne ClO/BrO resonance fluorescence detector (Anderson). An aircraft version also exists.

Measurement of OH

OH is a vital atmospheric species: it "cleanses" the troposphere as the main daytime oxidant (see Dr Kalberer's lectures), and plays a major role in stratospheric chemistry. It is very short-lived and difficult to measure.

Resonance fluorescence. Techniques similar to the stratospheric OH measurements described above have been employed in the troposphere, exploiting the $A^2\Sigma^+ \leftarrow X^2\Pi$ transition around 308 nm. A complication is that this radiation can itself produce OH, via photolysis of O_3 in the presence of water vapour; this contamination can be reduced by measuring at low pressure.

Differential optical absorption spectroscopy. The strongly banded structure of the $A^2\Sigma^+ \leftarrow X^2\Pi$ transition makes it an excellent candidate for DOAS measurements. OH has been measured this way using multipass cells (total path ~2 km) or open double-pass instruments with a source and retro-reflector some kilometres apart. Sensitivity tends to be moderate (~10^6 cm^{-3}), owing to reflective losses and the "averaging" of concentration along long paths — but this is adequate for daytime measurements in the presence of significant OH sources.

Global inferral. OH concentrations have also been derived by estimating the lifetimes of species — such as CH_3CCl_3 — whose only sink is reaction with OH. A key question here is exactly what a "global" OH concentration means, since OH is very short-lived and expected to show significant spatial and temporal variability.

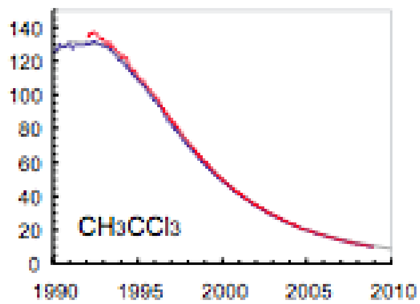


Figure 5b.14 — Observed decline in atmospheric CH_3CCl_3 (methyl chloroform) mixing ratio (ppt).

The figure shows the observed decline in atmospheric CH_3CCl_3 mixing ratios, reflecting the impact of controls under the Montreal Protocol. Emissions into the atmosphere are now essentially zero, so the observed change is entirely due to chemical loss by reaction with OH — making the decay rate itself a global-average OH "clock".

This concludes the Michaelmas term (Lectures 1–5, 5b). Next: Module 6 — Atmospheric Composition: Sources, Sinks and Lifetimes, opening the Lent term (Lectures 7–12).

Lent Term

Atmospheric Composition: Sources, Sinks and Lifetimes

Module 6 · Atmospheric Composition: Sources, Sinks and Lifetimes

Learning objectives By the end of this module you should be able to: 1. Outline the basic processes controlling trace gas and particle concentrations in the troposphere. 2. List the main natural and anthropogenic sources of tropospheric trace gases. 3. Describe the physical and chemical processes that remove trace constituents from the atmosphere. 4. Explain acid deposition and its effects on ecosystems. 5. Define and calculate the lifetime of an atmospheric trace gas.

1.1 Introduction

Chemistry in the Atmosphere is a two-part story. The first part of this course covered the stratosphere; this module opens the second part, which deals with the troposphere — the lowest part of the atmosphere, extending from the surface up to roughly 8–18 km depending on latitude.

Two regions matter here:

- Planetary boundary layer (PBL) — the lowest ~1 km, into which almost all surface emissions are injected. Under still (anticyclonic) conditions, pollutants can build up in the PBL, producing air pollution episodes.
- Free troposphere — the region above the PBL, extending to the tropopause. Mixing within the troposphere is relatively fast, but the tropopause is a fairly effective barrier to exchange with the stratosphere.

The troposphere is also where weather happens (clouds, precipitation), and radiative processes here dominate surface climate.

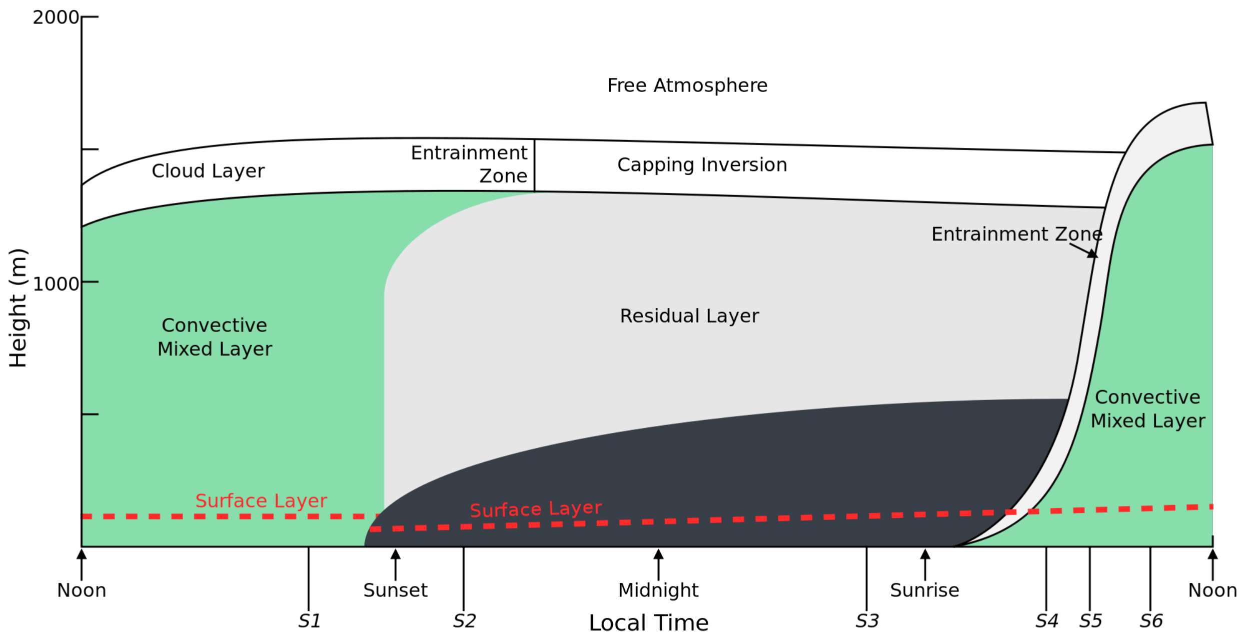


Figure 1.0 — Schematic of the diurnal variation in the planetary boundary layer (after Stull et al., 1990).

Historically, interest in tropospheric chemistry has been driven by air pollution — the effects of trace gases and particles on human health and ecosystems. Two classic pollution regimes are distinguished:

	Winter Smog	Photochemical (Summer) Smog
First recognised	Late 18th century	Mid-1940s, Los Angeles basin
Conditions	Winter, often foggy	High temperature, strong sunlight
Cause	Local pollutant accumulation (topography, temperature inversions)	High NO and organic emissions
Key emissions	High SO_2 and soot (coal burning)	Organics + NO_x + sunlight → O_3 + other products (Haagen-Smit, 1952)
Ozone formed?	No	Yes

This module — and the rest of the course — develops a quantitative understanding of the chemistry behind photochemical smog, so that we can predict how to minimise human exposure to harmful pollutants.

1.2 Atmospheric Composition

The composition of the atmosphere at any point in time reflects a balance between sources and sinks:

- Sources — primary emissions from atmospheric, land, or ocean processes; natural or anthropogenic.
- Sinks — removal by chemical or physical processes, operating on a wide range of timescales.

Because sources and sinks are spatially non-uniform, the resulting balance is complex, coupling chemistry with atmospheric transport. Quantifying this coupled system is a central aim of atmospheric chemistry.

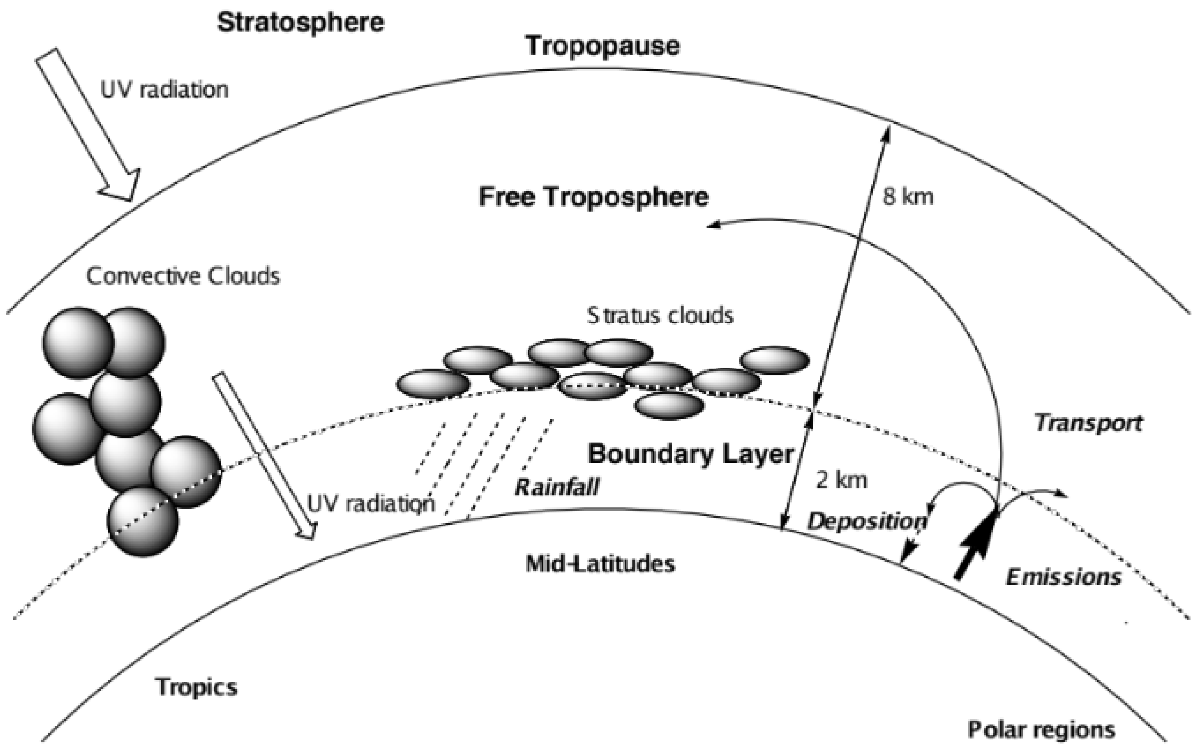


Figure 1.1 — Key features and processes in the troposphere.

About 90% of the total atmospheric mass resides in the troposphere, and most trace-gas emission (and a good deal of trace-gas formation) happens here. Chemically, the troposphere performs two important jobs:

1. Destruction — preventing surface-emitted gases from accumulating to toxic levels or reaching the stratosphere and damaging the ozone layer.
2. Transport and transformation — acting as the conduit for natural biogeochemical cycles.

Troposphere vs. stratosphere

Property	Stratosphere	Troposphere
Temperature	200–250 K	~288 K ("room temperature")
Pressure	1–100 mbar	≥1000 mbar
UV reaching this altitude	< 200 nm	> 290 nm
Mixing time	Months–years	Seconds–days
Dynamics	Large-scale	Turbulent to large-scale
Number of compounds	Limited	> 10^6
Dominant chemistry	Mostly inorganic (HO☐, NO☐, …)	Inorganic and organic, roughly equal importance
Water abundance	Low	High — clouds, precipitation

1.3 Sources of the Minor Constituents

Most trace gases in the troposphere originate as surface emissions.

Compound	Main sources	Emission rate (Tg yr^{-1})
CH_4	Wetlands; natural gas leakage; combustion	400–500
CO	Atmospheric VOC oxidation; combustion	800
Isoprene	Natural vegetation	500
VOC*	Solvents; combustion; fermentation; vegetation	≫100
NO	Soil microorganisms; lightning; combustion	40
N_2O	Soil/marine microorganisms; industry; combustion	4.4–10.5
NH_3	Animal waste breakdown; soil microorganisms	82
SO_2	DMS** oxidation; volcanoes; combustion; smelting	110
DMS	Marine microorganisms	40
CH_3Cl	Marine/terrestrial microorganisms	1.5
CH_3Br	Marine microorganisms; agriculture	0.1
CFCs/HCFCs	Solvents and refrigerants	1.1

* VOC = volatile organic compounds (hydrocarbons, halocarbons, oxygenated organics). ** DMS = dimethyl sulphide.

Natural emissions are mostly biogenic, though volcanism contributes significantly to atmospheric sulphur. Anthropogenic emissions arise from energy production, industry, and agriculture.

Chemical transformation of these primary emissions generates secondary products that can themselves be important — notably H_2SO_4, HNO_3, HCl, and a wide range of organic products from the oxidation of S-, N-, Cl-containing, or organic trace gases. Ozone (O_3), formed photochemically in the troposphere, is a particularly important example — central to tropospheric chemistry as well as the stratosphere.

1.4 Sinks of the Minor Constituents

Removal processes fall into two broad classes:

Sink process	Species removed
Physical — dry deposition to water/land	SO_2, O_3, HNO_3, CO_2, H_2O_2, aerosol particles
Physical — wet deposition in precipitation	HCl, H_2SO_4, NH_3, SO_2, HNO_3, H_2O_2, aerosol particles
Chemical — oxidation by OH	VOCs, CO, SO_2, NO_2, H_2O_2, H_2S, DMS
Chemical — direct photolysis	O_3, HCHO, CH_3I, H_2O_2, NO_2, NO_3
Chemical — cloud/aerosol reactions	H_2SO_4, NH_3, SO_2, HNO_3, VOCs

These sink strengths vary considerably with time of day, season, and geography.

1.4.1 Dry deposition

Removal at the Earth's surface (water, soil, vegetation) in the absence of precipitation. Governed by transport through the boundary layer and by uptake at the surface. The flux to the surface, F (molecules cm^{-2} s^{-1}), relates to the near-surface concentration c (molecules cm^{-3}) via the deposition velocity v_g (cm s^{-1}):

$$F = v_g \cdot c$$

(1.1)

The lifetime of a species with respect to dry deposition depends on v_g and the mixing height h:

$$\tau = \frac{h}{v_g}$$

(1.2)

Worked comparison:

- SO_2: $v_g \approx 0.8$ cm s^{-1} over NW Europe, mixed through $h \approx 1$ km → $\tau \approx 1.4$ days.
- O_3: similar v_g over land, but well-mixed up to the tropopause ($h \approx 12$ km) → $\tau \approx 17$ days.

The large difference is entirely down to the mixing height — the same deposition velocity gives a very different lifetime depending on how deep an atmospheric column the species is spread through.

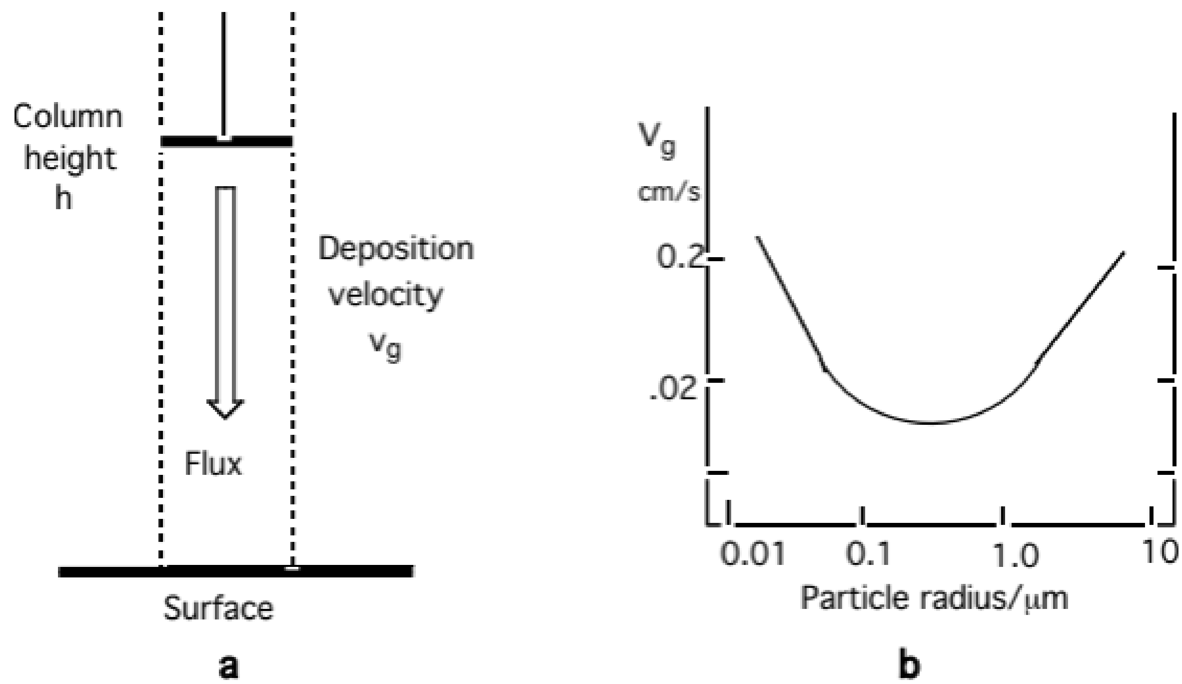


Figure 1.2 — Schematic illustration of surface deposition.

1.4.2 Wet deposition

Removal via precipitation (cloud, rain, snow):

- Rain-out / in-cloud scavenging — gases absorbed into droplets (possibly followed by reaction); particles act as condensation nuclei or are captured by coagulation.
- Wash-out / below-cloud scavenging — uptake into falling precipitation; removal times of order hours for soluble gases in moderate rain.

Most of these processes are reversible (e.g. droplet evaporation can release particles or gases back to the gas phase), and because precipitation is highly variable, quantitative estimates of wet deposition rates are difficult.

As an approximation, the wet deposition rate is taken as first-order in concentration:

$$\text{Rate} = \lambda c$$

where λ, the washout (scavenging) coefficient, scales with mean precipitation and is typically 0.12–0.5 day^{-1} (lifetime 2–8 days).

1.4.3 Acid deposition (wet and dry)

Oxidising conditions in the atmosphere (Modules 2–3) generate inorganic acids (H_2SO_4, HNO_3, HCl) and organic acids (formic, acetic). These lower the pH of precipitation and are also removed by dry deposition.

"Acid rain" is not a modern discovery — Robert Boyle noted sulphur and acidity in rain in the 17th century — but it was only widely recognised as a fossil-fuel-driven pollution problem in the 1970s.

Ecological effects:

- Sulfate/nitrate deposition leaches nutrient cations (Ca, Mg, K) from foliage and soils, reducing buffering capacity and nutrient availability for trees.
- Mobilises soil aluminium, interfering with root nutrient uptake; combined with O_3 toxicity, this leaves trees more vulnerable to drought, temperature extremes and disease.
- Leached aluminium and low pH are toxic to fish and other aquatic species.
- SO_2/NO_x oxidation products damage buildings and cultural heritage — here dry deposition is usually the dominant pathway.

Regulation has cut SO_2 and NO_x emissions substantially in recent decades (SO_2 more than NO_x), but both remain major anthropogenic air-quality pollutants.

1.5 Lifetimes — putting it together

The atmospheric lifetime (or residence time) τ of a species is the average time a molecule spends in the atmosphere before removal. For a species at steady state with a single first-order loss process of rate constant k_{loss}:

$$\tau = \frac{1}{k_{\text{loss}}}$$

More generally, for a reservoir of total burden N (molecules) with a total removal rate R (molecules s^{-1}):

$$\tau = \frac{N}{R}$$

When several independent loss processes act in parallel (e.g. OH oxidation and dry deposition), the loss rates add, so the lifetimes combine as reciprocals:

$$\frac{1}{\tau_{\text{total}}} = \frac{1}{\tau_1} + \frac{1}{\tau_2} + \cdots$$

— the fastest process dominates the overall lifetime.

> Key points — Module 1 - Atmospheric composition is set by the balance of sources and sinks. - Sources: mainly surface emission, but some species (notably O_3) are produced photochemically in the atmosphere itself. - Sinks: physical (dry/wet deposition) and chemical (oxidation, photolysis)

removal, with widely varying efficiency and timescale depending on the species. - Lifetime ties source/sink strength to the standing concentration, and is essential for comparing how "long-lived" different pollutants are.

Worked Example 1 — Steady-state concentration of compound X

Problem. Compound X is emitted at the surface at a rate of 1 tonne yr^{-1} (molar mass 44 g mol^{-1}). It is lost by reaction with OH, with rate constant $k = 1 \times 10^{-11}$ cm^3 $molecule^{-1}$ s^{-1}, and by dry deposition with $v_g = 1$ cm s^{-1}. Assume $[\text{OH}] = 10^6$ cm^{-3} and a boundary-layer mixing height of $h = 1$ km.

Method. 1. Convert the emission rate to a molecular flux (molecules cm^{-2} s^{-1}) using the molar mass and an assumed emission area, or work in a well-mixed box of height h to get a volumetric production rate (molecules cm^{-3} s^{-1}). 2. Write the two loss rate constants: chemical, $k_{\text{OH}} = k[\text{OH}]$; physical, $k_{\text{dep}} = v_g/h$. 3. At steady state, production = total loss: $P = (k_{\text{OH}} + k_{\text{dep}})[\text{X}]_{ss}$. 4. Solve for $[\text{X}]_{ss}$, and use $\tau = 1/(k_{\text{OH}} + k_{\text{dep}})$ to sanity-check against the emission rate.

This is exactly the kind of steady-state / pseudo-first-order problem you'll meet repeatedly through the course — it's worth being fluent doing it both by hand and by setting up the balance numerically. The companion notebook (`06-sources-sinks-lifetimes.ipynb`) walks through this example step by step, and lets you explore how $[\text{X}]_{ss}$ and τ respond to changing k, $[\text{OH}]$, v_g and h.

Try it yourself

Open `notebooks/06-sources-sinks-lifetimes.ipynb` to:

- Solve Worked Example 1 numerically and check it against the by-hand answer.
- Build a simple one-box steady-state / time-dependent model with combined chemical + deposition loss.
- Explore how lifetime and steady-state concentration change as you vary k_{OH}, $[\text{OH}]$, v_g, and mixing height h.
- Reproduce the SO_2 vs. O_3 dry-deposition lifetime comparison from §1.4.1.

Next: Module 7 — Tropospheric Photochemistry and the Hydroxyl Radical

Tropospheric Photochemistry and the Hydroxyl Radical

Module 7 · Tropospheric Photochemistry and the Hydroxyl Radical

Aims of this module 1. To describe the basic characteristics of photolysis in the troposphere. 2. To outline the processes that result in a steady-state concentration of OH.

2.1 Introduction

In this module we introduce the basic photochemistry of the troposphere and show how photochemical reactions are the main driver leading to oxidation of trace gases — preventing their accumulation and pollution of the atmosphere.

As discussed in the first part of the lecture course, the rate of photolysis is given by:

$$J_{(i)} = \int_{\lambda=0}^{\infty} \phi_{(i,\lambda)} F_{(\lambda)} \sigma_{(i,\lambda)} d\lambda$$

(2.1)

where J is the photolysis frequency (s^{-1}), obtained by integrating the product of the solar flux F, the absorption cross-section σ, and the quantum yield for dissociation ϕ, over all wavelengths λ where the molecule absorbs. Determination of the solar flux in the atmosphere requires either direct measurement, or a calculation including absorption, scattering and reflection of solar radiation entering the atmosphere.

2.2 Tropospheric Photolysis

The altitude dependence of photolysis rates was discussed in previous lectures. In spectral regions where atmospheric absorption by O_2 and O_3 is strong (λ shorter than ~290 nm), virtually no solar photons penetrate to the troposphere, while at longer (visible) wavelengths, where absorption is weak, the atmosphere is nearly transparent. The result: the photolysis environment in the troposphere differs dramatically from that of the stratosphere. Only molecules that absorb light in the near-UV/visible can be photo-dissociated in the troposphere — with important implications for tropospheric photochemistry.

One consequence: photolysis of O_2, which requires photons with wavelength below the threshold for breaking the O–O bond,

$$O_2 + h\nu\,(\lambda < 242\text{ nm}) \rightarrow O + O$$

(R2.1)

cannot occur in the troposphere, so ozone cannot be produced there via the Chapman mechanism.

However, species such as O_3, NO_2, H_2O_2, HCHO (and other organic peroxides and aldehydes), which absorb at longer wavelengths, can be photo-dissociated in the troposphere — with important consequences.

For photolysis to occur there must be spectral overlap with the solar spectrum, and dissociation must occur with non-zero quantum yield — usually meaning the photon energy $h\nu$ must exceed the dissociation energy ε_0. Some important tropospheric photolysis processes are given in Table 2.1, with representative photolysis coefficients for the lower atmosphere. Ozone photolysis is particularly important and is considered in detail below.

Table 2.1 — Important photolysis rates in the lower atmosphere

Photochemical process	ε_0/kJ mol^{-1}	$\lambda_{threshold}$/nm	J_x (Z=30°)*/s^{-1}
$O_2 \rightarrow 2\ O(^3P)$	494	242	0
$O_3 \rightarrow O(^3P) + O_2(^3\Sigma)$	101	1180	4.17×10^{-4}
$O_3 \rightarrow O(^1D) + O_2(^1\Delta)$	386	310	2.73×10^{-5}
$SO_2 \rightarrow O(^3P) + SO$	552	216	0
$NO_2 \rightarrow O(^3P) + NO$	300	398	1.6×10^{-2}
$H_2O_2 \rightarrow 2\ OH$	215	557	6.80×10^{-6}
$HONO \rightarrow NO + OH$	208	572	1.83×10^{-3}
$HCHO \rightarrow H + HCO$	364	329	2.77×10^{-5}
$HCHO \rightarrow H_2 + CO$	−1.9	–	4.41×10^{-5}
$CH_3CHO \rightarrow HCO + CH_3$	349	343	3.84×10^{-6}
$CH_3COCHO \rightarrow HCO + CH_3CO$	284	421	1.18×10^{-4}
$CH_3OOH \rightarrow OH + CH_3O$	188	637	5.02×10^{-6}

*J values calculated for the Earth's surface, summer N. mid-latitude.

2.3 Photolysis of ozone in the troposphere

Above 240 nm, O_2 photolysis is irrelevant, and in the 240–320 nm range O_3 is the dominant absorber. Absorption by ozone occurs in two spectral regions in the troposphere: the visible (450–600 nm) and the UV (290–330 nm). Photolysis of O_3 occurs by two pathways:

$$O_3 + h\nu \rightarrow O(^3P) + O_2 \qquad \lambda < 1150 \text{ nm}$$

(R2.2)

$$O_3 + h\nu \rightarrow O(^1D) + O_2 \qquad \lambda < 310 \text{ nm}$$

(R2.3)

Photolysis at the longer wavelengths (R2.2) occurs in the visible, yielding ground-state $O(^3P)$, followed by recombination with O_2 to reform O_3:

$$O(^3P) + O_2 + M \rightarrow O_3 + M$$

(R2.4)

— this process has no net effect on ozone concentration.

Photolysis process R2.3 yields excited oxygen atoms, $O(^1D)$. The flux of photons in the troposphere at wavelengths shorter than 310 nm is rather low, due to attenuation of UV radiation by ozone absorption higher up — this decreases the photolysis rate and $O(^1D)$ production.

At the high pressures found in the troposphere, O(¹D) is rapidly quenched to ground-state O(³P). If these were the only processes, there would be no net destruction of ozone by tropospheric photolysis, and the troposphere was initially thought to be rather chemically inert — with tropospheric ozone believed to result from direct transport from the stratosphere followed by dry deposition. Subsequent research showed this is incorrect.

The photolysis process R2.3 is in fact central to the chemistry of the troposphere, as it is the primary step in generating OH radicals — which, as we shall see, initiate oxidative chemistry.

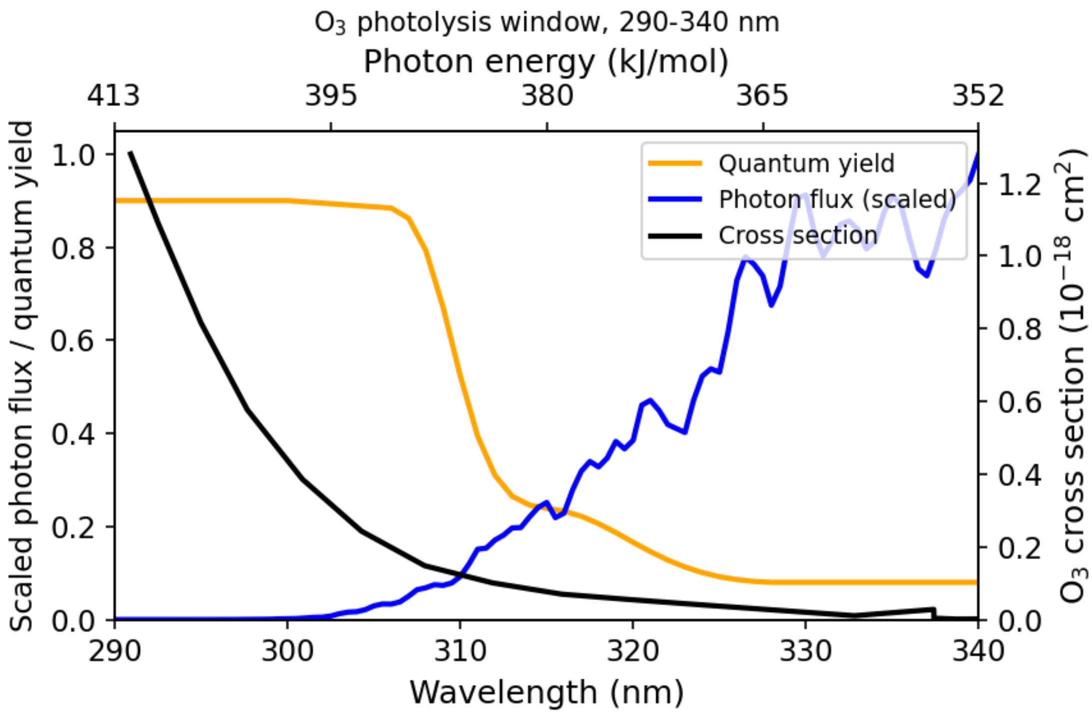


Figure 2.2a — Cross-section, quantum yield for reaction R2.3, and solar photon flux at the Earth's surface.

2.3.1 Fate of O(¹D) in a dry atmosphere

A small amount of highly reactive excited atomic oxygen O(¹D) is produced in the troposphere by photolysis of O_3 at $\lambda \lesssim 310$ nm. The rate of photolysis of ozone to produce O(¹D) is typically written $J_{O(^1D)}$, incorporating the summation over all wavelengths and the wavelength dependence of quantum yield.

O(¹D) is quenched to the ground state by reaction with N_2 or O_2:

$$O_3 + h\nu \rightarrow O(^1D) + O_2 \qquad J_{2.3} = 2.5 \times 10^{-6}\ \text{s}^{-1}$$

$$O(^1D) + M \rightarrow O(^3P) + M \qquad k_{2.5} = 3 \times 10^{-11}\ \text{cm}^3\text{s}^{-1}$$

where $M = [N_2] + [O_2] = 2.4 \times 10^{19}$ cm⁻³ at the Earth's surface.

Assuming steady state, $[O(^1D)] = \dfrac{J_{2.3}[O_3]}{k_{2.5}[M]}$.

2.4 Generation of OH radicals

Water vapour is abundant throughout most of the troposphere, its mixing ratio falling from ~1–4% at the surface to ~20 ppmv near the tropopause. H_2O competes effectively with quenching by M for reaction with $O(^1D)$, giving a branching ratio, f, between quenching and OH formation of about 10%:

$$O_3 + h\nu \rightarrow O(^1D) + O_2$$

(R2.3)

$$O(^1D) + M \rightarrow O(^3P) + M$$

(R2.5)

$$O(^1D) + H_2O \rightarrow OH + OH$$

(R2.6)

Reaction R2.6 is exothermic ($\Delta H^0 = -119$ kJ/mol), while the corresponding reaction of H_2O with $O(^3P)$ does not proceed (it's endothermic). The difference in enthalpy is largely because the electronically excited state of atomic oxygen lies ~190 kJ/mol above the ground state.

R2.6 represents the principal (though not sole) mechanism for generating hydroxyl radicals in the troposphere, initiating further reaction and degradation of VOCs.

The diurnal behaviour of [OH] closely tracks $J_{O(^1D)}$ over the course of a day, measured in the unpolluted conditions typical of the Southern Hemisphere marine environment — demonstrating the fast conversion of $O(^1D)$ into OH and the dominance of R2.3/R2.5–R2.6 in OH production. Under sunny conditions, $J_{O(^1D)}$ reaches a maximum of about 3×10^{-5} s^{-1}.

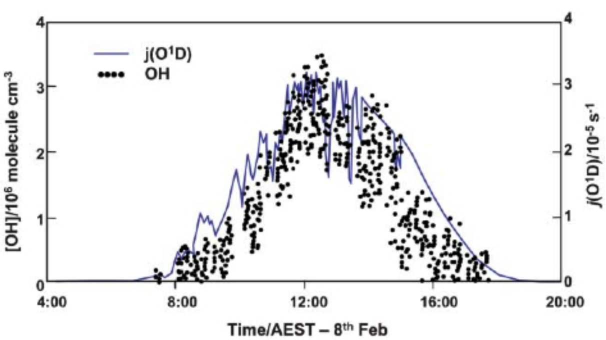


Figure 2.2b — Typical diurnal variation of [OH] and J(O¹D) in the clean marine atmosphere.

2.4.1 Other sources of OH in the troposphere

Although O_3 photolysis is the main OH source during the day, other sources matter in the early morning and at night. Photolysis of HONO (which builds up overnight) is an important OH source in the early morning:

$$HONO + h\nu \rightarrow NO + OH \qquad (\lambda < 400 \text{ nm})$$

Photolysis of organic compounds such as formaldehyde, and reaction of alkenes with O_3 and NO_3, also contribute to overall OH production (covered in Module 9).

2.5 The HOx family

OH does not react with the major atmospheric components (N_2, O_2, CO_2) but reacts with most trace gases.

Following its formation from O(^{1}D), OH is rapidly converted into HO_2 — primarily (~70% of the time) by reaction with CO, followed by fast reaction of H with O_2:

$$OH + CO \rightarrow CO_2 + H$$

(R2.7)

$$H + O_2 + M \rightarrow HO_2 + M$$

(R2.8)

About 30% of OH is converted to HO_2 via a more complex chain of reactions initiated by:

$$OH + CH_4 \rightarrow CH_3 + H_2O$$

(R2.9)

(details in Module 9). The lifetime of OH is less than a second — R2.9 sets an upper limit, since methane is widespread and well-mixed in the troposphere.

In relatively unpolluted (low-NOx) regimes, HO_2 reacts slowly with ozone to regenerate OH:

$$HO_2 + O_3 \rightarrow 2O_2 + OH$$

(R2.10)

R2.10 is an interconversion reaction between the two HOx forms, letting us meaningfully define the odd-hydrogen family $HO_x = [OH] + [HO_2]$.

Alternatively, HO_2 can recombine to form hydrogen peroxide:

$$HO_2 + HO_2 \rightarrow H_2O_2 + O_2$$

(R2.11)

H_2O_2 is a reservoir and sink for HOx: being highly soluble, it's removed from the atmosphere fairly rapidly by absorption into cloud water and rainout. H_2O_2 can also be photolysed ($\lambda < 400$ nm), releasing HOx, though this is rather slow in the troposphere; alternatively it reacts (slowly) with OH to form HO_2 and water.

The lifetime of HO_2 is about a minute in clean air, depending on the rates of R2.10 and R2.11. HOx concentrations react very rapidly to changes in the ozone photolysis rate.

In summary, reactions R2.3–R2.11 (representing the unpolluted atmosphere) lead to loss of O_3 and formation of peroxides like H_2O_2.

2.5.1 Steady-state concentration of OH and HO_2

Approximate values for [OH], [HO_2] and total [HOx] can be derived from R2.3–R2.11 together with $J_{O(^1D)}$.

Globally, $J_{O(^1D)}$ has a rough average (over solar zenith angle etc.) of 2.5×10^{-6} s^{-1}. With ozone at 40 ppbv, this gives an O(^{1}D) production rate from ozone photolysis such that O(^{1}D) is being formed at about 2.5×10^6 molecule cm^{-3} s^{-1}. As above, in the lower troposphere ~10% reacts with H_2O to form two OH (R2.6) — please do not commit this figure to memory! — giving an OH production rate of around 5×10^5 cm^{-3} s^{-1}.

OH is produced primarily from ozone photolysis, since the O_3 + HO_2 reaction (R2.10) is generally too slow to contribute significantly. OH is lost primarily by reaction with CO (R2.7). At steady state, production and destruction balance. Since $[CO] \sim 75$ ppbv on average and $k_{2.7} = 2.0 \times 10^{-13}$ at 1 atm, this leads to a steady-state OH concentration in the sunlit atmosphere of:

$$[\mathrm{OH}] = 1.4 \times 10^{6} \text{ molecule cm}^{-3}$$

In the clean troposphere, where [NO_2] is very small, it's safe to assume R2.11 is the main HO_2 loss route, with $k_{2.11} = 6.0 \times 10^{-12}$ cm^3 $molecule^{-1}$ s^{-1}. Putting HOx into steady state:

$$\frac{d[HO_x]}{dt}(\text{steady state}) = 0 = R_p(OH) - 2k_{2.11}[HO_2]^2$$

gives:

$$[\mathrm{HO_2}] = 2 \times 10^{8} \text{ molecule cm}^{-3}$$

Thus HO_2 is the dominant component of HOx. The ratio $[HO_2]/[OH]$ can be found from the interconversion reactions, e.g.:

$$k_{2.10}[HO_2][O_3] = k_{2.7}[OH][CO]$$

In general, the precise ratio depends on the composition of the air parcel — [O_3], [CO], etc.

Example 2 — steady-state [OH] in the tropical atmospheric boundary layer

In the tropics, [H_2O] is much higher than at mid-latitudes. The reduced overhead ozone column (see Michaelmas Module 1) also leads to increased rates of ozone photolysis. Work through the same steady-state derivation as above, but substituting tropical values for [H_2O], the overhead O_3 column (and hence $J_{O(^1D)}$), and [CO]/[O_3] — see the companion notebook.

2.6 Trace gas oxidation

OH is highly reactive with respect to many species, and is responsible for oxidising a range of tropospheric gases including NO_2, SO_2, CH_4 and nearly all VOCs, as well as CO. Reactivity varies greatly by hydrocarbon (Table 2.2). Saturated molecules react by H-atom abstraction, while unsaturated molecules react via electrophilic addition at the double/triple bond.

Oxidation of halocarbons (CH_3Cl, $CHCl_3$, C_2Cl_4, CH_3CCl_3) also occurs by H-abstraction and addition; rate constants tend to be slower than the analogous hydrocarbons, but this remains an important removal process for these potentially ozone-depleting substances in the troposphere. Other H-containing molecules (H_2S, NH_3) are similarly removed by abstraction reactions with OH.

Fully halogenated hydrocarbons do not react with OH (the abstraction reaction to form HOX, X = Cl, F, is endothermic). These and other OH-resistant gases (e.g. N_2O, OCS) are, absent energetic short-wavelength photons, inert in the troposphere, and so pass unimpeded into the stratosphere, where they are destroyed by UV photolysis.

Table 2.2 — Examples of trace gas reaction rates with OH

Molecule	Rate coefficient / cm^3 $molecule^{-1}$ s^{-1}	Atmospheric lifetime*
CH_4	~7×10^{-15}	~4.5 years
CH_3Cl	~4×10^{-14}	~0.8 year
CO	~2×10^{-13}	~2 months
C_2H_6	~3×10^{-13}	~1.5 months
C_3H_8	~1×10^{-12}	~11 days
C_2H_4	~8×10^{-12}	~1.5 days
C_5H_8 (isoprene)	~7×10^{-11}	~4 hours
CH_3SCH_3 (DMS)	~9×10^{-12}	~1.4 days
SO_2	~1×10^{-12}	~11 days
NO_2	~9×10^{-12}	~1.4 days
NH_3	~1.6×10^{-13}	~3 months
CCl_4	~1×10^{-17}	**

* Based on mean [OH] of 1×10^6 molecule cm^{-3}. ** Destroyed in the stratosphere — lifetime set by transport.

2.7 Observations of HOx radicals

Direct atmospheric measurements have convincingly confirmed the presence of OH in the daytime troposphere. Measurements in very clean atmospheres over the Southern Ocean confirm the basic mechanism, in which HOx radicals are predicted to have a square-root dependence on the ozone photolysis rate. The observations also show that ozone is slowly destroyed through photolysis, and that peroxides like H_2O_2 are formed.

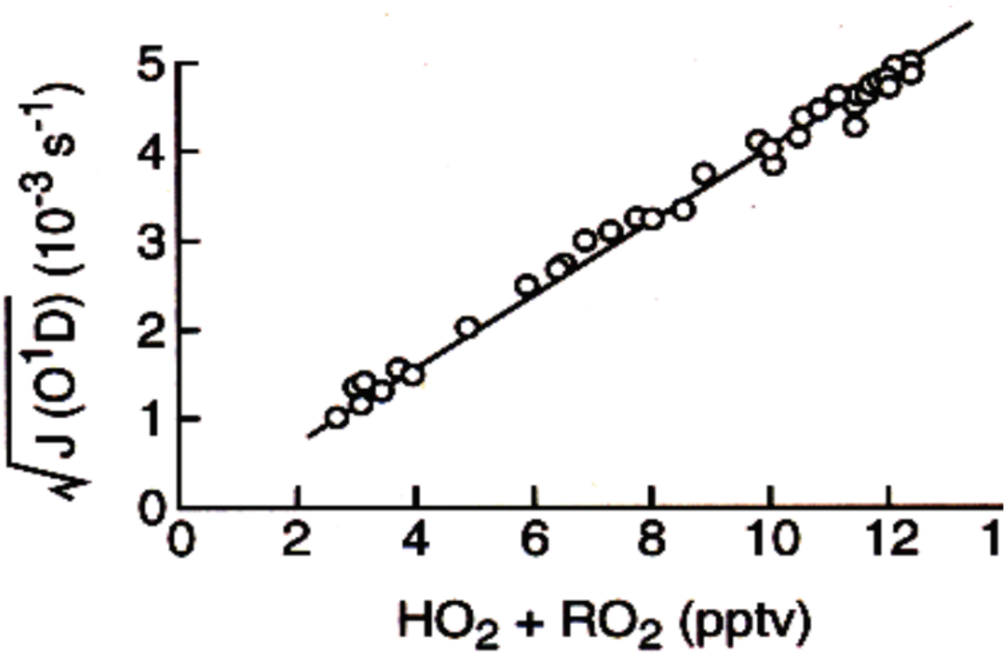


Figure 2.5 — Correlation plot of measured HO_2 (+RO_2) radical concentrations vs. calculated $J(O^1D)$, showing the dependence of peroxy radical concentration on the square root of $J(O^1D)$. Data from the Cape Grim observatory, Tasmania.

Key points — Module 7 - Photolysis of trace gases in the troposphere occurs only at wavelengths longer than ~300 nm. - OH and HO_2 form the HOx family. Key formation/loss processes and steady-state conditions involve reactions with CO, CH_4, NO and NO_2. - OH reacts with many trace gases, driving their oxidation. Its presence provides the oxidizing capacity of the atmosphere, and is used to calculate lifetimes of tropospheric trace gases.

Try it yourself

Open `notebooks/07-oh-steady-state.ipynb` to:

- Reproduce the [OH] = 1.4×10^6 cm^{-3} and [HO_2] = 2×10^8 cm^{-3} steady-state results from §2.5.1 numerically.
- Solve Example 2 (tropical boundary layer OH) by adjusting [H_2O], the O_3 column/$J_{O(^1D)}$, and [CO] to tropical values, and compare the resulting [OH] to the mid-latitude case.
- Explore the predicted square-root dependence of [HOx] on $J_{O(^1D)}$ (Fig 2.5) from the steady-state equations, and check it against the observational correlation.
- Rank trace gases by OH-reaction lifetime using Table 2.2, and see how lifetime scales with rate coefficient for fixed [OH].

Next: Module 8 — Methane Oxidation and NOx-Driven Ozone Production

Methane Oxidation and NOx-Driven Ozone Production

Module 8 · Methane Oxidation and NOx-Driven Ozone Production

Aims of this module 1. To describe the oxidation of atmospheric methane. 2. To understand, qualitatively, the effect of methane on tropospheric ozone. 3. To describe the role of NOx in production of tropospheric ozone. 4. To describe the processes controlling the concentration of nitrogen oxides in the troposphere.

3.1 Methane oxidation in the troposphere

Hydroxyl radicals react with methane, constituting the major removal process for atmospheric methane:

$$OH + CH_4 \rightarrow H_2O + CH_3$$

(R3.1)

The lifetime of CH_4, based on the R3.1 rate coefficient of 6×10^{-15} cm^3 molecule^{-1} s^{-1} at 288 K and the average tropospheric [OH], is approximately 7 years. This means methane is well mixed in the troposphere, and its oxidation occurs wherever OH is present. Indeed, the concentration of methane sets an upper limit on the lifetime of OH in the troposphere of about one second.

In clean, unpolluted air, about 30% of OH is removed by reaction with CH_4 (the remainder reacts with CO). The methyl radical formed reacts instantly with O_2 to form the methyl peroxy radical, CH_3O_2:

$$CH_3 + O_2 + M \rightarrow CH_3O_2 + M$$

(R3.2)

In a low-NOx environment (unpolluted air), the principal fate of CH_3O_2 is reaction with HO_2 to produce methyl hydroperoxide, CH_3OOH:

$$HO_2 + CH_3O_2 \rightarrow CH_3OOH + O_2$$

(R3.3)

As in Module 7, peroxides are reservoirs for radicals, and methyl hydroperoxide — like H_2O_2 — is soluble and readily washed out, acting as a sink for HOx. Since HOx is formed from ozone photolysis, this washout makes the overall conversion of O_3 to peroxides (via reaction with methane) into a sink for O_3.

Photolysis of methyl hydroperoxide is also possible, releasing methoxy, CH_3O, and OH:

$$CH_3OOH + h\nu \rightarrow CH_3O + OH$$

(R3.4)

CH_3O reacts with O_2 to form formaldehyde plus HO_2:

$$CH_3O + O_2 \rightarrow HCHO + HO_2$$

(R3.5)

and formaldehyde is itself photolysed to produce more HO_2:

$$HCHO + h\nu \rightarrow HCO + H$$

(R3.6)

$$HCO + O_2 \rightarrow CO + HO_2$$

(R3.7)

This recycles, and in certain circumstances (such as the dry upper troposphere) may even amplify [HOx].

Recall HO_2 converts back to OH by reaction with ozone:

$$HO_2 + O_3 \rightarrow OH + 2O_2$$

(R3.8 / R2.10)

However, the efficiency of this OH recycling is rather low in the clean lower troposphere: both CH_3OOH and HCHO are soluble in water droplets, leading to a net loss of HOx through rainout. R3.8 is competitive with the HO_2 self-reaction at ozone concentrations typical of the clean troposphere, and will also contribute to O_3 removal.

To recap: R3.3, together with loss of peroxides (or HCHO) in rain, leads to a net loss of HOx radicals. As ozone is lost in the process of generating HOx from $O(^1D)$, there is a net loss of ozone overall — in the unpolluted (low-NOx) troposphere.

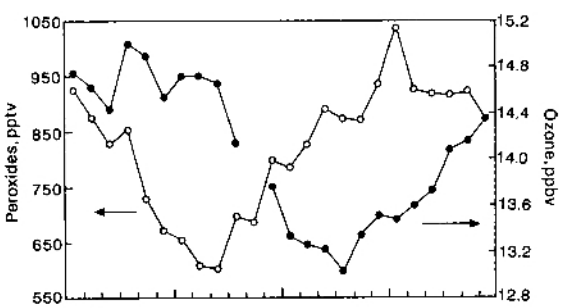


Figure 3.2 — Diurnal variation of peroxides ROOH (open symbols) and ozone (filled symbols) in the clean marine troposphere, showing production of peroxides and destruction of ozone during daytime (highlighted region).

3.2 Nitrogen oxides and production of tropospheric ozone

Much of the troposphere contains traces of NO and NO_2, which play an important role in tropospheric photochemistry. We've seen that in the unpolluted troposphere, oxidation of CO and CH_4 to CO_2 leads to a loss of both ozone and OH. The presence of nitrogen oxides changes this picture dramatically. In a polluted atmosphere, additional reactions must be considered to determine the overall oxidant budget.

NOx is emitted/generated from a number of surface and tropospheric processes. Major anthropogenic sources: fossil fuel combustion (mostly ground-level, with a small aviation contribution) and agricultural activities. Dominant natural sources: biomass burning (though much is human-caused, especially in the tropics) and lightning (associated with deep convection, largest sources in the tropics). Only lightning and aircraft are non-surface sources.

NOx is removed from the atmosphere through dry and wet deposition, with wet deposition of nitrate (nitric acid) the dominant process.

3.2.1 Photostationary state in the absence of peroxy radicals

Consider first the reactions of NO, NO_2 and O_3 in isolation. NO_2 is photolysed to $O(^3P)$, which reacts almost exclusively with O_2 to form O_3 (O-atom reactions with other pollutants don't compete under normal tropospheric conditions). O_3 then reacts with NO, reforming NO_2:

$$NO_2 + h\nu \rightarrow NO + O$$

(R3.10)

$$O + O_2 + M \rightarrow O_3 + M$$

(R3.11, very fast)

$$NO + O_3 \rightarrow NO_2 + O_2$$

(R3.12)

This coupled cycle establishes a photostationary state, with $[O_3]$ set by the NO_2/NO ratio via the Leighton relationship:

$$\frac{d[O_3]}{dt} = J_{3.10}[NO_2] - k_{3.12}[NO][O_3]$$

At steady state:

$$[O_3]_{ss} = \frac{J_{3.10}[NO_2]}{k_{3.12}[NO]}$$

Example 3 — timescale for establishment of the photostationary state

As in Modules 2 and 4, the timescale to reach photostationary state between NO, NO_2 and O_3 is set by the sum of the pseudo-first-order forward and backward rates linking the species — here approximately $(J_{3.10} + k_{3.12}[O_3])^{-1}$. Work through this calculation with typical boundary-layer $J_{3.10}$ and $[O_3]$ in the notebook: the PSS is normally established on a timescale of order 100 seconds — fast compared to the hours/days over which NOx and O_3 budgets otherwise evolve.

3.2.2 Reaction of peroxy radicals with NO leads to O_3 production

In the real atmosphere, the NO:NO_2 ratio is perturbed by other oxidants — mostly hydroperoxyl and organic peroxide radicals — which also convert NO to NO_2 and lead to net ozone production.

In the daytime, NO reacts with HO_2, regenerating OH (R3.13) and producing NO_2:

$$HO_2 + NO \rightarrow OH + NO_2$$

(R3.13)

$$k_{3.13} = 8 \times 10^{-12}\ \text{cm}^3\text{molec}^{-1}\text{s}^{-1}$$

NO also reacts with CH_3O_2, leading to HO_2 formation:

$$CH_3O_2 + NO \rightarrow CH_3O + NO_2$$

(R3.14)

$$CH_3O + O_2 \rightarrow HCHO + HO_2$$

(R3.15)

Although R3.13/R3.14 are much slower than NO + O_3 ($[HO_2]$ and $[CH_3O_2] \ll [O_3]$ in the troposphere), they significantly perturb the photostationary state. This is because reaction of NO with peroxy radicals RO_2 (e.g. HO_2, CH_3O_2) oxidises NO to NO_2 without consuming O_3. Ozone production follows when the resulting NO_2 is photolysed and the O product combines with O_2.

3.3 Rate of production of ozone in the presence of alkanes and NOx

We've seen NO is oxidised by RO_2 radicals to NO_2 — a key ingredient of photochemical smog: conversion of NO to NO_2 via hydrocarbon oxidation drives ozone generation.

In the absence of hydrocarbons, the Leighton relationship tells us ozone and the NO/NO_2 ratio stay constant: every $NO \rightarrow NO_2$ conversion costs an O_3 molecule, and every $NO_2 \rightarrow NO$ conversion generates one. But when hydrocarbon oxidation is occurring, $NO \rightarrow NO_2$ conversion proceeds faster, shifting the NO/NO_2 ratio and increasing $[O_3]$. Increasing $[O_3]$ increases [OH], which, to a first approximation, increases the rate of VOC oxidation — a positive feedback.

We can derive the rate of O_3 production in the presence of RO_2 and NOx in two steps.

Step 1 — steady state for [O], from R3.10 and R3.11 (neglecting O_3 photolysis):

$$\frac{d[O]}{dt} = J_{3.10}[NO_2] - k_{3.11}[O][O_2][M] = 0 \quad \Rightarrow \quad [O]_{ss} = \frac{J_{3.10}[NO_2]}{k_{3.11}[O_2][M]}$$

Since essentially all O atoms proceed to O_3 via R3.11, the rate of O_3 formation from this branch is simply $J_{3.10}[NO_2]$.

Step 2 — steady state for [NO], now including the peroxy-radical oxidation pathway (R3.13/R3.14) alongside R3.12:

$$\frac{d[NO]}{dt} = J_{3.10}[NO_2] - k_{3.12}[NO][O_3] - k_{3.13}[RO_2][NO] = 0$$

(taking $[RO_2] = [HO_2] + [CH_3O_2] + \ldots$, and $k_{3.13}$ as a representative rate constant for all NO + peroxy-radical reactions.) Solving for [NO] and substituting back gives the net ozone production rate:

$$P(O_3) = k_{3.13}[RO_2][NO]$$

— i.e. the rate of ozone production is set by the rate at which peroxy radicals oxidise NO to NO_2. Thus, in the presence of NOx, ozone is produced rather than destroyed by the overall CH_4/CO oxidation cycle involving HOx radicals.

3.4 Formation of nitric acid and NOx lifetime

Depending on [NOx], the OH radical that initiates VOC oxidation (and hence ozone production) may have another fate: reaction with NO_2 to form nitric acid:

$$OH + NO_2 + M \rightarrow HNO_3 + M$$

(R3.16)

This reaction matters for two reasons: HNO_3 is a reservoir species for both HOx and NOx, and its formation terminates the free-radical chain leading to ozone production. HNO_3 also has a high deposition velocity over land and is highly soluble, so it's readily lost by wet and dry deposition. Thus, at high NOx levels, the OH + NO_2 reaction controls the net loss of HOx: a decrease is observed in the steady-state [HOx] at high NOx levels.

R3.12 ($NO + O_3 \rightarrow NO_2 + O_2$) is temperature-dependent: $k_{3.12} = 2 \times 10^{-12}$ exp$(-1400/T)$ cm^3 $molecule^{-1}$ s^{-1}. This, together with changes in [O_3] and [OH] between the boundary layer and the upper troposphere, affects the lifetime of NOx with altitude. Assuming only R3.10, R3.12 and R3.16 set the NOx lifetime:

$$\tau_{NOx} = \frac{[NO_x]}{k_{3.16}[NO_2][OH]}$$

Fig 3.3 shows this lifetime increasing to about 15 days in the tropical upper troposphere, versus around one day in the boundary layer.

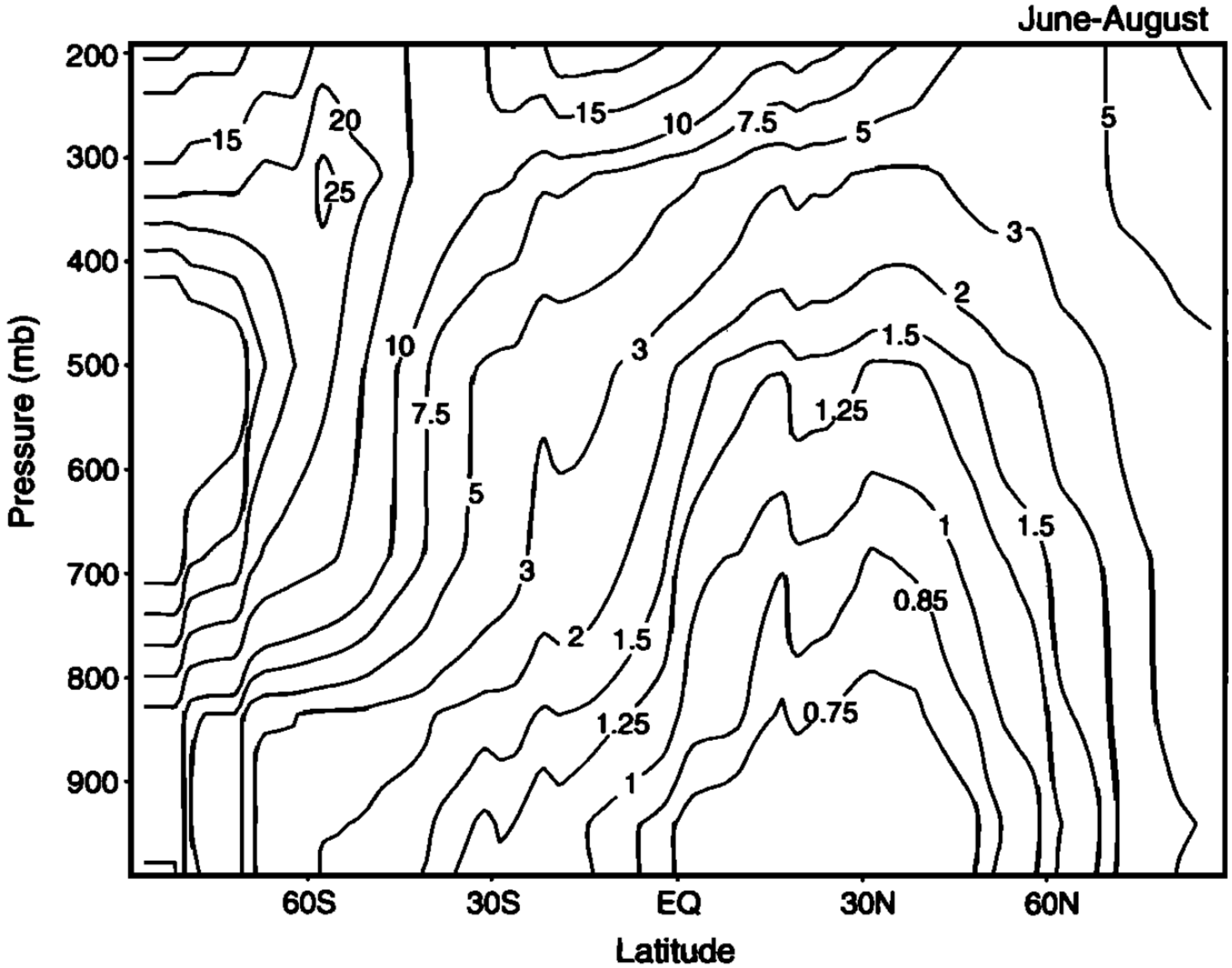


Figure 3.3 — Lifetime of NOx in the troposphere (days), as a function of latitude and height.

3.5 Ozone production under low and high NOx conditions

Consider CO oxidation once more. In the absence of NOx, overall O_3 destruction results:

$$OH + CO \rightarrow H + CO_2 \, (R2.7) \qquad H + O_2 + M \rightarrow HO_2 + M \, (R2.8) \qquad HO_2 + O_3 \rightarrow 2O_2 + OH \, (R2.10)$$

Overall: $O_3 + CO \rightarrow CO_2 + O_2$

Compare with ozone formation in the presence of NOx:

$$OH + CO \rightarrow H + CO_2 \, (R2.7) \qquad H + O_2 + M \rightarrow HO_2 + M \, (R2.8) \qquad HO_2 + NO \rightarrow OH + NO_2 \, (R3.13)$$

$$NO_2 + h\nu \rightarrow NO + O \, (R3.10) \qquad O + O_2 + M \rightarrow O_3 + M \, (R3.11)$$

Overall: $CO + 2O_2 + h\nu \rightarrow CO_2 + O_3$

Clearly, increasing [NOx] increases the rate of the ozone-production cycle. There's a critical NOx concentration at which production outweighs destruction.

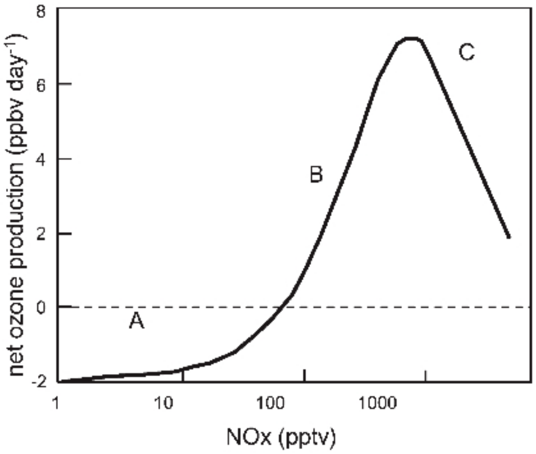


Figure 3.4 — Schematic representation of the dependence of net ozone production on NOx mixing ratio.

The figure divides into three regions, illustrating a strongly non-linear dependence of ozone production on NOx:

- Region A — ozone destruction dominates. As NOx increases, the production rate rises until it exceeds the (NOx-independent) destruction rate, and net production begins.
- The transition — where does it occur? To first approximation, treat HO_2 as the only peroxy radical. The transition occurs when enough NO is present for NO + HO_2 ($k_{3.13} \approx 8 \times 10^{-12}$ cm³ s⁻¹) to compete with the much slower O_3 + HO_2 ($k_{3.8} \approx 2 \times 10^{-15}$ cm³ s⁻¹). Equating these rates ([HO_2] cancels) gives $[NO] = k_{3.8}[O_3]/k_{3.13}$. For a typical [$O_3$] ~ 40 ppbv, this gives $[NO] \sim 0.01$ ppbv (10 pptv). Anthropogenic NO emissions (largely motor vehicles) significantly exceed this over much of the northern-hemisphere land area, so the ozone-production cycle dominates there. Even in clean oceanic air (Cape Grim, Tasmania), small amounts of NO suffice to cause net ozone production — photochemical ozone production is thus a global phenomenon. Only in remote marine areas ([NOx] < 10 pptv) does photochemistry act as a net O_3 sink.

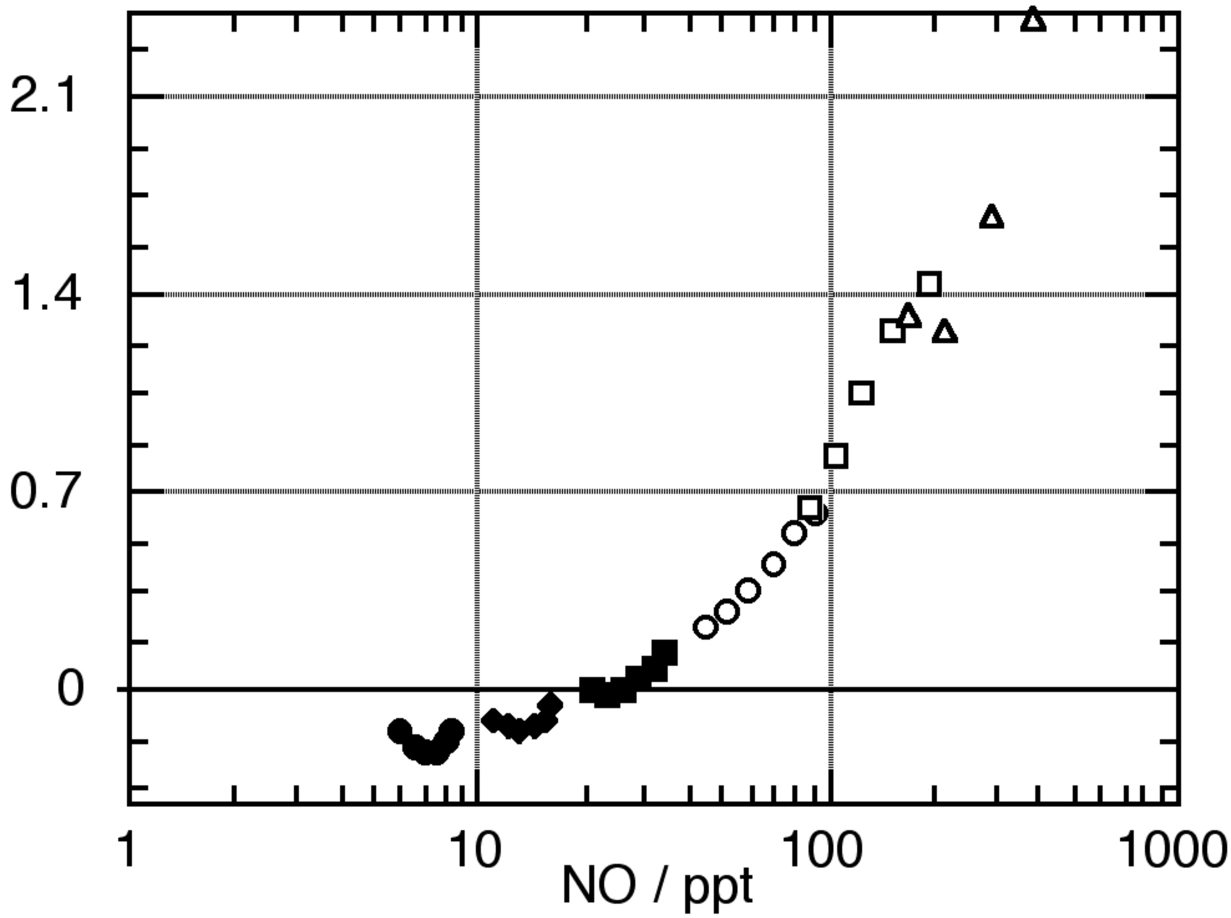
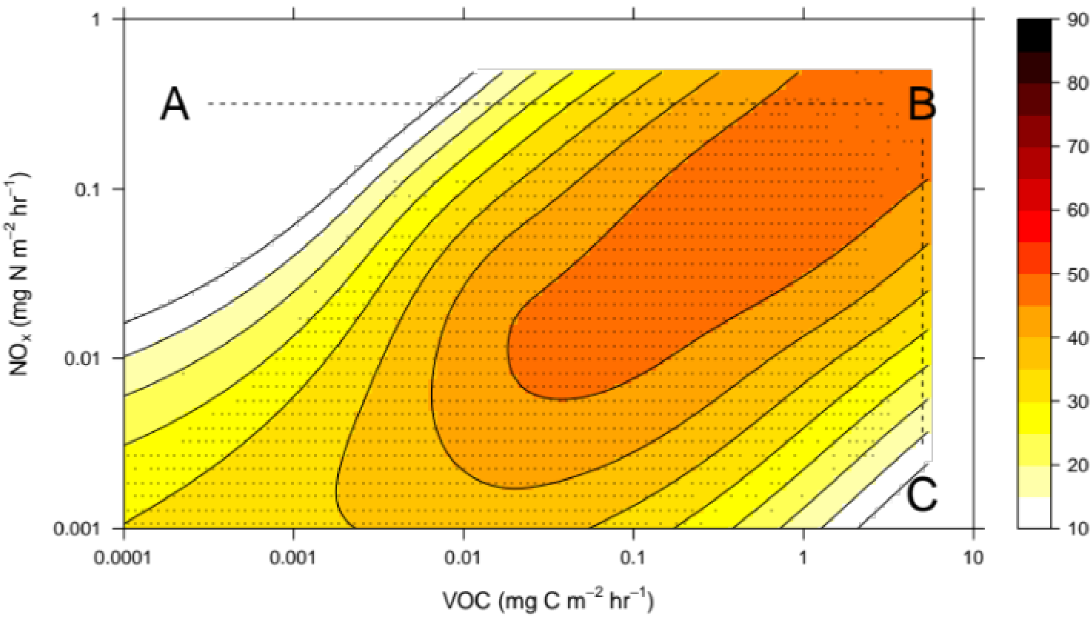


Figure 3.5 — Net ozone production as a function of [NO] (ppt), in clean oceanic air at Cape Grim, Tasmania.

- Ozone production does not keep increasing with NOx indefinitely — at high enough NOx, production peaks and then decreases. This turnover is caused by the increasing rate of the termination reaction R3.16: more NOx means more $OH + NO_2 \rightarrow HNO_3$, making the free-radical-catalysed production cycle less efficient.

This complex mechanism produces a strongly non-linear relationship between ozone amounts and precursor emission rates — an ozone “isopleth” surface in NOx–VOC emission space:



A
B
C
1
0.1
0.01
0.001
0.0001
0.001
0.01
0.1
1
10
NO_x (mg N m^{-2} hr^{-1})
VOC (mg C m^{-2} hr^{-1})
90
80
70
60
50
40
30
20
10

Figure 3.6 — Modelled ozone mixing ratio (ppb) as a function of NOx emission rate and VOC emission rate.

Quite different regulatory responses are needed depending on where a region sits on this isopleth surface. For example: at point B, reducing NOx emissions without reducing VOC (path B→C) has limited effect; reducing VOC alone (path B→A) would require ~3 orders of magnitude reduction to see significant effects. There is no magic bullet — the chemistry is complex! More detailed analysis suggests the best results come from tighter regulation of the most reactive hydrocarbons combined with NOx reduction. (Note: not all VOCs are anthropogenic — isoprene, for example, is biogenic, emitted from certain plants and trees, at an emission rate comparable to CH_4.)

An increase in NOx at point C leads to ozone production — this regime is 'NOx-limited'. At point A, adding VOCs leads to ozone production — this regime is 'VOC-limited'. Much of the industrialised northern hemisphere is VOC-limited (high ambient NOx); many tropical and southern-hemisphere locations are NOx-limited. Predicting the atmospheric response to emissions policy is consequently very complex.

Try it yourself

Open `notebooks/08-photostationary-state.ipynb` to:

- Solve Example 3 (photostationary-state establishment timescale) and confirm it's fast (~100 s) relative to NOx/O_3 budget timescales.
- Implement the Leighton relationship and reproduce how $[O_3]_{ss}$ depends on the NO_2/NO ratio.
- Compute the low-NOx/high-NOx transition [NO] (§3.5) and reproduce the qualitative shape of Fig 3.4/3.5 as a function of [NO], including the high-NOx turnover from R3.16.
- Build a simple ozone-isopleth calculator as a function of NOx and VOC emission rates, and locate NOx-limited vs. VOC-limited regimes.

Next: Module 9 — The Tropospheric Ozone Budget, PAN and NOx Reservoirs

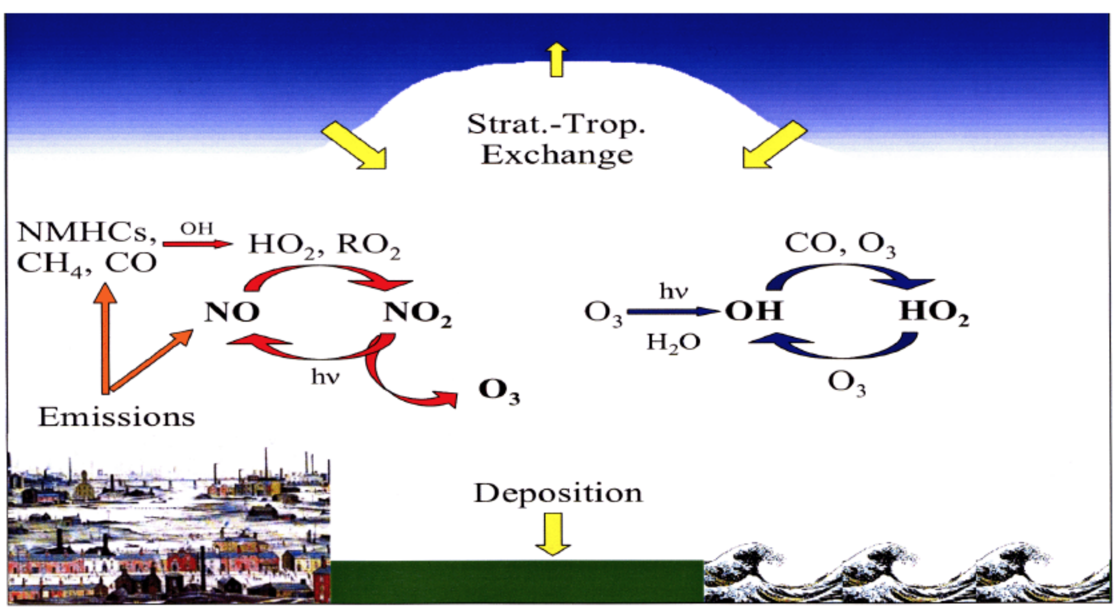
The Tropospheric Ozone Budget, Carbonyl Formation and NOx Reservoirs

Module 9 · The Tropospheric Ozone Budget, Carbonyl Formation and NOx Reservoirs

Learning aims By the end of this module you should be able to: 1. Describe the main terms in the global tropospheric ozone budget (stratospheric input, chemical production/loss, surface deposition) and explain why the net budget is so sensitive to the large gross chemical terms. 2. Describe how formaldehyde and other carbonyls are formed during NOx-driven VOC oxidation, and their role in recycling HOx. 3. Describe the formation, stability and atmospheric role of the main NOx reservoir species: HONO, PAN, organic nitrates, and N_2O_5/NO_3. 4. Explain why NO_3 chemistry is particularly important at night.

3.6 The budget of tropospheric ozone

The main processes controlling tropospheric O_3 are shown schematically in Fig 3.7. There are four main terms: transport from the stratosphere (where the ozone mixing ratio is much higher than in the troposphere — a genuinely important source); in-situ photochemical production and destruction; and deposition at the surface. These terms approach balance in the annual mean, but are not independent — increased stratospheric flux, or increased photochemical production, might for example lead to additional O_3 deposition. Precursors of photochemical ozone production originate from both anthropogenic and biogenic processes.



Figure

3.7 — Schematic of the processes controlling tropospheric O_3.

Table 3.2 — approximate global budget of tropospheric ozone (Tg/year)

Stratospheric origin	Photochemical production − loss	Surface loss
400	800 (4000 − 3200)	−1200

Although the chemical production and destruction terms are individually large, their net effect is comparable in magnitude to the stratospheric influx and surface deposition. Despite these comparable net terms, photochemical O_3 production and loss are by far the largest gross terms — so any small trend (or error) in either has a disproportionately large impact on the tropospheric ozone burden (about 300 Tg).

O_3 destruction via $O(^1D) + H_2O$ dominates in the tropics and near the surface, where $[H_2O]$ is highest. O_3 surface deposition (expressed as a deposition velocity) depends strongly on surface type and is quite uncertain — some models use a deposition term half the size of others. Deposition to vegetated surfaces is much stronger than to ocean or snow/ice, but the dependence on vegetation type is still not well quantified.

Ozone concentrations in Europe and other industrialised regions increased strongly over roughly the last hundred years (Fig 3.8), due to increased anthropogenic activity and associated NOx/VOC emissions.

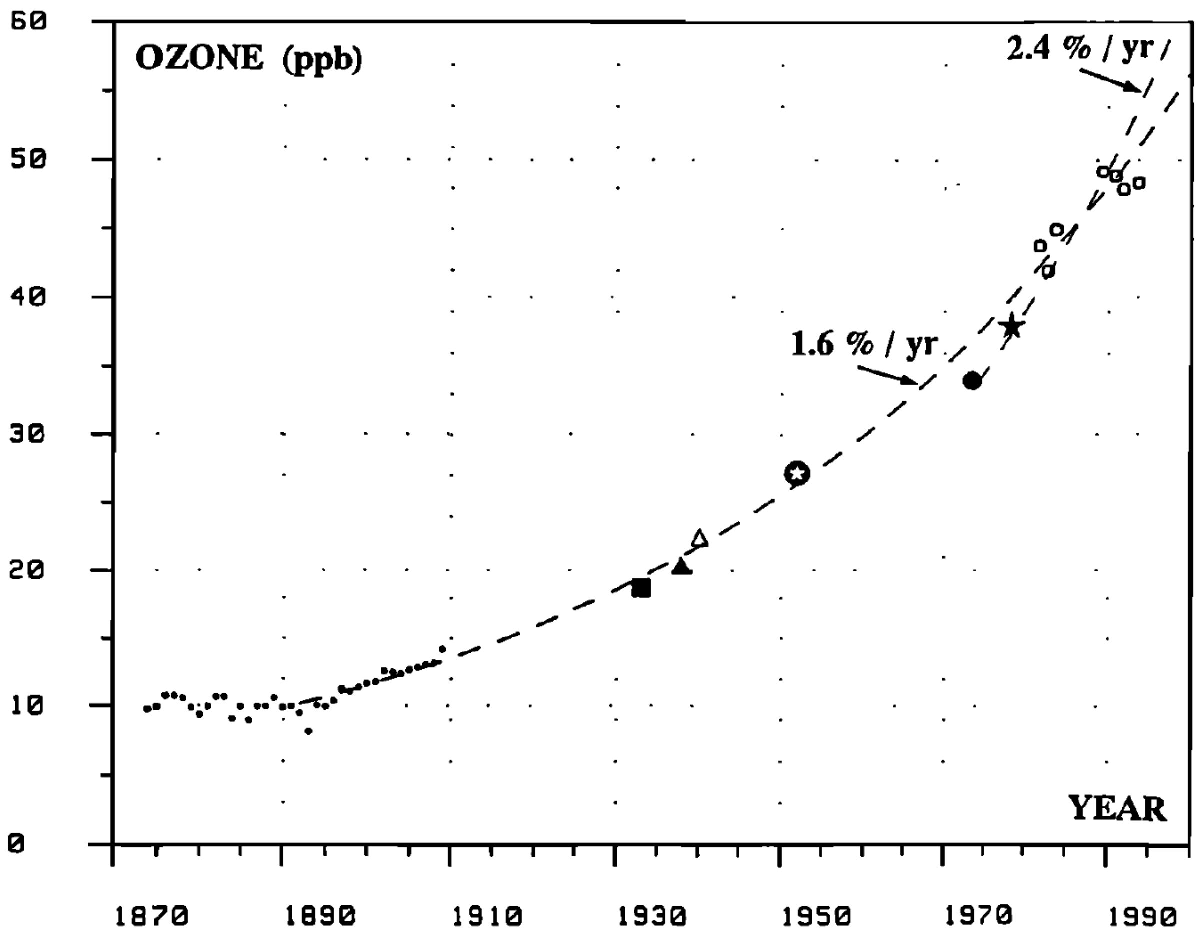


Figure 3.8 — Historical ozone concentrations in Paris since 1870. Annual average concentrations increased by a factor of 5–10 over this period.

3.7 Formation of carbonyl compounds

The reaction of NO with methyl peroxy radicals produces both NO_2 and formaldehyde (R3.15). The fate of this formaldehyde matters both for HOx concentrations and for the rate of VOC oxidation, since formaldehyde is photolysed in the troposphere to liberate HOx:

$$CH_3O_2 + NO \rightarrow CH_3O + NO_2$$

(R3.14)

$$CH_3O + O_2 \rightarrow HCHO + HO_2$$

(R3.5)

$$HCHO + h\nu \rightarrow HCO + H$$

(R3.6)

$$HCO + O_2 \rightarrow CO + HO_2$$

(R3.7)

$$H + O_2 \rightarrow HO_2$$

(R3.8)

Radicals from R3.5 and R3.7 re-enter the HOx family, which can react further (e.g. R3.1) leading to more O_3 formation. Thus, the presence of NO fundamentally shifts the oxidation process in favour of producing both ozone and HOx.

3.8 NOx reservoirs

The importance of NOx to tropospheric chemistry is clear from the preceding discussion of its influence on ozone production. It follows that any mechanism removing NOx from the atmosphere, or transporting it from polluted to clean regions, could be highly significant for ozone photochemistry — particularly given how little NO is needed for net ozone production (§3.5).

An overview of the important NOx reservoirs, that comprise NOy in the troposphere, are shown in Figure 3.9.

We already discussed the role of HNO_3 as an important NOx sink (§3.4) and its ecosystem effects (Module 6, §1.4.3).

Nitrous acid, HONO (HNO_2), is mostly formed in heterogeneous reactions from NO_2, and represents a night-time reservoir for both HOx and NOx. HONO is quickly photolysed at sunrise, regenerating NO and OH:

$$HONO + h\nu \rightarrow OH + NO$$

(R3.17)

This generates a pulse of OH into the early-morning atmosphere, partly responsible for kick-starting daytime photochemistry (see Module 7, §2.4.1).

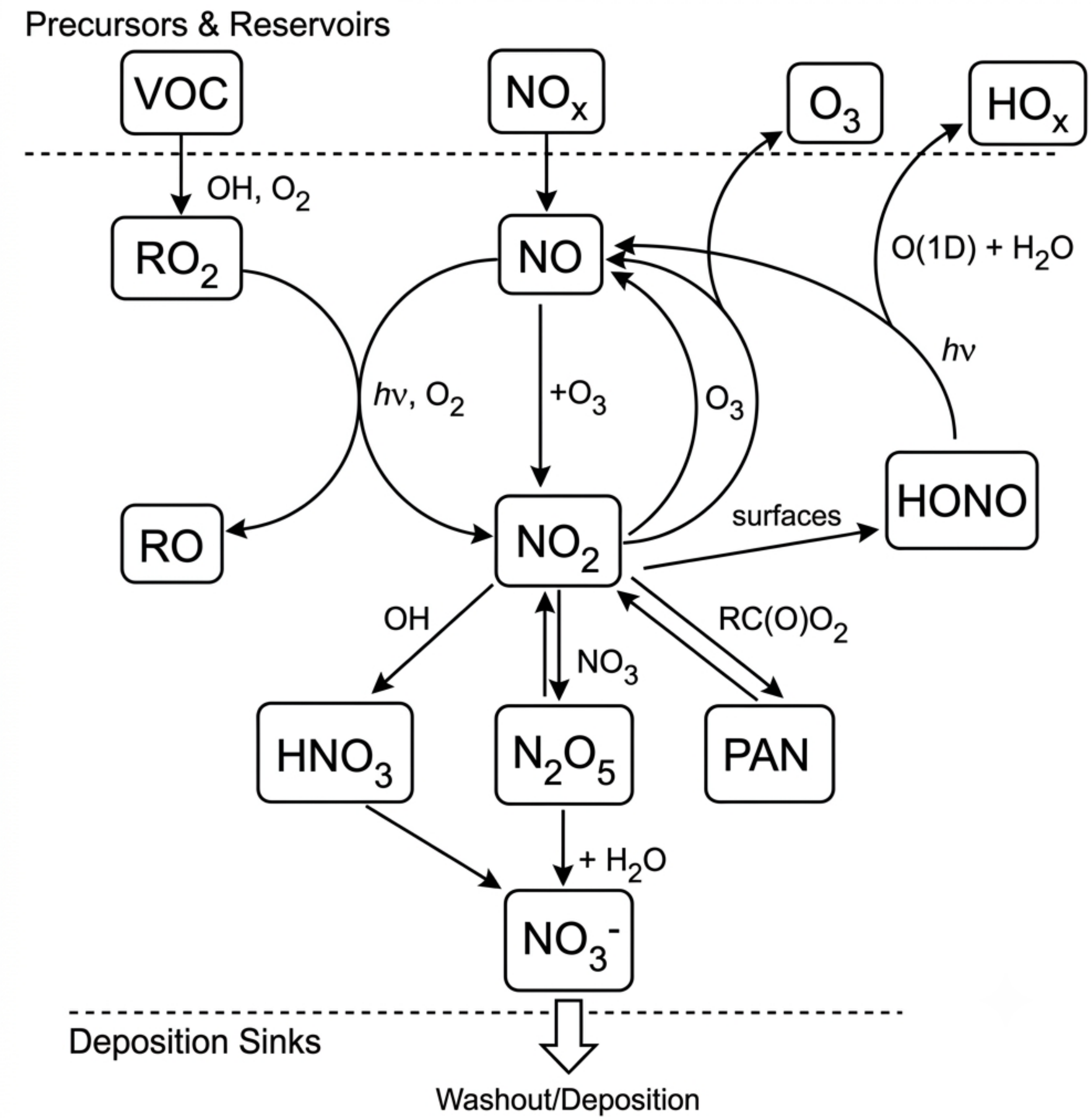


Figure 3.9 — Schematic illustration of tropospheric NOx chemistry emphasising the major NOx reservoir molecules.

3.8.1 Formation of the NOx reservoir peroxyacetyl nitrate (PAN)

Peroxyacetyl nitrate (PAN: $CH_3C(O)O_2NO_2$) forms readily in the troposphere as a degradation product of C_2+ hydrocarbons in the presence of NO_2. For example, OH + acetaldehyde (a typical VOC degradation product) forms acetylperoxy radicals:

$$CH_3CHO + OH \rightarrow CH_3CO + H_2O$$

(R3.18)

$$CH_3CO + O_2 + M \rightarrow CH_3C(O)O_2 + M$$

which react with NO_2 to form PAN. At ambient temperature, PAN has a relatively high rate of reverse decomposition, maintaining equilibrium with its precursors:

$$CH_3C(O)O_2 + NO_2 \rightleftharpoons CH_3C(O)O_2NO_2$$

(R3.19 / R3.20)

In the presence of NO, the acetylperoxy radical instead decomposes:

$$CH_3C(O)O_2 + NO \rightarrow CH_3 + CO_2 + NO_2$$

(R3.21)

(the CH_3 radical is then oxidised to HCHO by the reactions covered earlier in this course). Steady-state analysis gives the net rate constant for PAN decomposition:

$$k = k_{3.20}\left\{1 - \frac{1}{1 + \dfrac{k_{3.21}[NO]}{k_{3.19}[NO_2]}}\right\}$$

The decomposition rate depends on the relative amounts of NO and NO_2, and on temperature — $k_{3.20}$ has a large activation energy: $k_{3.20} = 4.0 \times 10^{15}$ exp$\{-108/RT\}$ s^{-1} (1 atm, E_a in kJ mol^{-1}). In the daytime boundary layer, NO and NO_2 are in photostationary state and PAN's lifetime is a few hours; at night, [NO] → 0 and the decomposition rate falls to zero.

Above the boundary layer, lower temperatures give PAN a much longer lifetime (up to several weeks) — long enough for significant transport to other parts of the atmosphere, where, on reaching warmer regions, it can dissociate, releasing NO_2 and affecting ozone chemistry there.

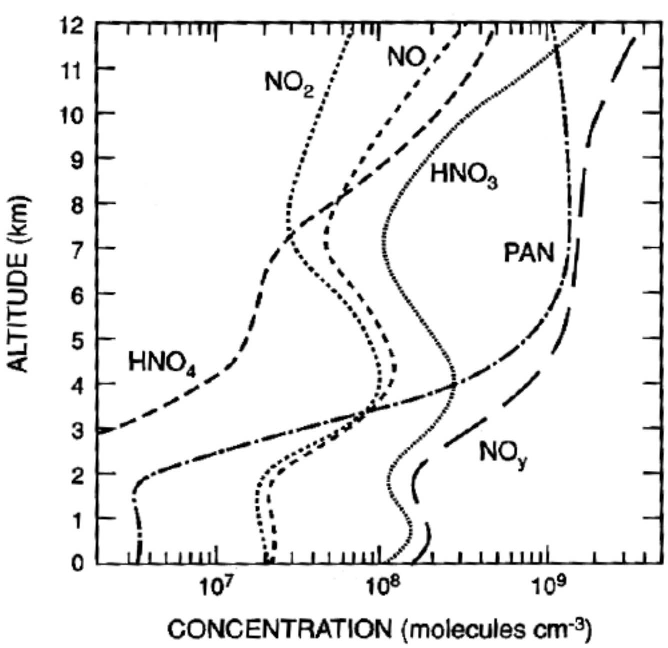


Figure 3.10 — Modelled concentrations of nitrogen compounds as a function of altitude throughout the troposphere. In the upper troposphere, PAN is the dominant nitrogen species.

PAN has indeed been detected in significant quantities in parts of the atmosphere remote from direct pollution, implying it affects the global ozone budget.

PAN should not be confused with the organic nitrates, $RONO_2$, formed in small amounts from the RO_2 + NO reaction (R = alkyl, aryl or allyl group). Organic nitrates are thermally stable, removed slowly by photolysis and reaction with OH — these too act as reservoirs for long-range NOx transport.

3.8.2 Formation of N_2O_5 from NO_3 — at night — can lead to NOx removal

NO_2 + ozone forms nitrogen pentoxide via the intermediate NO_3 radical:

$$O_3 + NO_2 \rightarrow NO_3 + O_2$$

(R3.22)

$$NO_3 + NO_2 + M \rightleftharpoons N_2O_5 + M$$

(R3.23)

N_2O_5 is thermally unstable and decomposes back to NO_3, making R3.23 reversible. NO_3 undergoes very rapid photolysis in visible light:

$$NO_3 + h\nu\ (\lambda < 580\,\text{nm}) \rightarrow NO_2 + O$$

(R3.24)

$$NO_3 + h\nu\ (\lambda < 630\,\text{nm}) \rightarrow NO + O_2$$

(R3.25)

so N_2O_5 is not formed in daytime. Night-time steady-state [NO_3] (independent of [NO_2], since production and loss are both proportional to it) is of order 1 pptv — some 30 times [OH]. Under these conditions, oxidation by NO_3 at night may rival OH oxidation for some VOCs (e.g. DMS, isoprene, terpenes).

NO_3 also reacts with NO, rapidly converting back to NO_2:

$$NO_3 + NO \rightarrow 2NO_2$$

(R3.26)

which competes with N_2O_5 formation (R3.23) near NO sources.

N_2O_5 can undergo heterogeneous reaction with water on aerosol, cloud or fog droplet surfaces to form HNO_3:

$$N_2O_5 + H_2O_{(surface)} \rightarrow 2HNO_3$$

(R3.27)

Formation of N_2O_5 thus sequesters two molecules of NOx into reservoir form, and acts as a sink for tropospheric NOx (since the resulting HNO_3 can be wet- or dry-deposited) — a significant removal process affecting both the nitrogen budget and ozone.

> Key points — Module 8/9 - Atmospheric methane is oxidised to formaldehyde, then CO, then ultimately CO_2, following reaction with photochemically generated OH. The average CH_4 lifetime is about 7 years. - The CH_4/CO oxidation mechanism, via peroxy radicals, leads to loss of tropospheric ozone when only very small [NOx] are present. - At higher NOx levels, the NO_2/O_3 photostationary state is perturbed by reaction of peroxy radicals with NO, and net ozone production occurs. - HNO_3, HONO, N_2O_5 and PAN are the major NOx sinks and reservoir species. - NO_3 drives night-time chemistry.

Try it yourself

Open `notebooks/09-nox-reservoirs.ipynb` to:

- Build the global tropospheric ozone budget (Table 3.2) as a simple flux-balance diagram, and explore how sensitive the net trend is to small percentage changes in the large gross production/loss terms.
- Implement the PAN steady-state decomposition rate constant formula and explore its dependence on $[NO]/[NO_2]$ and temperature, reproducing the "few hours in the boundary layer, weeks aloft" contrast.
- Model night-time NO_3/N_2O_5 steady state from R3.22–R3.27, and estimate what fraction of NOx gets sequestered as HNO_3 overnight under different [NO] (i.e. R3.26 competition) scenarios.

Next: Module 10 — Oxidation of Non-Methane Organic Compounds I: Alkanes and Alkenes

Oxidation of Non-Methane Organic Compounds I: Alkanes and Alkenes

Module 10 · Oxidation of Non-Methane Organic Compounds I: Alkanes and Alkenes

> Aims of this module 1. The basic reaction schemes of VOCs with OH. 2. Ozonolysis reactions of alkenes. 3. (Aromatic oxidation follows in Module 11.)

4.0 Introduction

The reactions of larger organic compounds are similar to those of methane discussed in Modules 8–9 — leading, in the presence of sufficient NOx, to the generation of ozone.

Non-methane volatile organic compounds (NMVOCs) include, by definition, all gaseous hydrocarbons and partially oxidised hydrocarbons in the atmosphere, but exclude methane, CFCs, CO and CO_2.

There are many anthropogenic and biogenic NMVOC sources, but on a global scale biogenic, natural sources dominate — only about 10–20% of NMVOC emissions are anthropogenic (fossil fuel and biomass combustion, agriculture, industry). Over 10,000 NMVOCs have been found in the atmosphere — but the structure of most of them is unknown! Urgent work is needed in analytical/experimental atmospheric chemistry to fill this knowledge gap.

Governmental environmental policy (partly aimed at reducing ozone formation) has significantly reduced anthropogenic NMVOC emissions in Europe over recent decades — the opposite trend holds in developing nations like China and India.

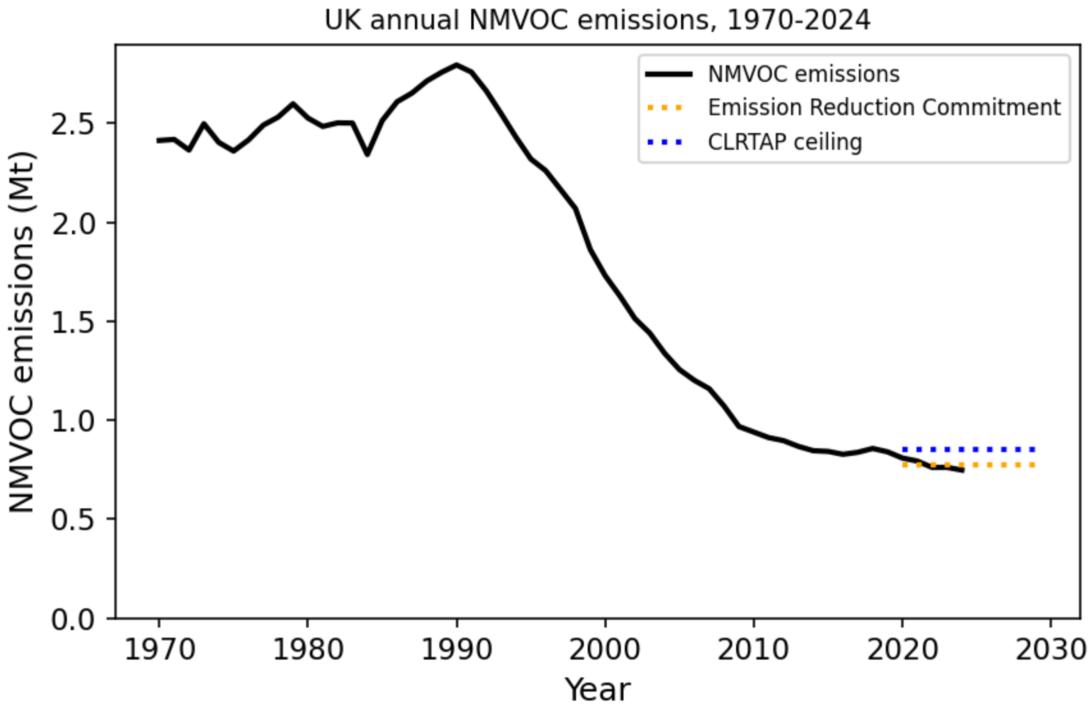


Figure 4.1 — Historical UK NMVOC emissions since 1970. Current UK and European legislated values are being adhered to.

4.1 The role of NMVOCs in ozone production

Whilst fascinating in their own right, the real reason we care about NMVOCs in the troposphere is that their photo-oxidation can generate ozone and other secondary pollutants harmful to human health and crops. There are countless possible NMVOC structures, but the capacity of different hydrocarbons to produce ozone (and HOx) depends roughly on the number of C–H bonds in the molecule — for example, C_2H_6 oxidation in the presence of NO produces around double the O_3 molecules of CH_4 oxidation to CO_2 (mole/mole basis). Generally, each $–CH_2–$ group oxidised to CO_2 in the presence of NO yields a maximum of 3 O_3 molecules.

The general mechanism of ozone formation in the presence of VOC and NOx centres on the reactions of organic (RO_2) and inorganic (HO_2) peroxy radicals, which facilitate the interconversion of NO to NO_2 and so enable tropospheric ozone production (as in Modules 8–9).

As NMVOC complexity increases, the reaction mechanism becomes more complicated. In reality, reaction with NO_3, O_3, and even direct photolysis, may also be important initiation steps — the relative importance of each pathway is set by the respective (pseudo-)first-order rate constants.

This module (and Module 11) outlines the oxidation schemes for the most abundant organic compound classes in the troposphere. Based on this you should be able to construct plausible oxidation mechanisms for many different NMVOCs.

4.2 Prediction of rate coefficients for initiation of NMVOC oxidation

The first step in NMVOC oxidation is reaction with an oxidant — suitable oxidants include OH, Cl, O_3 and NO_3 (and, to a very small extent, Br). Which dominates depends on both the abundance of the oxidant and the rate constant for its reaction with the specific NMVOC. Knowledge of these rate constants is vital for evaluating the atmospheric lifetime and degradation mechanisms of organic compounds.

Rate constants have been measured in the laboratory for many reactions, and patterns of reactivity can be recognised within a given reaction type (e.g. H-abstraction, addition to C=C bonds) related to molecular structure. There are clear trends in rate constants with carbon number for many oxidants, but also significant deviations from these trends — carbon number alone is not ultimately the best predictor of the initiation rate constant.

This nonetheless enables prediction of rate constants for molecules of known structure that are experimentally difficult to measure directly, and similar relationships can help predict products and their yields for reactions with multiple possible pathways.

4.3 Oxidation of alkanes

Under tropospheric conditions, alkanes react mostly with OH, and to a minor extent (~10%) with Cl radicals. Both proceed via H-abstraction, generating an alkyl radical plus water or HCl respectively. This reaction is generally faster for tertiary H atoms (>CH) than secondary (–CH_2–) or primary (–CH_3) H atoms.

Alkane + OH rate constants are typically 10^{-12}–10^{-11} cm³ molecule⁻¹ s⁻¹ (increasing slightly with carbon number) — about 10^3 times faster than CH_4 + OH ($k = 6.2 \times 10^{-15}$ cm³ molecule⁻¹ s⁻¹). Alkane + Cl rate constants are around 10^{-10} cm³ molecule⁻¹ s⁻¹, but tropospheric [Cl] is very low (though uncertain, estimated at 10^2–10^5 molecules cm⁻³), so Cl oxidation doesn't affect the overall alkane lifetime — though these reactions do represent a significant sink for Cl itself.

As with CH_4 oxidation, the alkyl radical reacts with O_2 to form a peroxy radical:

$$R + O_2 + M \rightarrow RO_2 + M$$

(R4.1)

$k_{4.1}$ is fairly similar across measured R groups, spanning 5×10^{-12}–2×10^{-11} cm³ molecule⁻¹ s⁻¹ (independent of M). Under standard tropospheric conditions, R has a lifetime of order 10^{-8} s.

Fate of RO_2

RO_2 can react with several tropospheric species. Reaction with NO gives two products:

$$RO_2 + NO \rightarrow RO + NO_2$$

(R4.2)

$$RO_2 + NO + M \rightarrow RONO_2 + M$$

(R4.3)

For most peroxy radicals, R4.2 dominates. For larger radicals, R4.3 becomes increasingly important, with yields up to 30%.

RO_2 can also react with HO_2 to form hydroperoxides:

$$RO_2 + HO_2 \rightarrow ROOH + O_2$$

(R4.4)

Reactions between different alkyl peroxy radicals are rather complex, with several observed overall pathways:

$$RO_2 + RO_2 \rightarrow 2RO + O_2$$

(R4.5)

$$RO_2 + RO_2 \rightarrow ROH + RCHO + O_2$$

(R4.6)

$$RO_2 + RO_2 \rightarrow ROOR + O_2$$

(R4.7)

Reaction with NO_3,

$$RO_2 + NO_3 \rightarrow RO + NO_2 + O_2$$

(R4.8)

is unimportant during daytime but may become significant at night, when $[NO_3]$ is highest.

Reaction of RO_2 with NO_2 forms peroxynitrates, which are highly unstable at tropospheric temperatures and decompose back to RO_2 + NO_2 — not a significant pathway overall.

Among all RO_2 reactions, reaction with NO is by far dominant in the polluted atmosphere. Reactions with HO_2 and other RO_2 matter mainly in the clean/remote atmosphere, where NO is in the low-ppt range.

Auto-oxidation

In recent years, hydrogen atom-shift (H-shift) reactions have attracted much attention, as they can generate highly oxidised compounds bearing multiple hydroperoxy (–OOH) groups. This chemistry requires a sufficiently long carbon chain to enable 5/6-membered-ring transition states, so it's more important for biogenic NMVOC (terpenes, C10; isoprene, C5) than for anthropogenic NMVOC (dominated by C2–C4 compounds). Over the last decade it has become established that these highly functionalised molecules not only form rapidly in the atmosphere but can lead to new aerosol particle formation, influencing clouds and climate. This process of H-shift, followed by O_2 addition and subsequent internal H-shift, is termed "auto-oxidation", since it requires no bimolecular reaction beyond addition of O_2.

Fate of RO

RO reacts via three channels: reaction with O_2, isomerization, and decomposition. The isomerization channel is only possible for compounds with more than 4 carbon atoms (able to form a 6-membered transition state). The O_2 and isomerization channels (where available) are generally more important than decomposition. Reaction of RO with NO and NO_2 to form nitro-compounds/organic nitrates is generally too slow to be significant.

Example 4 — the fate of the 2-pentoxy radical in the urban boundary layer

Using the RO reaction channels above (O_2-addition, isomerization via 6-membered TS, and decomposition), sketch the competing pathways for 2-pentoxy and estimate which dominates in a polluted urban boundary layer, given typical $[O_2]$, [NO] and temperature — see the companion notebook for a quantitative rate-constant comparison.

4.4 Oxidation of alkenes

Most alkenes are emitted from biogenic sources — trees (isoprene, terpenes, sesquiterpenes) — with a smaller anthropogenic contribution, particularly for smaller carbon-number alkenes. Besides contributing to the atmosphere's oxidative capacity (they're a direct/primary oxidant sink), alkenes are also important organic aerosol precursors. Over many regions, biogenic hydrocarbons are more abundant than anthropogenic ones.

Alkene oxidation proceeds mostly via OH-addition rather than H-abstraction:

$$C_2H_4 + OH \rightarrow HOCH_2CH_2$$

(R4.9)

The resulting hydroxy alkyl radical then reacts like the alkyl radical from alkane oxidation (e.g. with O_2 to form a hydroxy peroxy radical, $ROHO_2$).

NO_3, the main night-time oxidant, reacts across the C=C double bond, forming an excited nitro-oxy alkyl radical, which either forms an epoxide or nitro-oxy peroxy radicals (following the RO_2 scheme above). Terpenes — the most abundant atmospheric alkenes — react very rapidly with NO_3 ($k \sim 10^{-12}$–10^{-10} cm^3 molecule^{-1} s^{-1}), so NO_3 oxidation remains important even though [NO_3] rarely exceeds ppt levels.

In contrast to alkanes, alkenes also react significantly with ozone. Initial O_3 addition across the C=C bond forms a primary ozonide, which is unstable and decomposes via two pathways to form Criegee intermediates (Criegee biradicals). Though postulated over 50 years ago (with first atmospheric rate estimates over 30 years ago), it's only in the last few years that their structure and kinetics have been unambiguously identified.

In the gas phase, the Criegee intermediate is either stabilised or decomposes via various channels — one, via a hydroperoxide intermediate, yields a carbonyl and OH. This OH-formation route can be especially important at night, when other major OH sources are inactive.

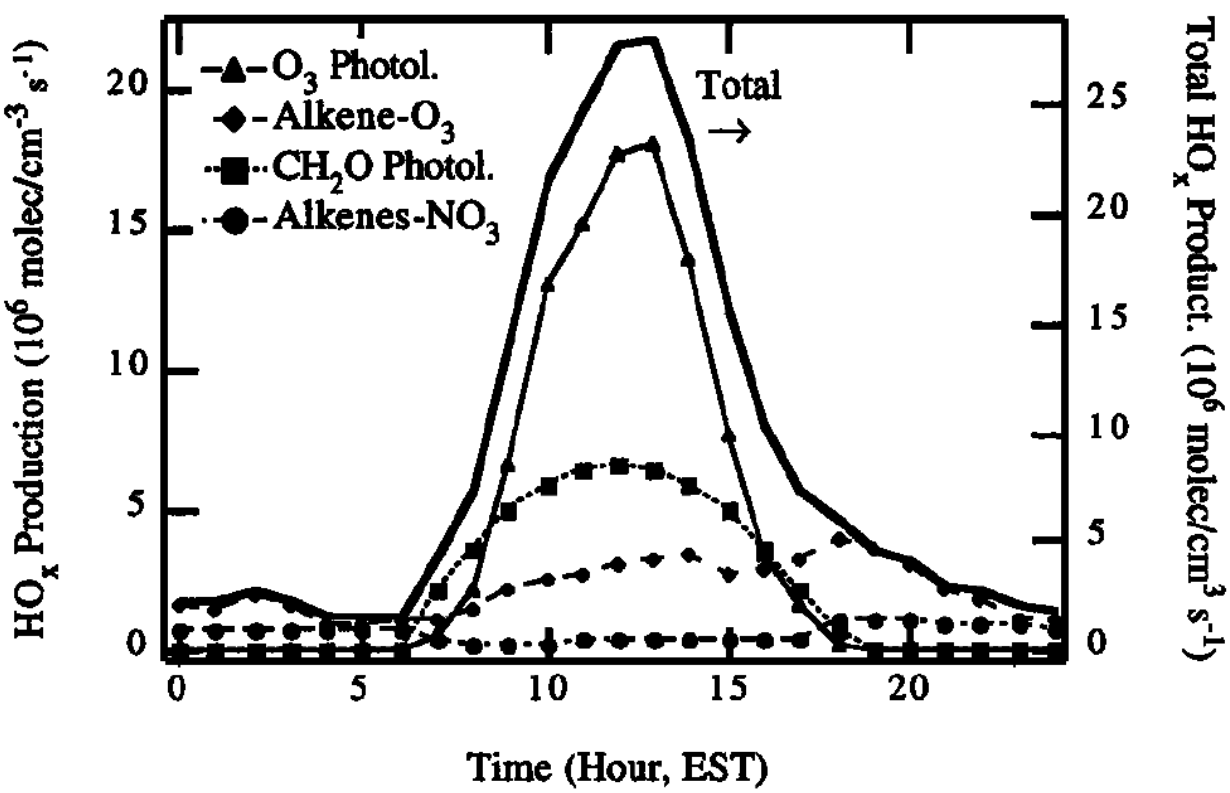


Figure 4.6 — Calculated rates of HOx generation from various sources at a rural site (Paulson and Orlando,

Geophysical Research Letters, 1996).

The major fate of the stabilised Criegee intermediate is believed to be reaction with H_2O, leading to carboxylic acids and hydroxy-hydroperoxides. Other reactions include those with NO_2, SO_2 and organics — the relative importance of these is not fully characterised.

Try it yourself

Open `notebooks/10-nmvoc-alkanes-alkenes.ipynb` to:

- Compare alkane OH- vs. Cl-oxidation lifetimes given realistic [OH] and [Cl] ranges, and confirm Cl oxidation doesn't control the overall alkane lifetime.
- Solve Example 4 (2-pentoxy radical fate) by comparing rate constants for the O_2, isomerization and decomposition channels at representative urban boundary-layer [O_2], [NO], T.
- Estimate O_3 yield per NMVOC carbon atom from the "max 3 O_3 per $-CH_2-$ group" rule, and compare ethane vs. a longer-chain alkane.
- Build a simple RO_2 fate calculator (R4.2–R4.8) and see how the dominant RO_2 loss pathway shifts between polluted and remote-marine [NO] regimes.

Next: Module 11 — Oxidation of Non-Methane Organic Compounds II: Aromatics and Oxygenated Organics

Oxidation of Non-Methane Organic Compounds II: Aromatics and Oxygenated Organics

Module 11 · Oxidation of Non-Methane Organic Compounds II: Aromatics and Oxygenated Organics

> Aims of this module 3. The oxidation scheme of aromatic compounds (continuing from Module 10).

4.5 Oxidation of aromatics

Most aromatics are emitted by anthropogenic processes — motor vehicle emissions, solvent use. They are also the most important anthropogenic organic aerosol precursors. Atmospherically relevant aromatics include benzene, toluene, xylenes and other methylated/ethylated benzene derivatives. OH is the most important oxidant for aromatics, with two competing pathways:

(a) H-abstraction (yield < 10%). Further reactions follow the alkane pathway, ultimately leading to aldehydes (e.g. benzaldehyde from toluene oxidation) or aromatic nitrates.

(b) OH addition to the aromatic ring (yield > 90%). Further reactions of the resulting OH–aromatic adduct are rather unclear, with many possible pathways proposed. The oxidation of aromatic compounds in the atmosphere is one of the least understood areas in tropospheric chemistry.

One of the more likely reaction routes (illustrated for toluene) generates mainly small dicarbonyls, like butenedial and methylglyoxal.

4.6 Oxidation of oxygenated organics

The alkane, alkene and aromatic reactions above all yield oxygenated compounds, which react further — ultimately to CO_2, or are removed by wet and dry deposition.

- Aldehydes react mostly with OH via abstraction of the weakly bound aldehyde H-atom. Further reactions involve O_2 addition and reaction with NO/NO_2, as described in Module 10.
- Ketones react similarly to alkanes, via H-abstraction at the alkyl chain.
- Alcohols also react analogously to alkanes (the alcohol O–H bond is stronger than the alkyl C–H bond, so abstraction of the alcohol H is unlikely).
- Carboxylic acids also undergo H-abstraction, but the main removal mechanism for most acids is wet and dry deposition.

Aldehydes (especially formaldehyde), ketones, and organic hydroperoxides are also readily photolysed in the troposphere — feeding back into HOx production, as discussed throughout Modules 8–10.

> Key points — Module 10/11 (Chapter 4) - The OH-oxidation mechanism for many VOCs involves peroxy radicals, and leads, in the presence of NOx, to net production of ozone and HOx. - Alkenes

react with OH but also efficiently with ozone, forming ozonides and ultimately hydroperoxides and carboxylic acids. - Tropospheric oxidation of aromatic compounds is highly complex and only partly understood; small dicarbonyls are abundant oxidation products. - Alkenes and aromatics are important biogenic and anthropogenic precursors of organic aerosols, respectively.

Example 5 — propose a mechanism for the oxidation of 2-methyl-2-butene

Using the alkene oxidation framework from Module 10 (§4.4) — OH addition across the C=C bond, the competing O_3 + alkene → Criegee pathway, and the general RO_2 fate scheme from §4.3 — sketch a plausible oxidation mechanism for 2-methyl-2-butene, and identify the expected major carbonyl products. The companion notebook sets up the rate-constant comparison needed to judge which initiation pathway (OH vs. O_3 vs. NO_3) dominates under different conditions.

Try it yourself

Open `notebooks/11-aromatics-oxygenates.ipynb` to:

- Compare OH-abstraction vs. OH-addition branching for a toluene-like aromatic, and see how the >90%/<10% split shapes the product distribution.
- Solve Example 5 by comparing pseudo-first-order initiation rates ($k_{OH}[OH]$, $k_{O_3}[O_3]$, $k_{NO_3}[NO_3]$) for a representative alkene across day/night and clean/polluted scenarios, to identify which oxidant dominates initiation.
- Build a simple multi-generation oxidation tracker (alkane → alkyl radical → peroxy radical → carbonyl → further oxidation), and estimate how many generations of oxidation are needed to reach CO_2 for a chosen starting NMVOC.
- Wrap up the course: combine Module 10/11's rate-constant machinery with Module 7's OH steady-state calculator to estimate whole-atmosphere NMVOC oxidation lifetimes for a few compounds of your choice, and compare to Table 2.2 (Module 7).

This concludes the Lent term (Lectures 7–12) and the B4 course. See the course README for the full module list.

Problem Sets

Suggested Exercises and Sample Tripos Questions for Part II Chemistry in the Atmosphere (Lectures 1–6)

Students will find past tripos papers from 2015 onwards helpful.

Chapter 1: Basic Physical and Chemical Structure

1. Calculate the mass of the atmosphere. Mean radius of earth = 6371 km and acceleration due to gravity (g) is 9.8 m s^{-2}.

2. What is the difference between the mass mixing ratio of CO_2 in air and the volume mixing ratio of CO_2 in air?

3. By what % will the mass change by doubling of the amount of CO_2 in the atmosphere from the present mixing ratio of 400 ppm?

4. Calculate how the scale heights for Venus and Mars compare with Earth? Mars and Venus have atmospheres dominated by CO_2 and gravitational acceleration (g) of 3.7 and 8.9 m s^{-2} respectively. The mean atmospheric temperatures of Mars and Venus are 210 K and 600 K respectively.

5. Give a rough account of the timescale for transport and mixing in the atmosphere.

6. Explain why mixing ratios are a useful concept in atmospheric chemistry?

7. Calculate the number of molecules cm^{-3} at: (i) the earth's surface (ii) 30 km (iii) 50 km. Take H, the scale height, = 7 km.

8. If the O_3 mixing ratio is (i) 50 ppb at the surface (ii) 18 ppm at 30 km and (iii) 2 ppm at 50 km. Calculate the respective concentrations (cm^{-3}) using the results from question 7.

Chapter 2: Chemical Kinetics in the Atmosphere

9. Calculate the bond dissociation energies for O_2 and O_3 given the following heats of formation: O_3 = 142 kJ mol^{-1}, O = 249 kJ mol^{-1}. Calculate the values in kJ mol^{-1} and as the equivalent wavelength in nm.

10. Calculate the time scale for equilibrium between O and O_3 and between NO and NO_2 at 30 km and comment on the validity of steady state under these conditions.

Data: At 30 km Temperature = 250 K. NO = 3 ppb, NO_2 = 4 ppb, O_3 = 8 ppm and O = 90 ppt ([M] = 4×10^{17} cm^{-3}).

Reaction	Rate/photolysis coefficient
$NO_2 + h\nu \rightarrow NO + O$	$J\_NO2 = 10^{-2}\ s^{-1}$
$NO + O_3 \rightarrow NO_2 + O_2$	$k_2 = 2.07 \times 10^{-12} \exp(-1400/T)\ cm^3\ s^{-1}$
$O_3 + h\nu \rightarrow O + O_2$	$J\_O3 \approx 2 \times 10^{-3}\ s^{-1}$
$O + O_2 + M \rightarrow O_3 + M$	$k_4 = 6 \times 10^{-34}(300/T)^2\ cm^6\ s^{-1}$

11. We saw in Exercise 5 that we can relate bi-molecular and ter-molecular rate constants by considering the role of pressure quenching. The rate coefficients for the following reactions exhibit a pressure dependence over the range 0.001–1 bar.

- $OH + HNO_3 \rightarrow NO_3 + H_2O$, $\Delta H° = -75$ kJ mol^{-1}
- $OH + NO_2 + M \rightarrow HNO_3 + M$, $\Delta H° = -207$ kJ mol^{-1}

Explain in broad terms the origin of these pressure dependencies. Sketch the potential energy curves for the reaction pathway in each case, considering the different nature of the products formed. Why do both rate constants increase as the temperature decreases?

Chapter 3: Atmospheric Photochemistry

12. Give a brief account of the factors which determine the photolysis rates of different constituents in the earth's atmosphere.

13. The following photolysis reaction occurs at wavelengths less than approximately 240 nm:

$O_2 + h\nu \rightarrow O + O$, J_1

Calculate the ratio of the contribution to J_1 at $\lambda = 220$ nm at altitudes of 40 km and 20 km, given the following information:

Absorption cross section of O_2 at 220 nm = 6×10^{-24} cm^2. Absorption cross section of O_3 at 220 nm = 2×10^{-18} cm^2. The overhead columns (i.e. $\int$[concentration] dz) of molecular oxygen and ozone above 20 km and 40 km are given in the table below.

	Column above 20 km (cm^{-2})	Column above 40 km (cm^{-2})
$\int[O_2]\,dz$	2.5×10^{23}	1.4×10^{22}
$\int[O_3]\,dz$	6.3×10^{18}	1.7×10^{17}

Chapter 4: Stratospheric Ozone Chemistry

14. The Chapman reactions can be written as:

Reaction	Rate/photolysis coefficient
$O_2 + h\nu \rightarrow O + O$	$J_2 \approx 5 \times 10^{-10}\ s^{-1}$
$O + O_2 + M \rightarrow O_3 + M$	$k_3 = 6 \times 10^{-34}(300/T)^2\ cm^6\ s^{-1}$
$O_3 + h\nu \rightarrow O + O_2$	$J_3 \approx 2 \times 10^{-3}\ s^{-1}$
$O + O_3 \rightarrow O_2 + O_2$	$k_4 = 1.0 \times 10^{-11} \exp(-2100/T)\ cm^3\ s^{-1}$

Use this reaction scheme to derive an expression for the equilibrium ozone concentration. State any assumptions that you make.

15. How does this expression account for the layer-like structure of ozone in the atmosphere?

16. Derive the equilibrium O_3 concentrations (i) assuming an overhead sun, and (ii) for the sun at 70° from the vertical.

Data: At 30 km, [M], the number density, = 3.7×10^{17} cm^{-3}, and the overhead O_3 column is 1.7×10^{18} cm^{-2}. J_2 is determined by photolysis at 200 nm, where the O_2 cross-section is 10^{-23} cm^2 and the O_3 cross-section is 5×10^{-19} cm^2, and J_3 is determined by photolysis at 500 nm where the O_3 cross-section is 2×10^{-21} cm^2. The photon fluxes at the top of the atmosphere at the two wavelengths are 2×10^{12} and 2.8×10^{17} cm^{-2} s^{-1}, respectively. The scale height, H = 7.5 km. Temperature = 250 K.

17. During the spring equinox, the equilibrium concentrations calculated in question 16 would be representative of those at 30 km above the Equator and at 70°N. The observed mixing ratios at these two latitudes during the spring equinox are 7 ppm and 5 ppm, respectively. Comment on these observations.

§ 4.3 NOx

18. Nitrogen oxides (NOx = NO + NO_2) give rise to odd oxygen loss in the stratosphere and production in the free troposphere. Odd oxygen loss depends on the following reactions:

Reaction	Rate/photolysis coefficient
$O_3 + h\nu \rightarrow O_2 + O$	$J_1 = 10^{-3}$ s^{-1}
$O + O_2 + M \rightarrow O_3 + M$	$k_2 = 10^{-33}$ cm^6 s^{-1}
$NO + O_3 \rightarrow NO_2 + O_2$	$k_3 = 4 \times 10^{-15}$ cm^3 s^{-1}
$NO_2 + O \rightarrow NO + O_2$	$k_4 = 10^{-11}$ cm^3 s^{-1}
$NO_2 + h\nu \rightarrow NO + O$	$J_5 = 10^{-2}$ s^{-1}

Derive an expression for the odd oxygen loss rate, identifying the rate limiting step and the ‘null’ cycle. Explain how the loss cycle depends on altitude.

19. Satellite measurements of HNO_3 and NO_2 are available at 40 km, where $[NO_2] = 1 \times 10^9$ cm^{-3}, and $[HNO_3] = 4 \times 10^8$ cm^{-3}. Derive the concentration of OH at 40 km given the following important reactions.

Reaction	Rate/photolysis coefficient
$OH + NO_2 + M \rightarrow HNO_3 + M$	$k_1[M] = 8.0 \times 10^{-12}$ cm^3 s^{-1}
$HNO_3 + h\nu \rightarrow OH + NO_2$	J_2
$HNO_3 + OH \rightarrow H_2O + NO_3$	$k_3 = 1.8 \times 10^{-13}$ cm^3 s^{-1}

Data: In the spectral region where there are incident solar photons at 40 km, the mean absorption cross section of HNO_3 is 1×10^{-21} cm^2, the quantum yield for OH formation is unity, and the photon intensity averaged over the same wavelength region is 3.5×10^{16} cm^{-2} s^{-1}.

Justify the use of any steady state relationship in your derivation by an analysis of the time scale for steady state.

§ 4.4 HOx

20. Discuss the role of OH in controlling the distribution of stratospheric ozone. Your answer should include both direct and indirect effects.

21. In the upper stratosphere hydrogen radicals destroy odd-oxygen by the following reactions:

Reaction	Rate/photolysis coefficient
$H + O_2 + M \rightarrow HO_2 + M$	$k_3 = 8.0 \times 10^{-32}$ cm^6 s^{-1}
$HO_2 + O \rightarrow OH + O_2$	$k_4 = 6.0 \times 10^{-11}$ cm^3 s^{-1}
$OH + O_3 \rightarrow HO_2 + O_2$	$k_5 = 3.0 \times 10^{-11}$ cm^3 s^{-1}
$OH + O \rightarrow H + O_2$	$k_6 = 3.0 \times 10^{-11}$ cm^3 s^{-1}

Show that the rate of odd-oxygen destruction by these reactions is given by $d[Ox]/dt = -2k_6[OH][O]$.

Calculate the rate of odd-oxygen destruction for the altitude where $[O] = [O_3] = 2 \times 10^{10}$ cm^{-3} and $[HOx] = [H] + [OH] + [HO_2] = 6 \times 10^6$ cm^{-3}. $[M] = 8 \times 10^{15}$ cm^{-3}.

22. HOx in the stratosphere is mainly produced, removed and interconverted by the following series of reactions:

#	Reaction	Rate/photolysis coefficient
1	$O(^1D) + H_2O \rightarrow OH + OH$	$k_1 = 2.2 \times 10^{-10}$ cm^3 s^{-1}
2	$HO_2 + OH \rightarrow H_2O + O_2$	$k_2 = 1.0 \times 10^{-10}$ cm^3 s^{-1}
3	$OH + O_3 \rightarrow HO_2 + O_2$	$k_3 = 3.0 \times 10^{-11}$ cm^3 s^{-1}
4	$HO_2 + O \rightarrow OH + O_2$	$k_4 = 6.0 \times 10^{-11}$ cm^3 s^{-1}
5	$HO_2 + NO \rightarrow NO_2 + OH$	$k_5 = 1.3 \times 10^{-13}$ cm^3 s^{-1}
6	$NO_2 + h\nu \rightarrow NO + O$	J_6
7	$O + O_2 + M \rightarrow O_3 + M$	k_7

Show that the rate of odd oxygen loss resulting from reactions [3]–[7] is given by $2k_4[HO_2][O]$ and calculate the rate of odd oxygen loss from these reactions at 40 km.

Data: $[H_2O] = 4 \times 10^{11}$ cm^{-3}, $[O_3] = 5 \times 10^{11}$ cm^{-3}, $[O] = 4 \times 10^8$ cm^{-3}, $[O(^1D)] = 160$ cm^{-3} and $[NO] = 8 \times 10^8$ cm^{-3}.

23. Calculate the lifetime of odd oxygen based on your answer above. Comment on this.

§ 4.5 Halogens

24. The following reactions play a role in the low stratosphere:

Reaction	Rate/photolysis coefficient
$Cl + O_3 \rightarrow ClO + O_2$	k_1
$ClO + O \rightarrow Cl + O_2$	$k_2 = 4 \times 10^{-11}$ cm^3 s^{-1}
$Br + O_3 \rightarrow BrO + O_2$	$k_3 = 1.2 \times 10^{-12}$ cm^3 s^{-1}
$BrO + O \rightarrow Br + O_2$	$k_4 = 4 \times 10^{-11}$ cm^3 s^{-1}
$ClO + BrO \rightarrow Cl + Br + O_2$	$k_5 = 1 \times 10^{-12}$ cm^3 s^{-1}
$ClO + NO \rightarrow Cl + NO_2$	$k_6 = 2 \times 10^{-11}$ cm^3 s^{-1}
$NO + O_3 \rightarrow NO_2 + O_2$	$k_7 = 7 \times 10^{-15}$ cm^3 s^{-1}
$NO_2 + O \rightarrow NO + O_2$	$k_8 = 1 \times 10^{-11}$ cm^3 s^{-1}
$NO_2 + h\nu \rightarrow NO + O$	$J_9 = 1 \times 10^{-2}$ s^{-1}

Show that the rate of odd-oxygen loss can be expressed as:

$d[Ox]/dt = -2k_2[ClO][O] - 2k_4[BrO][O] - 2k_8[NO_2][O] - 2k_5[ClO][BrO]$

Explain why the reaction between ClO and BrO appears in the expression.

Evaluate the odd-oxygen loss given the following data: $[O_3] = 10^{12}$ cm^{-3}; $[O] = 10^7$ cm^{-3}; $[ClO] = 5 \times 10^8$ cm^{-3}; $[NO] = 10^9$ cm^{-3}; $[Br] = 10^5$ cm^{-3}.

Comment on the result of your calculation.

25. Calculate the rate of odd oxygen destruction from the following mechanism:

Reaction	Rate/photolysis coefficient
$Cl + O_3 \rightarrow ClO + O_2$	$k_3 = 1.0 \times 10^{-11}$ cm^3 s^{-1}
$ClO + O \rightarrow Cl + O_2$	$k_4 = 4.0 \times 10^{-11}$ cm^3 s^{-1}
$ClO + NO \rightarrow NO_2 + Cl$	$k_5 = 2.0 \times 10^{-11}$ cm^3 s^{-1}
$NO_2 + h\nu \rightarrow NO + O$	J_6
$O + O_2 + M \rightarrow O_3 + M$	$k_2[M] = 8.0 \times 10^{-17}$ cm^3 s^{-1}
$Cl + CH_4 \rightarrow HCl + CH_3$	$k_7 = 4.0 \times 10^{-13}$ cm^3 s^{-1}
$HCl + OH \rightarrow Cl + H_2O$	$k_8 = 5.0 \times 10^{-13}$ cm^3 s^{-1}

Data: At 40 km $[O_3] = 5 \times 10^{11}$ cm^{-3}; $[OH] = 1.6 \times 10^7$ cm^{-3}; $[CH_4] = 2 \times 10^{10}$ cm^{-3}; $[NO] = 1 \times 10^9$ cm^{-3}; $[HCl] = 8 \times 10^8$ cm^{-3} and $[O_2] = 1.5 \times 10^{16}$ cm^{-3}. $[O] = 3.33 \times 10^9$ cm^{-3}.

26. How does your answer from question 25 compare to that from destruction by the reaction:

$O + O_3 \rightarrow O_2 + O_2$, $k_9 = 2.0 \times 10^{-15}$ cm^3 s^{-1}

Exercises — Part II Chemistry in the Atmosphere: Tropospheric Chemistry (Lent 2024)

Students will also find past tripos questions since 2015 relevant to the content of this course.

Chapter 1

1. The deposition velocity of ozone to the ground is 10^{-2} m s^{-1} and the mixed layer height is 500 m.
   (a) If the rate of photochemical ozone production is approximately constant at 10 ppb/hr during the day, what is the maximum possible $[O_3]$?
   (b) Is this concentration likely to be achieved?

2. Methyl Chloroform (1,1,1-trichlorethane) is an Ozone Depleting Substance formerly used widely as a solvent. Under the provisions of the Montreal Protocol its production and release ceased in 1992. The observed mean concentrations of Methyl Chloroform in the troposphere in recent years are given below:

Year	1993	1994	1995	1996	1997	1998	1999
$[CCl_3CH_3]$ / pptv	152	129	112	99	82	70	61

   (a) Use this data to determine the atmospheric lifetime of Methyl Chloroform. The release rate prior to the control was 1.2×10^{12} g yr^{-1}.
   (b) What would the steady-state volume mixing ratio of Methyl Chloroform have been if emissions had continued at this rate? Assume a uniformly mixed atmosphere.

3. Methyl Chloroform is removed from the atmosphere by reaction with OH radicals:

$OH + CCl_3CH_3 \rightarrow H_2O + CCl_3CH_2$

Estimate the global mean concentration of OH using the results from the analysis above ($k = 1.2 \times 10^{-14}$ cm^3 $molecule^{-1}$ s^{-1}) and comment on your result.

Chapter 2

1. At wavelengths below 398 nm the quantum yield for the photolysis of NO_2 is unity:

- $NO_2 + h\nu \rightarrow O + NO$, J_NO2
- $O + O_2 + M \rightarrow O_3 + M$
- $NO + O_3 \rightarrow NO_2 + O_2$

The average values for the NO_2 absorption cross section (cm^2 molecule^{-1}) and the daytime solar flux (photons cm^{-2} s^{-1}) in the 300–398 nm region at the earth's surface are 3.0×10^{-19} and 2.75×10^{16} respectively.

(a) What is the value of the photolysis constant for NO_2, J_NO2?

(b) Assume photostationary state for O_3, NO and NO_2, i.e., for the three reactions outlined above. If [O_3] = 60 ppbv and [NOx] = 15 ppbv, what is the concentration of NO_2 in the sunlit atmosphere? (Rate constant for NO + O_3 reaction = 2.0×10^{-14} cm^3 molecule^{-1} s^{-1}, 1 bar = 2.46×10^{19} molecule cm^{-3}.)

2. Photolysis of O_3 in the troposphere leads to the production of hydroxyl radicals.

(a) Outline the sequence of reactions that lead to OH formation and derive an expression for the rate of OH formation in terms of the photolysis rate of ozone at wavelengths less than approximately 310 nm (J_2), the concentrations of O_3, H_2O and the total gas concentration, M.

(b) What is the mean rate of production of OH from this process for the surface atmospheric conditions (1 bar and 298 K) where the relative humidity is 50% and the local ozone concentration is 1×10^{12} molecules cm^{-3}?

$J(O^1D) = 3 \times 10^{-5}$ s^{-1}, $k(O(^1D) + H_2O)/k(O^1D + air) = 7.1$, saturated vapour pressure of water is 30 mbar at 298 K.

(c) How do you expect OH production to vary with altitude in the troposphere?

(d) In clean air at the earth's surface (pressure = 1 bar, temperature 298 K), OH is removed primarily by its reaction with CO:

$OH + CO \rightarrow H + CO_2$, k_CO = 2×10^{-13} cm^3 s^{-1}

Write an expression for the steady state concentration of OH and evaluate it using the following information:

$J_2 = 2.7 \times 10^{-5}$ s^{-1}; [O_3] = 40 ppbv; [CO] = 100 ppbv; [H_2O] = 0.02 bar

$k(O(^1D) + M) = 3 \times 10^{-11}$ cm^3 s^{-1}, $k(O(^1D) + H_2O) = 2.2 \times 10^{-10}$ cm^3 s^{-1}

3. The following simplified reaction scheme provides a description of the chemistry governing concentrations of OH and HO_2 in the troposphere:

Reaction	Rate coefficient (cm^3 s^{-1} unless noted)
$O_3 + h\nu\ (\lambda<315\text{ nm}) \rightarrow O(^1D) + O_2$	$k_1 = 2.5 \times 10^{-6}$ s^{-1}
$O(^1D) + M \rightarrow O(^3P) + M$	$k_2 = 3.0 \times 10^{-11}$
$O(^1D) + H_2O \rightarrow 2OH$	$k_3 = 2.2 \times 10^{-10}$
$CO + OH \rightarrow CO_2 + H$	$k_4 = 2.1 \times 10^{-13}$
$H + O_2 + M \rightarrow HO_2 + M$	k_5 (very fast)
$HO_2 + O_3 \rightarrow OH + 2O_2$	$k_6 = 2.0 \times 10^{-15}$
$HO_2 + HO_2 \rightarrow H_2O_2 + O_2$	$k_7 = 3.5 \times 10^{-12}$

Temperature = 298 K; pressure = 1 bar ([M] = 2.46×10^{19} cm^{-3}).

(a) Derive an expression for the steady state concentration of HOx radicals (OH + HO_2).

(b) Evaluate [OH] in the lower troposphere for [CO] = 100 ppbv; [O_3] = 40 ppbv; and [H_2O] = 0.02 bar. You may assume that HOx interconversion by reactions 4 and 6 is rapid.

(c) How would the steady state of HOx radicals vary with solar intensity?

(d) What is the effect of these processes on tropospheric ozone?

In the presence of NOx (NO + NO_2), the following reactions occur:

Reaction	Rate coefficient ($cm^3\ s^{-1}$)
$HO_2 + NO \rightarrow OH + NO_2$	$k_8 = 8.6 \times 10^{-12}$
$NO_2 + OH \rightarrow HNO_3$	$k_9 = 1.1 \times 10^{-11}$

(e) Assuming NO = 0.1 ppb and NO_2 = 0.36 ppb, derive a new expression for the HOx steady state.

(f) Evaluate [OH].

(g) Comment on its dependence on solar intensity.

Chapter 3

1. The dominant loss process for tropospheric CH_4 is by the reaction:

$OH + CH_4 \rightarrow CH_3 + H_2O$, $k_1 = 7.7 \times 10^{-15}$ cm^3 $molecule^{-1}$ s^{-1}

The production rate of OH from the $O(^1D) + H_2O$ reaction P = 5×10^5 molecule cm^{-3} s^{-1} and OH is lost only by reaction with CH_4 and with CO.

$OH + CO \rightarrow H + CO_2$, $k_2 = 2.0 \times 10^{-13}$ cm^3 $molecule^{-1}$ s^{-1}

(a) Evaluate the steady state concentration of OH ([CO] = 100 ppb and [CH_4] = 1.7 ppm).

(b) Estimate the mean chemical lifetime of CH_4 in the troposphere.

(c) What vertical profile would you expect for CH_4 in the troposphere?

2. The ratio of OH to HO_2 in the lower atmosphere is governed mainly by the following reactions:

- $OH + CO \rightarrow CO_2 + H$, $k_1 = 2 \times 10^{-13}$ cm^3 $molec.^{-1}$ s^{-1}
- $H + O_2 + M \rightarrow HO_2 + M$ (very fast)
- $HO_2 + NO \rightarrow NO_2 + OH$, $k_2 = 8 \times 10^{-12}$ cm^3 $molec.^{-1}$ s^{-1}

(a) Evaluate the steady state concentration of HO_2 for the following trace gas concentrations (in molec. cm^{-3}): [OH] = 4×10^6, [NO] = 1.2×10^9, [CO] = 4×10^{12}, [O_3] = 40 ppb (p = 1 atm, T = 298 K).

The steady state of NO is controlled by reaction (2) plus the following reactions:

- $NO_2 + h\nu \rightarrow NO + O$, $J_3 = 1 \times 10^{-2}$ s^{-1}
- $O + O_2 + M \rightarrow O_3 + M$ (very fast)
- $O_3 + NO \rightarrow NO_2 + O_2$, $k_4 = 2 \times 10^{-14}$ cm^3 $molec.^{-1}$ s^{-1}

(b) Derive an expression for the rate of ozone production and evaluate it for the above concentrations. You may assume that the ratio [NO_2]/[NO] is approximated by the photostationary state, i.e. reactions 2, 3 and 4.

3. Peroxy acetyl nitrate (PAN) is a characteristic product of atmospheric photochemical degradation of non-methane hydrocarbons. It is formed by the addition reaction of acetyl peroxy radicals with NO_2. It reacts only very slowly with OH radicals, is not photolysed but thermally decomposes by the following mechanism:

$CH_3C(O)O_2NO_2 \rightleftharpoons CH_3C(O)O_2 + NO_2$, k_1/k_{-1}

$CH_3C(O)O_2 + NO \rightarrow CH_3 + CO_2 + NO_2$, k_2

followed by oxidation of CH_3 to formaldehyde.

(a) Derive an expression for the overall rate of thermal decomposition of PAN in an atmosphere containing NO and NO_2 and also an expression for its lifetime.

(b) In daylight, the ratio [NO_2]/[NO] is governed primarily by the photostationary state involving the following reactions:

- $NO_2 + h\nu \rightarrow NO + O$, J4
- $O + O_2 + M \rightarrow O_3 + M$, k3 (v. fast)
- $NO + O_3 \rightarrow NO_2 + O_2$, k5

Derive an expression for the ratio $[NO_2]/[NO]$.

(c) Evaluate the lifetime of PAN at 298 K in sunlit boundary layer, with $[O_3]$ = 50 ppb and J4 = 1×10^{-2} s^{-1}.

(d) A parcel of polluted boundary layer air containing PAN is transferred by convection to 5 km altitude where the ambient daytime temperature is 260 K. Assuming that J4 and the mixing ratio of O_3 are independent of altitude, calculate the chemical lifetime of PAN at 5 km.

Data: $k1 = 4.0 \times 10^{15} \exp(-13000/T)$ s^{-1}; $k_{-1}/k2 = 0.5$ (independent of temperature); $k5 = 2.0 \times 10^{-12} \exp(-1400/T)$ cm^3 $molecule^{-1}$ s^{-1}; 1 ppb = 2.46×10^{10} molecule cm^{-3} at 298 K and 1 bar; atmospheric pressure at 5 km = 0.542 bar.

Chapter 4

1. Outline, in brief, the role that organic peroxy radicals play in the formation of tropospheric ozone.

2. What is the main difference in the oxidation schemes of alkanes and alkenes?

3. Two oxidation products of limonene have been observed. Show mechanisms that would explain these products. Consider all relevant atmospheric oxidants.

4. This question wants you to think about the atmospheric fate of 1-pentene.

(a) What are the atmospherically important oxidants for 1-pentene?

(b) Show the reaction mechanism for each important oxidant following through to the appearance of the first stable (closed shell) products. (Consider conditions with and without nitrogen oxide present in the atmosphere.)

5. Explain the mechanism of formation of these two products from oxidation of alpha-pinene and explain which oxidants are going to dominate alpha-pinene destruction in a NOx-free tropospheric environment. [2.5+2.5+2, 2.5 points for each mechanism and 2 for the oxidants] (from Tripos 2021, Paper 3 Q44)

6. Explain the mechanism of formation of these two products from oxidation of isoprene (2-methyl-1,3-butadiene) in a polluted environment at nighttime. State which oxidants are relevant in these conditions. [7] (from Tripos 2022, Paper 3 Q43)